

REGION X



EMS

May 25, 2026
Advanced Life Support
Standard Operating Procedures

Version 202605.1

SIGNATURES

Region X Standard Operating Procedures

These protocols have been developed and approved through a collaborative process involving the five Emergency Medical Service (EMS) Systems located in the EMS/Trauma Region X of the Illinois Department of Public Health (IDPH).

- Northwestern Medicine North Region EMS System, Lake Forest, IL
- Advocate Condell Medical Center EMS System, Libertyville, IL
- Endeavor Health Highland Park Hospital EMS System, Highland Park, IL
- Ascension Saint Francis Hospital EMS System, Evanston, IL
- Vista Health North Lake County EMS System, Waukegan, IL

These protocols shall be used:

- As written practice guidelines and pre-hospital standing medical orders approved by the EMS Medical Directors and to be initiated by the System EMS personnel for off-line medical control,
- As the standing medical orders to be used by Emergency Communication Radio Nurses (ECRNs) when providing on-line medical control
- in disaster situations, given that the usual and customary forms of communication are contraindicated as specified in the Region X Multiple Patient Management Plan (MPMP)

The signatures of the EMS System Medical Directors listed below officially authorize the provision of emergency medical care by Region X EMS personnel and hospital-based Emergency Communication Registered Nurses. These protocols have been approved by the Illinois Department of Public Health.



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USE OF STANDARD OPERATING PROCEDURES

The following Standard Operating Procedures (SOPs) are to be employed for all patients requiring pre-hospital medical treatment within the IDPH designated EMS/Trauma Region X.

IMPORTANT POINTS:

- Care is to be initiated consistent with these SOP's upon arrival of EMS or at the earliest possible time after EMS determines the scene is considered safe.
- EMS System providers are authorized to carry out these protocols to the extent indicated by the patient's condition.
- ECRN's may give only orders outlined in these protocols. Any deviations from SOP's are to be made in collaboration with the Emergency Department (ED) Physician.
- An alternate order of listed interventions may be appropriate based upon patient assessment.
- If a patient's situation is not covered by the SOP's, providers should initiate routine medical/trauma care and Contact OLMC at the appropriate hospital for further direction from an ED physician. In all circumstances, ED physicians have the latitude to deviate from these protocols if it is in the best interest of the patient.
- Under no circumstances shall emergency pre-hospital care be delayed while attempting to establish contact with OLMC.
- EMS personnel may withhold or withdraw resuscitative efforts in accordance with Withdrawing/Withholding Resuscitative Efforts
- Unless otherwise specified, the pediatric patient is considered to be under the age of 16.
- Pediatric medication dosages should not exceed adult dosages unless specifically indicated.
- It is understood that during multiple patient incidents altered standards of care may be necessary in order to provide the greatest good to the greatest number of patients.
- Procedures/Interventions are dictated by individual System permissions.
- System entered Pre-Hospital Registered Nurse, Pre-Hospital Physician Assistant, Pre-Hospital Advanced Practice Registered Nurse may operate at the highest level of patient care, as approved by the System.

USING THE CHARTS:

Level of Care Columns

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - support life threatening problems with airway, breathing circulation and disability.	•	•	•
2. ASPIRIN : 81 mg x 4 chewable PO (if already taken then augment to 324 mg total)	•	•	•
3. Obtain ECG and 12 lead, transmit it to receiving facility, and monitor Capnography.		•	•
Unstable → UN STABLE: Altered Mental Status, MAP<65 or SBP <90			
→ 4. NORMAL SALINE in 500mL increments, titrate to MAP ≥ 65			•
Stable care → STABLE: GCS 15, MAP >65, SBP ≥ 90, skin warm/dry			
→ 5. NITROGLYCERIN : 0.4 mg SL, repeat every 5 min., max 1.2mg			•

- Scope of practice is denoted by level of care columns:
EMR: Emergency Medical Responder
EMT: Emergency Medical Technician
PM: Paramedic
Utilize vertical column to identify skills/care within appropriate scope of practice.
- System entered Pre-Hospital Registered Nurse, Pre-Hospital Physician Assistant, Pre-Hospital Advanced Practice Registered Nurse may operate at the highest level of patient care, as approved by the System.



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PRE-ARRIVAL REPORT

SEQUENCE FOR TRANSMISSION OF PATIENT INFORMATION

1. Identify provider name and vehicle number
2. Alert: STEMI, Sepsis, Stroke, Trauma I/II
3. Identify protocol used
4. Age, gender, approximate weight if pertinent
5. History/Subjective
 - Chief complaint - degree of distress/pain
 - History of present illness or injury (HPI), Mechanism of injury (MOI), Onset time
 - past medical history, medications, allergies
6. Exam/Objective
 - Level of consciousness, vitals, pain scale, temperature, blood glucose if indicated
 - skin parameters - color, temperature, moisture
 - if indicated:
 - Resp: Lung sounds, Pulse Oximetry, Capnography
 - CV: ECG/12 Lead
 - CNS: Glasgow Coma Scale, Stroke screen, Pupils
7. Treatment and response
8. ETA and destination

ABBREVIATED REPORT

Used in situations where resources are limited and/or patient's condition is critical

1. Identify provider name and vehicle number
2. State: "This is an abbreviated report"
3. Alert: STEMI, Sepsis, Stroke, Trauma I/II
4. Identify protocol used
5. Age and gender
6. History: chief complaint, brief HPI/MOI
7. Exam: vitals, GCS
8. Treatment and response
9. ETA and destination

NOTE:

1. See [High Risk Refusals](#) for guidance.

AEROMEDICAL TRANSPORTATION

Circumstances that may warrant Aeromedical Transportation include, but are not limited to, high acuity patients (e.g., Level-1 trauma, hyperbaric center) when:

1. Time is critical for patient survival and/or distances are long, i.e., need for prompt transport when helicopter response time to scene, and flight time to appropriate hospital, is faster than ground transport.
2. Extrication/rescue is prolonged and significantly greater than helicopter estimated time of arrival (ETA) at scene.
3. Patients or hospitals are inaccessible or transport times are significantly delayed due to weather, traffic, or disaster/mass-casualty situation.
4. Ground transport to a Level-1 trauma center will be significantly greater than 25 minutes.
5. Special skills or equipment needed at the scene.

Patient Care/Scene Responsibilities

1. Consider aeromedical transportation, after completing an appropriate patient assessment.
2. EMS personnel at the scene may contact a helicopter service directly.
3. Continue ongoing assessment, treatment, and patient packaging until transfer of patient care to aeromedical transportation medical team.

Contact MEDICAL CONTROL, with information that aeromedical transport service has been requested, with a patient report and hospital destination.

ADULT ROUTINE MEDICAL CARE

	EMR	EMT	PM
1. Assess for scene safety and use of standard precautions	•	•	•

PRIMARY ASSESSMENT

2. Assess airway, breathing, and circulation	•	•	•
3. Assess AVPU and Glasgow Coma Scale	•	•	•
4. Expose and examine as indicated	•	•	•
5 Identify priority of patient and make transport decision	•	•	•

SECONDARY ASSESSMENT: SAMPLE, FOCUSED PMH, PHYSICAL EXAM

6. Obtain vital signs, pain scale, temp., blood glucose, weight (as indicated)	•	•	•
7. Administer supplemental OXYGEN to maintain SpO ₂ >94%. Monitor and record SpO ₂ reading before/during oxygen administration. Nasal cannula: 2-6 LPM Non-rebreather mask: 12-15 LPM Bag-valve mask: 15 LPM	•	•	•
8. Monitor and record Capnography before/during oxygen administration.		•	•
9. Perform focused physical and obtain PMH	•	•	•
10. Evaluate ECG (PM only), obtain 12-lead as indicated, transmit and report interpretation		•	•

INITIAL TREATMENT:

11. Establish IV/IO as indicated by patient condition	•
12. Administer NORMAL SALINE and adjust flow rate as indicated by patient condition. See link for dosages based on etiology of shock.	•

ONGOING ASSESSMENT:

13. Reassess condition, vital signs, pain scale, and Capnography as frequently as condition indicates, and following every treatment.	•	•	•
14. Contact OLMC to give report	•	•	•
15. Transport to Closest Appropriate Facility	•	•	•

NOTE:

- Closest appropriate facility is defined as the comprehensive ED of the patient's choice within the department's transport area or nearest hospital in cases of life threatening emergencies.
- For best practice, obtain at least 2 sets of vitals for each patient.

ROUTINE TRAUMA CARE

SCENE SIZE-UP	EMR	EMT	PM
1. Standard Precautions Scene Hazards Mechanism of Injury Number of Patients Need for Additional Resources	•	•	•

INITIAL ASSESSMENT/PRIMARY SURVEY			
2. Airway/Spinal Precaution Breathing Circulation/Hemorrhage Management AVPU and Glasgow Coma Scale	•	•	•
3. Management of immediate life threats/airway management: Control bleeding	•	•	•
If open pneumothorax, apply occlusive dressing and tape on three sides		•	•
If tension pneumothorax, needle decompression to affected side			•
If traumatic arrest associated with chest trauma, perform bilateral needle decompression			•

IDENTIFY PRIORITY OF TRANSPORT: Complex Multisystem			
4. Perform rapid trauma assessment	•	•	•
5. Continue management of life threats		•	•
6. Examine head, neck, chest, abdomen, pelvis, extremities, back		•	•
7. Vital signs, pulse oximeter & Capnography		•	•
8. Pain scale, Neuro exam, Blood glucose		•	•
9. Obtain history		•	•
10. Package patient and transport		•	•
11. If patient has unstable BP (MAP<65 or SBP<90), NORMAL SALINE in 500 mL increments, max. 1L, titrate to MAP ≥ 65. Consider TXA .			•
12. Manage other serious injuries	•	•	•
13. Perform Detailed Exam as time permits enroute	•	•	•
14. Ongoing assessment every 5 minutes	•	•	•
15. Contact OLMC enroute; Abbreviated Radio Report may be appropriate	•	•	•

IDENTIFY PRIORITY OF TRANSPORT: Single System			
4. Perform focused assessment	•	•	•
5. Examine areas where trauma is expected per mechanism of injury and patient complaint	•	•	•
6. Vital signs, pulse oximeter & Capnography		•	•
7. Pain scale, Neuro exam, Blood glucose		•	•
8. Obtain history		•	•
9. Manage problems with airway, manage injuries		•	•

10. Consider need for IV and pain management			•
11. Package patient and transport		•	•
12. Perform Detailed Exam as time permits enroute	•	•	•
13. Ongoing assessment as patient condition indicates	•	•	•
14. Contact OLMC enroute; Abbreviated Radio Report may be appropriate	•	•	•

PEDIATRIC ROUTINE MEDICAL CARE
ROUTINE TRAUMA CARE

	EMR	EMT	PM
1. Assess for scene safety and use of standard precautions	•	•	•

PEDIATRIC ASSESSMENT TRIANGLE

2. Establish a level of severity, determine urgency for life support and identify key physiologic problems	•	•	•
3. Assess appearance, work of breathing, circulation to skin	•	•	•

PRIMARY ASSESSMENT

4. Assess airway, breathing, and circulation	•	•	•
5. Assess need for spinal motion restriction	•	•	•
6. Assess AVPU and Pediatric Glasgow Coma Scale	•	•	•
7. Expose and examine as indicated	•	•	•
8. Identify priority of patient and make transport decision	•	•	•

SECONDARY ASSESSMENT: SAMPLE, FOCUSED, PMH, PHYSICAL EXAM

9. Obtain vital signs, pain scale, temp., blood glucose	•	•	•
10. Determine weight (utilize length-based tape) and age	•	•	•
11. Administer supplemental OXYGEN to maintain SpO ₂ >94%. Monitor and record SpO ₂ reading before/during oxygen administration. Nasal cannula: 2-6 LPM Non-rebreather mask: 12-15 LPM Bag-valve mask: 15 LPM	•	•	•
12. Perform physical (toe-to-head on 0-4 years) and obtain PMH	•	•	•
13. Monitor and record Capnography before/during oxygen administration.		•	•

INITIAL TREATMENT:

14. Establish IV/IO as indicated by patient condition	•
15. Administer NORMAL SALINE and adjust flow rate as indicated by patient condition. See link for dosages based on etiology of shock.	•

ONGOING ASSESSMENT:

16. Reassess condition, vital signs, pain scale, and Capnography as frequently as condition indicates, and following every treatment.	•	•	•
17. Contact OLMC to give report	•	•	•
18. Transport to Closest Appropriate Facility	•	•	•

NOTE:

- 1. All medication dosage should be weight based. Reference [Pediatric Reference Chart](#).
- 2. A patient under the age of 16 is considered to be a pediatric patient
- 3. If less than 5 kg, Contact OLMC for medication guidance.

BEHAVIORAL EMERGENCIES

HPI	Signs & Symptoms	Differential Diagnosis
• Situational crisis • Psychiatric illness/meds • Injury to self or threats to others • Detailed plan • Substance abuse or overdose	• Anxiety, agitation, confusion • Affect change • Hallucinations • Delusional thoughts, bizarre behavior • Combative, violent	• AMS • Diabetic emergency • Stroke • Overdose Hypoxia • Head trauma •

	EMR	EMT	PM
1. Ensure scene safety	•	•	•
2. Contact Law Enforcement when appropriate	•	•	•
3. Protect patient from harm to self and/or others if EMS safety ensured	•	•	•
4. Attempt verbal de-escalation	•	•	•
5. Do not participate in patient's delusions/hallucinations	•	•	•
6. Warn acutely psychotic/unpredictable patient before making physical contact (e.g., VS measurement)	•	•	•
7. Document all pertinent behavior (e.g., tearful, flat affect, verbally combative, physically threatening posture, obvious internal stimuli like auditory hallucinations)	•	•	•
8. Consider medical etiology for abnormal behavior and cognition Hypoxia Metabolic Disorder (hypoglycemia) Stimulant overdose (amphetamines) Depressant withdrawal (alcohol or benzodiazepines) Neurologic disease (stroke , seizure , delirium, dementia) Trauma (head injury , blood loss)	•	•	•
9. If can confirm medical etiology and it is safe to approach the patient, treat medical causes per protocol	•	•	•
10. If cannot confirm medical etiology, or it is NOT safe to approach the patient, refer to below	•	•	•

INVOLUNTARY PATIENT TRANSPORT: danger to themselves or others, refusing transport to hospital - LE participation

11. Ensure Law Enforcement presence	•	•	•
12. Obtain Law Enforcement <u>participation</u> in involuntary	•	•	•
13. Coordinate approach and restraint procedure with Law Enforcement	•	•	•
14. Secure all 4 extremities using commercial restraints. NEVER leave patient prone after secured.	•	•	•
15. Continue to monitor patient throughout transport to include PulseOx, capnography, and cardiac monitor	•	•	•
16. Document incident thoroughly.	•	•	•

SEVERELY AGITATED, AGGRESSIVE, OR VIOLENT BEHAVIOR

17. Provide Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
18. > 5 years old and ≤ 65 years old: KETAMINE : IM dosing, use ideal body weight chart below , max. single dose 300mg If severe agitation, aggression, or violent behavior continues after 5 minutes: KETAMINE : IM dosing, use ideal body weight chart below , max. total dose 450mg >65 years old: MIDAZOLAM : 2mg IN/IM/IV/IO, may repeat as needed, every 2 min., max total dose 10mg			•

NOTE:

1. If emergence reaction occurs following Ketamine administration, contact OLMC to consider Midazolam for agitation or hallucinations and/or ondansetron for nausea or vomiting
2. Ketamine should not be used as a first line medication for post-ictal confusion or AMS due to brain injury. Midazolam should be considered. Contact OLMC for questions or guidance.

INVOLUNTARY PATIENT TRANSPORT: danger to themselves or others, refusing transport to hospital - LE **NOT** able to participate

11. NO forcible/involuntary restraint is to be used	•	•	•
12. If safe - obtain AMA/refusal signature	•	•	•
13. Document the following: EMS crew safety NOT ensured Police department declining assistance Discussion with OLMC	•	•	•
14. Contact OLMC and make the following statement: "Police have determined this individual has NOT met involuntary detention criteria. EMS crew safety cannot be ensured. We cannot safely transport this patient against their will."	•	•	•
15. OLMC will make the following statement in response "We cannot grant your refusal request but understand you are unable to safely transport the patient at this time" OLMC may decline the refusal but crews are permitted to leave the scene as a non-transport.	•	•	•

KETAMINE 500MG/10mL

For IM dosing ONLY

IDEAL BODY WEIGHT		
ADULT MALE:		
	Initial Dose:	
height	dose	vol.
< 5'6"	200 mg	4 mL
5'6"-6"	250 mg	5 mL
> 6'	300 mg	6 mL
	Repeat Dose:	
height	dose	vol.
< 5'6"	100 mg	2 mL
5'6"-6"	125 mg	2.5 mL
> 6'	150 mg	3 mL

PEDIATRIC PATIENTS

Initial: 4 mg/kg | Repeat: 2 mg/kg

Use calculator for dosing

ADMINISTRATION LOCATION:

anterolateral thigh: max 5mL

shoulder: max 2mL

ADULT FEMALE:		
	Initial Dose:	
height	dose	vol.
< 5'	180 mg	3.6 mL
5'-5'6"	200 mg	4 mL
> 5'6"	250 mg	5 mL
	Repeat Dose:	
height	dose	vol.
< 5'	90 mg	1.8 mL
5'-5'6"	100 mg	2 mL
> 5'6"	125 mg	2.5 mL

CHECK VIAL CONCENTRATION PRIOR TO ADMINISTRATION

MIDAZOLAM 10MG/2ML (5MG/1ML)

dose	volume
	5mg/mL
2 mg	0.4 mL

MIDAZOLAM 1MG/1ML

dose	volume
	1mg/mL
2 mg	2 mL

ACUTE CORONARY SYNDROME

HPI	Signs & Symptoms	Differential Diagnosis
- Quality: crushing, sharp - PMH of angina, previous MI, diabetes, HTN - Severity, onset - Recent physical exertion - Radiation, referred pain	- Chest pain - pressure, aching, tightness - Location - sub-sternal, epigastric, arm, jaw, neck, shoulder - Shortness of breath - Nausea/vomiting - Pale/diaphoretic	- Angina vs. MI - Pericarditis - GI reflux, esophageal spasm Asthma/COPD - Aortic dissection - Pulmonary embolism - Trauma vs. Medical

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - support life threatening problems with airway, breathing circulation and disability.	•	•	•
2. ASPIRIN : 81 mg x 4 chewable PO (if already taken then augment to 324 mg total)	•	•	•
3. Obtain ECG and 12 lead, transmit to receiving facility, and monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP<65 or SBP <90

4. NORMAL SALINE in 500mL increments, titrate to MAP ≥ 65	•
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STABLE: GCS 15, MAP >65, SBP ≥ 90, skin warm/dry

5. NITROGLYCERIN : 0.4 mg SL, repeat every 5 min., max 1.2mg	•
6. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg	•

PEARLS:

1. If positive ST elevation then contact **STEMI** receiving center and call Alert at earliest possible time. Computer assisted ECG interpretation may be used along with provider interpretation.
2. If ST Elevation noted in II, III, aVF: contact **OLMC** as **NITROGLYCERIN** may be contraindicated. Obtain V4R to evaluate right ventricular involvement. Do NOT transmit V4R.
3. **NITROGLYCERIN** should be withheld from patients who have taken erectile dysfunction or pulmonary hypertension medications in the last 48 hours such as Viagra, Levitra, Cialis.

ADULT RAPID ATRIAL FLUTTER/FIBRILLATION
(NARROW COMPLEX TACHYCARDIA)

HPI	Signs & Symptoms	Differential Diagnosis
- Medications: Digoxin, diet pills, decongestants, thyroid supplements - Drugs: caffeine, cocaine, amphetamines - PMH of palpitations	- HR >100 - Irregular narrow complex - Chest pain - Shortness of breath - AMS - Hypotension - Lack of P waves - Dizziness	SVT - Sinus tachycardia - Junctional tachycardia Acute Coronary Syndrome - Hypoxia - Hyper/o-kalemia Hypoglycemia - Toxins/Overdose Heart failure

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and 12 lead, transmit to receiving facility, and monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90

3. Consider sedation for conscious patients MIDAZOLAM : 2 mg IVP/IO, repeat 2 min., max. 10 mg	•
4. Synchronized cardioversion per manufacturer's guidelines.	•
5. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg	•

STABLE: GCS 15, MAP >65, SBP >90, skin warm/dry

3. Instruct patient to perform vagal maneuver	•	•
4. VERAPAMIL : 5 mg IVP, slowly over 5 min., may repeat 15 min. (if SBP ≥ 90), max. 10 mg		•

PEARLS:

- 1. Vagal maneuvers include: coughing, bearing down, attempting to blow the plunger out of a 10mL syringe.
- 2. As it is difficult to ascertain onset of atrial flutter/fibrillation rhythm, risk of stroke needs to be considered prior to cardioversion.

CARDIOVERSION DOSING CHART:

<u>Zoll</u>	<u>LifePak</u>	<u>Phillips</u>
120J	100J	100J
150J	200J	150J
200J	300J	200J
	360J	

ADULT BRADYCARDIA

AV BLOCKS

HPI	Signs & Symptoms	Differential Diagnosis
- Medications: - Beta blockers - Calcium channel blockers - Clonidine - Digoxin - Pacemaker	- HR <60 - Chest pain - Respiratory distress - Hypotension, MAP<65 - AMS - Syncope	- Acute MI - Hypoxia - Hypothermia - Toxins/Overdose - Head injury/ ICP - Neurogenic shock - AV block

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and 12 lead, transmit to receiving facility, and monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90

3. ATROPINE: 1 mg IVP/IO, rapid push (while preparing for pacing). Do not delay pacing for medication administration.	•
4. If patient remains unstable, Transcutaneous pacing	•
5. If needed, ATROPINE 1 mg IVP/IO, rapid push, may repeat 3 min., max. 3 mg	•
6. Consider sedation for conscious patients MIDAZOLAM: 2mg IVP/IO, may repeat 2 min., max. 10mg	•
7. For Pain Management : FENTANYL: 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg	•
8. If condition continues to deteriorate and MAP <65 despite NORMAL SALINE contact OLMC: PUSH DOSE EPINEPHRINE: 50 mcg (5mL) IVP/IO, repeat 5 min., max. 100mcg, target a MAP 65	•

NOTE:

- 1. Do not administer **ATROPINE** for 2nd Degree Type 2 or 3rd Degree Heart Block - skip straight to Transcutaneous pacing.

PEARLS:

- 1. Consider potential accidental/intentional overdose on [beta-blockers](#) and [calcium channel blockers](#). Refer to specific protocols.
- 2. Patients who have undergone cardiac transplant will not respond to **ATROPINE**.
- 3. Bradycardia with hypertension can be a sign of increased intracranial pressure. See [Head Injury](#).

ADULT CARDIOGENIC SHOCK

HPI	Signs & Symptoms	Differential Diagnosis
- Cardiac ischemia - Acute heart failure - Arrhythmia - Medications, OD, reaction - Post cardiac arrest - Congenital heart defect - Cardiac surgery	- Restlessness, confusion - Weakness, dizziness - Weak, rapid pulse - Pale, cool, diaphoretic - Peripheral edema - Hypotension - Dyspnea	- Hypovolemic shock - Septic shock - Medication reaction - Arrhythmia - Pulmonary embolus - Tension pneumothorax

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and 12 lead, transmit to receiving facility, and monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP<65 or SBP <90

3. Verify and check often that lung sounds are clear	•	•
4. NORMAL SALINE in 500mL increments, titrate to MAP ≥ 65		•
5. If condition continues to deteriorate and MAP <65 despite FLUID CHALLENGE contact OLMC : PUSH DOSE EPINEPHRINE : 50 mcg (5mL) IVP/IO, repeat 5 min., max. 100mcg, target a MAP ≥ 65		•

PEARLS:

1. Circulatory failure due to inadequate cardiac function may be caused by MI, congenital heart defect, acute heart failure, marked symptomatic tachycardia/bradycardia.
2. Hypotension indicates uncompensated shock, and may progress to cardiopulmonary collapse within minutes.

ADULT SUPRAVENTRICULAR TACHYCARDIA
(NARROW COMPLEX TACHYCARDIA)

HPI	Signs & Symptoms	Differential Diagnosis
- Medications: Digoxin, diet pills, decongestants, thyroid medication - Drugs: caffeine, cocaine, amphetamines - PMH of palpitations	- HR >100 - Regular narrow complex - Chest pain - Shortness of breath - AMS - Hypotension - Lack of P waves - Dizziness	- Atrial flutter/fibrillation - Sinus tachycardia - Junctional tachycardia - ACS - Hypoxia - Hyper/o-kalemia - Hypoglycemia Toxins/ - Overdose Heart failure

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and 12 lead, transmit to receiving facility, and monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90

3. Consider sedation for conscious patients MIDAZOLAM: 2 mg IVP/IO, repeat 2 min., max. 10 mg	•
4. Synchronized cardioversion per manufacturer's guidelines.	•
5. For Pain Management : FENTANYL: 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg	•

STABLE: GCS 15, MAP >65, SBP >90, skin warm/dry

3. Instruct patient to perform vagal maneuver	•	•
4. ADENOSINE: 6 mg rapid IVP, followed by rapid flush NORMAL SALINE: 20 mL		•
5. If no response in 2 min. - ADENOSINE: 12 mg rapid IVP, followed by rapid flush NORMAL SALINE: 20 mL		•
6. If no response in 2 min. - VERAPAMIL: 5 mg IVP, slowly over 5 min., may repeat 15 min. (if SBP ≥ 90), max. 10 mg		•

PEARLS:

- 1. Vagal maneuvers include: coughing, bearing down, attempting to blow the plunger out of a 10mL syringe.
- 2. Irregular narrow complex tachycardia is likely [Atrial Fibrillation](#) - **VERAPAMIL** may be considered if can confirm.
- 3. Risk of causing sustained asystole following **ADENOSINE** administration is increased if given slowly - always rapid push and rapid follow with **NORMAL SALINE**.

CARDIOVERSION DOSING CHART:

<u>Zoll</u>	<u>LifePak</u>	<u>Phillips</u>
70J	50J	50J
120J	100J	100J
150J		150J
200J		

ADULT VENTRICULAR TACHYCARDIA WITH A PULSE

(WIDE COMPLEX TACHYCARDIA)

HPI	Signs & Symptoms	Differential Diagnosis
- Medications: Digoxin, diet pills, decongestants, thyroid supplements - Drugs: caffeine, cocaine, amphetamines - PMH of palpitations	- HR >150 - Dizziness - Chest pain - Shortness of breath - AMS - Hypotension	QRS >0.12s - Wolff-Parkinson-White - SVT with aberrancy - Ventricular paced rhythm - Hyper/o-kalemia - Hypoglycemia - Toxins/Overdose - MI - Hyperthyroidism

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and 12 lead, transmit to receiving facility, and monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90

3. Consider sedation for conscious patients MIDAZOLAM : 2 mg IVP/IO, repeat 2 min., max. 10 mg	•
4. Synchronized cardioversion per manufacturer's guidelines.	•
5. If patient does not convert after first synchronized cardioversion: AMIODARONE : 150 mg diluted in 100 mL D5W IVPB, over 10 min., may repeat, max. 300 mg and continue synchronized cardioversion per manufacturer's	•
6. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg	•

STABLE: GCS 15, MAP >65, SBP >90, skin warm/dry

3. AMIODARONE : 150mg diluted in 100mL D5W IVPB, over 10 min., may repeat, max. 300 mg	•
---	---

NOTE:

1. If VT recurs, synchronized cardioversion at energy level that was previously successful.
2. If rhythm appears to be Torsade's de Pointes (polymorphic ventricular tachycardia), contact **OLMC** to consider [MAGNESIUM SULFATE](#) 2 g diluted in 100mL D5W IVPB, over 5 minutes. **Not to be given to renal failure or dialysis patients.
3. If rhythm appears to be Torsade's de Pointes (polymorphic ventricular tachycardia), it is not possible to synchronize on the monitor, therefore, defibrillate at maximum joules per manufacturer's guidelines.

CARDIOVERSION DOSING CHART:

<u>Zoll</u>	<u>LifePak</u>	<u>Phillips</u>
120J	100J	100J
150J	200J	150J
200J	300J	200J
	360J	

AED PULSELESS ARREST

TRANSITION OF CARE FROM AED TO ALS

HPI	Signs & Symptoms	Possible Causes	
• Witnessed vs unwitnessed • Last known well • PTA AED use • Bystander CPR • Suspected etiology • PMH	• Pulseless • Apneic • No signs of rigor or lividity • No findings of signs incompatible with life	• Hypovolemia • Hypoxia • Acidosis • Hyper/o-kalemia • Hypothermia • Hypoglycemia	• Toxins/Overdose • Cardiac tamponade • Acute coronary syndrome • Pulmonary embolism • Tension Pneumo

UNIVERSAL EMERGENCY CARDIAC CARE	EMR	EMT	PM
1. Assess responsiveness, pulse, and breathing for no more than 10 seconds.	•	•	•
2. If no pulse and not breathing/gasping - begin CPR 30:2 compressions to ventilations for a 2 min. cycle	•	•	•
3. Place monitor/defibrillator immediately as staffing allows.	•	•	•
4. Follow prompts to analyze rhythm. Follow prompts for shockable or non-shockable. Continue CPR 2 min.	•	•	•

ARRIVAL OF ALS PERSONNEL

5. Obtain report from AED personnel	•
6. Allow AED to finish cycle, attach monitor/defibrillator before disconnecting AED	•
7. Perform rhythm interpretation and refer appropriate protocol: Adult V.Fib/Pulseless V.Tach or Adult Asystole/PEA	•
8. Obtain vascular access IV/IO	•
9. Consider advanced airway: ETT	•
or i-Gel. Monitor placement with Capnography. Deliver 1 breath/6 sec.	•

NOTE:

1. Subsequent ALS defibrillation should be at maximum joules. ALS should consider the shocks delivered by the AED PTA to be part of the ALS protocol.
2. A pediatric dose-attenuator system is preferred for pediatrics age 1 to 8 years old, however, a standard AED may be used if none is available.
3. For infants <1 years old, manual selection defibrillation is preferred, however a pediatric dose-attenuator may be used. A standard AED may be used if neither is available.
4. **Patient movement during cardiac arrest resuscitation should only occur under the following circumstances:**
 - [ROSC](#)
 - environmental/provider safety concerns
 - extenuating circumstances (pediatric)**Contact OLMC for request to transport in any other situation.**
5. If arrest suspected to be opioid related, naloxone delivery should be considered during resuscitation. It should not delay/interrupt CPR and ventilation.

ADULT ASYSTOLE
PULSELESS ELECTRICAL ACTIVITY

HPI	Signs & Symptoms	Possible Causes	
• Witnessed vs unwitnessed • Last known well • PTA AED use • Bystander CPR • Suspected etiology • PMH	• Pulseless • Apneic • No signs of rigor or lividity • No findings of signs incompatible with life	• Hypovolemia • Hypoxia • Acidosis • Hyper/o-kalemia • Hypothermia • Hypoglycemia	• Toxins/Overdose • Cardiac tamponade • Acute coronary syndrome • Pulmonary embolism • Tension Pneumo

UNIVERSAL EMERGENCY CARDIAC CARE	EMR	EMT	PM
1. Assess responsiveness, pulse and breathing for no more than 10 seconds.	•	•	•
2. If no pulse and not breathing/gasping - begin CPR 30:2 compressions to ventilations for a 2 min. cycle	•	•	•
3. Place monitor/defibrillator immediately as staffing allows.	•	•	•
4. Analyze rhythm and continue CPR 2 min.			•
5. Obtain vascular access IV/IO			•
6. Consider advanced airway: ETT			•
or i-Gel.			
Monitor placement with Capnography. Deliver 1 breath/6 sec.		•	•

PERSISTENT NON-SHOCKABLE

7. Analyze rhythm. Continue CPR 2 min.			•
8. EPINEPHRINE 1 mg/10mL: 1 mg IVP/IO, repeat 3-5min., no max.			•
9. NORMAL SALINE in 500mL increments			•
10. Dialysis patients ONLY: SODIUM BICARBONATE : 50mEq IVP/IO			•
11. If return of spontaneous circulation see ROSC	•	•	•

NOTE:

1. Utilize High Performance CPR for best patient outcome.
2. Patient movement during cardiac arrest resuscitation should only occur under the following circumstances:
 - [ROSC](#)
 - environmental/provider safety concerns
 - extenuating circumstances (pediatric)

Contact **OLMC** for request to transport in any other situation.

PEARLS:

1. All attempts should be made to prevent avoidable interruptions in chest compressions, such as hovering over the chest during rhythm
2. The airway management strategy should not interrupts compressions.
3. Hyperventilation should be avoided because it decreases preload, cardiac output, coronary perfusion, and cerebral blood flow.

ADULT VENTRICULAR FIBRILLATION
PULSELESS VENTRICULAR TACHYCARDIA

HPI	Signs & Symptoms	Possible Causes	
• Witnessed vs unwitnessed • Last known well • PTA AED use • Bystander CPR • Suspected etiology • PMH	• Pulseless • Apneic • No signs of rigor or lividity • No findings of signs incompatible with life	• Hypovolemia • Hypoxia • Acidosis • Hyper/o-kalemia • Hypothermia • Hypoglycemia	• Toxins/OD • Cardiac tamponade • Acute coronary syndrome • Pulmonary embolism • Tension Pneumo

UNIVERSAL EMERGENCY CARDIAC CARE	EMR	EMT	PM
1. Assess responsiveness, pulse and breathing for no more than 10 seconds.	•	•	•
2. If no pulse and not breathing/gasping - begin CPR 30:2 compressions to ventilations for a 2 min. cycle	•	•	•
3. Place monitor/defibrillator immediately as staffing allows.	•	•	•
4. Analyze rhythm and defibrillate incrementally as chart indicates below. Continue CPR 2 min.			•
5. Obtain vascular access IV/IO			•
6. Consider advanced airway: ETT			•
or i-Gel.			
Monitor placement with Capnography. Deliver 1 breath/6 sec.		•	•

PERSISTENT V.FIB/PULSELESS V.TACH

7. Analyze rhythm and defibrillate incrementally as chart indicates below. Continue CPR 2 min.	•
8. EPINEPHRINE 1 mg/10mL: 1 mg IVP/IO, repeat 3-5min., no max.	•
9. NORMAL SALINE in 500mL increments	•
10. Analyze rhythm and defibrillate incrementally as chart indicates below. Continue CPR 2 min.	•
11. AMIODARONE : 300 mg IVP/IO	•
12. Dialysis patients ONLY: SODIUM BICARBONATE : 50mEq IVP/IO	•
13. Analyze rhythm and defibrillate incrementally as chart indicates below. Continue CPR 2 min.	•
14. AMIODARONE : 150 mg IVP/IO	•
15. Analyze rhythm. If persistent shockable Vfib/P-Vtach, perform vector change defibrillation: – ADD second set of pads in the anterior-posterior position. Defibrillate at maximum joules per chart.	•
15. If return of spontaneous circulation see ROSC	•

NOTE:

1. If rhythm appears to be Torsades de Pointes (polymorphic ventricular tachycardia), contact **OLMC** to consider [MAGNESIUM SULFATE](#) 2gm diluted in 100mL D5W IVPB, over 5 minutes. **NOT to be given to renal failure or dialysis patients.
2. Patients with Automatic Implantable Cardiac Defibrillators should be

defibrillated at the maximum joules per manufacturer's recommendations.

3. Utilize High Performance CPR for best patient outcome.

4. Patient movement during cardiac arrest resuscitation should only occur under the following circumstances:
- ROSC
 - environmental/provider safety concerns
 - extenuating circumstances (pediatric)

Contact **OLMC** for request to transport in any other situation.

PEARLS:

1. All attempts should be made to prevent avoidable interruptions in chest compressions, such as pre-charging the defibrillator and hovering over the chest during defibrillation.
2. The airway management strategy should not interrupt compressions.
3. Hyperventilation should be avoided because it decreases preload, cardiac output, coronary perfusion, and cerebral blood flow.

DEFIBRILATOR DOSING CHART:

Per manufacturer's recommendation, defibrillation dosages should be incrementally increased, starting from the lowest joules and increasing as the patient remains in a shockable rhythm.

<u>Zoll</u>	<u>LifePak</u>	<u>Phillips</u>
120J	200J	150J
150J	300J	150J
200J	360J	150J

RETURN OF SPONTANEOUS CIRCULATION (ROSC)

	EMR	EMT	PM
1. Confirm ROSC: Palpable carotid pulse Recordable blood pressure Capnography ≥ 35 mmHg		●	●
2. Administer OXYGEN with a target of achieving SpO ₂ 94-99%. Do not hyperoxygenate.		●	●
3. If apneic, ventilate to achieve Capnography 30-40 mmHg. Do not hyperventilate.		●	●
4. Reassess need for placement of advanced/supraglottic airway.		●	●
5. Consider etiology of arrest (treat causes), monitor closely for re-arrest			●
6. Perform neuro assessment, obtain vital signs, obtain 12 lead, transmit to receiving facility		●	●
7. For adult hypotension (SBP < 90 and/or MAP <65) despite NORMAL SALINE contact OLMC : <u>PUSH DOSE EPINEPHRINE</u> : 50 mcg (5mL) IVP/IO, repeat 5 min., max. 100mcg, target a MAP ≥ 65			●

NOTE:

1. Transport patient to PCI capable facility if appropriate.

POSSIBLE ETIOLOGIES:

- Hypovolemia
- [Poisoning/Overdose](#)
- Hypoxia
- Cardiac Tamponade
- Acidosis
- [Tension Pneumothorax](#)
- Hyper/o-kalemia
- [Acute Coronary Syndrome](#)
- [Hypothermia](#)
- Pulmonary Embolism
- [Hypoglycemia](#)
- [Trauma](#)

TERMINATION OF RESUSCITATIVE EFFORTS

	EMR	EMT	PM
1. Perform at least 20 minutes of ALS care			●
2. Contact OLMC while continuing patient care			●
3. Report events of call, including estimated time of cardiac arrest and all treatments rendered			●
4. Reaffirm all of the following: Non-traumatic arrest Normothermic ≥ 18 years old Patient experienced an unwitnessed arrest by EMS provider No respirations, pulse, heart sounds In asystole or wide complex PEA < 60 bpm			●
5. Call OLMC for termination of resuscitation or consultation if: No ROSC after 20 minutes or No evidence of neurological function: pupils non-reactive no response to pain no spontaneous movement EtCO ₂ < 20 mmHg All EMS clinicians involved in the patient's care agree that discontinuation of the resuscitation is appropriate			●
6. If physician orders termination of efforts, document the following: note time of withdrawal of efforts note physician's name providing OLMC			●
7. Notify Coroner/Medical Examiner			●

NOTE:

- 1. Only a physician can make the determination to withdraw resuscitative efforts.
- 2. Local Law Enforcement may assist in notification of Coroner/Medical Examiner.
- 3. Local department policy and/or logistical factors (collapse in public place, safety of the crew and public) may affect transportation consideration.
- 4. **Contact OLMC for request to transport in any other situation.**

WITHHOLDING RESUSCITATIVE EFFORTS

REASONS TO WITHHOLD CARE:				EMR	EMT	PM
1.	There is a risk to health and safety of personnel				•	•
2.	Resources are inadequate to treat all patients (multiple patients)				•	•
3.	Patient shows signs of irreversible death: Rigor mortis without profound hypothermia Profound dependent lividity Decapitation Transection Incineration Decomposition Injuries incompatible with life Mummification or putrefaction				•	•
4.	Death has been declared by a physician, coroner/medical examiner				•	•
5.	Valid "IDPH Uniform Practitioner Order For Life-Sustaining Treatment" (POLST) form or state approved "Do Not Resuscitate" (DNR) order that included: Name of patient The words "Do Not Resuscitate" or "Do Not Attempt Resuscitation/DNR" Dated document Signature of: Patient or legal representative Authorized practitioner (physician, resident ≥ 2 years, PA, APRN)				•	•

NOTE:

- 1. A living will by itself cannot be recognized by pre-hospital providers.

ADULT AIRWAY OBSTRUCTION

HPI	Signs & Symptoms	Differential Diagnosis
- Recent witnessed choking - Lack of teeth - Age	- Universal choking sign - Stridor - Inability to speak complete sentences - Increased respiratory rate/effort - Altered mental status	- Anaphylaxis - Aspiration/obstruction - Pulmonary embolism - Upper respiratory infection - Anxiety, hyperventilation

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability	•	•	•
2. Obtain ECG and 12 lead, monitor Capnography		•	•

UNRESPONSIVE:

3. Immediately begin CPR regardless of pulses.	•	•	•
4. Open the airway and check the mouth for foreign body	•	•	•
5. Assess for positive gag reflex. If present, utilize Adult Drug Assisted Intubation protocol to proceed.			•
6. If FBO is not able to be removed, utilize direct visualization of the airway via video laryngoscopy and remove with Magill forceps.			•
7. If unable to remove and FBO is passed the vocal cords, intubate. Attempt to push the FBO into the right mainstem bronchus, then pull ETT back and ventilate left lung			•
8. If unable to remove and FBO is supraglottic (obstructing the view of the vocal cords), remove the videoscope and attempt to suction ≤ 10 seconds If suction does not remove FBO, utilize appropriate surgical cricothyrotomy			•

CONSCIOUS: CANNOT Speak

3. Perform repeated cycles of 5 back blows followed by 5 abdominal thrusts until the foreign body obstruction is relieved or the patient becomes unresponsive. For pregnant or morbidly obese patients, substitute chest thrusts for abdominal thrusts.	•	•	•
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CONSCIOUS: Able to speak or cough

3. Allow patient to attempt to clear obstruction on their own.	•	•	•
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ADULT ASTHMA/COPD WITH WHEEZING

HPI	Signs & Symptoms	Differential Diagnosis
- Asthma, COPD, chronic bronchitis, emphysema - Home treatment: oxygen, inhaler, nebulizer - Medications: steroids, bronchodilators - Toxic exposure, smoke inhalation	- Shortness of breath - Pursed lip respirations - Inability to speak complete sentences - Increased respiratory rate/effort - Use of accessory muscles - Tripoding - Wheezing, rhonchi, rales	Heart failure - Aspiration/obstruction Anaphylaxis Acute Coronary Syndrome Pneumothorax - Pulmonary embolism - Upper respiratory infection - Anxiety, hyperventilation

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability	•	•	•
2. Obtain ECG and 12 lead, monitor Capnography		•	•

SEVERE/IMPENDING FAILURE: Altered Mental Status, pale/diaphoretic

3. DuoNeb: ALBUTEROL : 2.5 mg / IPRATROPIUM BROMIDE : 0.5 mg, via CPAP: 5 cm PEEP, may increase to max. 10 cm PEEP, may repeat DouNeb x 2	•	•
4. MAGNESIUM SULFATE : 2g diluted in 100 mL D5W IVPB, over 15 min., do not delay administration (≥65, OLMC to consider 1g dosage)		•
5. If patient continues to deteriorate contact OLMC : EPINEPHRINE 1 mg/mL: 0.3 mg IM in anterolateral thigh	•	•

MILD/MODERATE: GCS 15, MAP >65, SBP >90, skin warm/dry

3. DuoNeb: ALBUTEROL : 2.5 mg / IPRATROPIUM BROMIDE : 0.5 mg, may repeat x 2	•	•
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NOTE:

- MAGNESIUM SULFATE** is NOT to be given to renal failure or dialysis patients.
- ALBUTEROL** and **IPRATROPIUM BROMIDE** may be administered in-line CPAP or intubation.
- Do not delay transport while waiting for response.

PEARLS:

- MAGNESIUM SULFATE** may cause hypotension that will usually respond to a fluid bolus.
- When assessing for cause of respiratory distress, CHF tends to be associated with lower levels of Capnography compared to COPD.

CARBON MONOXIDE/SMOKE INHALATION

HPI	Signs & Symptoms	Differential Diagnosis
- Type of exposure - Duration of exposure - Associated airway injury	- Dizziness - Loss of consciousness - Nausea/vomiting - Confusion - Burns - Hoarseness, wheezing	Thermal burns Cyanide poisoning Altered mental status

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability	•	•	•
2. Remove from hazardous environment. Evaluate exposure.	•	•	•
3. Obtain ECG and/or 12 lead, monitor Capnography		•	•
4. Administer <u>OXYGEN</u> : 100% by NRB or support respirations with a BVM as indicated	•	•	•
5. Nausea/vomiting precautions		•	•

PEARLS:

1. Pulse oximetry may give falsely elevated readings in patients with methemoglobin or CO / CN exposure.
2. Have a high index of suspicion and a low intubation threshold when treating burn patients with possible airway involvement.
2. Consider cyanide poisoning in presence of smoke/fire situations.

NON-INVASIVE CO MEASUREMENT:

<u>MEASUREMENT:</u>	<u>SIGNS & SYMPTOMS:</u>
< 5%	None (Normal for non-smoker)
5 - 9%	Minor headache (May be normal for smoker)
10 - 19%	Headache, shortness of breath
20 - 29%	Headache, nausea, dizziness, fatigue
30 – 39%	Severe headache, vomiting, vertigo, altered LOC
40 - 49%	Confusion, syncope, tachycardia
50 - 59%	Seizures, shock, apnea
> 59%	Coma, death, cardiac dysrhythmias

ACUTE HEART FAILURE

(PULMONARY EDEMA)

HPI	Signs & Symptoms	Differential Diagnosis
- Congestive heart failure - Previous MI - Medications: bumex, lasix	- Shortness of breath - Peripheral pitting edema - Rales, ronchi - Pink, frothy sputum - Inability to speak complete sentences - Jugular vein distension - Chest pain	Acute coronary syndrome Asthma/COPD - Pneumonia - Aspiration/obstruction

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability	•	•	•
2. Obtain ECG and 12 lead, monitor Capnography		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 100

3. Treat dysrhythmias per protocol	•
4. See Cardiogenic Shock	•

UNSTABLE: SBP > 160

3. NITROGLYCERIN : 1.2 mg (3 tabs) SL initial dose 0.4 mg SL, repeat 5 min., no max.	•
4. CPAP: 5 cm PEEP, may increase, max. 10 cm PEEP	•

STABLE: GCS 15, MAP >65, SBP 100 - 160, skin warm/dry

3. CPAP: 5 cm PEEP, may increase, max. 10 cm PEEP	•	•
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NOTE:

1. **NITROGLYCERIN** should be withheld from patients who have taken erectile dysfunction or pulmonary hypertension medications in the last 48 hours such as Viagra, Levitra, Cialis.
2. If during CPAP patient deteriorates, remove CPAP and consider advanced airway placement.

PEARLS:

1. When assessing for cause of respiratory distress, CHF tends to be associated with lower levels of Capnography compared to COPD.
2. **NITROGLYCERIN** reduces left ventricular filling pressure primarily via venous dilation. At higher doses the drug variably lowers systemic afterload and increases stroke volume and cardiac output.

ADULT DRUG ASSISTED INTUBATION

HPI	Signs & Symptoms	
• Anaphylaxis • Asthma/COPD • CHF/Pulmonary edema • Trauma • Inhalation burns •	• Failure to oxygenate /ventilate • Failure to maintain airway • Imminent loss of airway • Expected clinical course • Respiratory failure	• RR < 10 or > 40 • Shallow/labored effort • SpO2 ≤ 92% on 100% O2 • Airway edema • Soot in mouth/nares •

Considerations:

1. When less-invasive methods (two-person BVM with adjuncts) are ineffective or inappropriate, consider drug assisted endotracheal intubation to maintain oxygenation and/or ventilation.
2. Respiratory failure not responding to BLS interventions/treatment, consider drug assisted endotracheal intubation

	EMR	EMT	PM
1. Provide Adult Routine Medical / Trauma Care - treat life threatening problems with airway, breathing, circulation, and disability.	•	•	•
2. Obtain ECG, monitor Capnography (obtain 12 lead as able)		•	•
3. Pre-oxygenate 100% OXYGEN for 3 minutes, goal SpO2 100%	•	•	•
4. Assist ventilations 1 breath/6 sec. (10 breaths/min.)	•	•	•
5. Support blood pressure to achieve MAP > 65			•
6. Assess for positive gag reflex		•	•
7. ***Use ideal body weight chart below KETAMINE: <u>Initial induction dose</u> , IVP/IO, slowly over 1 min, max single dose 300mg			•
8. After 1 minute, reassess gag reflex for therapeutic effects of Ketamine. If inadequate sedation ***Use ideal body weight chart below KETAMINE: <u>Second induction dose</u> , IVP/IO, slowly over 1 min., max. total dose 450mg			•
9. Intubate. Verify placement with Capnography (continue to monitor throughout) ***if unable to intubate, consider alternative airway devices			•
10. Secure tube. Place c-collar to maintain tube position			•
11. As needed, for post-intubation sedation: ***Use ideal body weight chart below KETAMINE: <u>Post-intubation sedation dose</u> , IVP/IO, slowly over 1 min., may repeat 5 min., no max. dosage. Titrate to RASS -2 to -3			•

PEARLS:

1. Patients with significant respiratory distress should have continuous pulse oximetry and waveform capnography monitoring for both assessment and for guiding therapy.
2. During respiratory failure, utilize BVM with patient's head elevated 30 degrees for optimal pre-oxygenation

- 3. Optimize patient for first-pass success with pre-procedure preoxygenation, sniffing positioning, and sedatives and as indicated by patient presentation
- 4. See [Drug Shortages](#) if Ketamine not stocked

KETAMINE 500MG/10mL

IDEAL BODY WEIGHT

ADULT MALE:			
Initial Induction Dose			
height	Kg	dose	vol.
< 5'6"	50 kg	100 mg	2 mL
5'6"-6"	63 kg	126 mg	2.5 mL
> 6'	75 kg	150 mg	3 mL

Second Induction Dose			
height	Kg	dose	vol.
< 5'6"	50 kg	50 mg	1 mL
5'6"-6"	63 kg	63 mg	1.3 mL
> 6'	75 kg	75 mg	1.5 mL

Post-Intubation Sedation			
height	Kg	dose	vol.
< 5'6"	50 kg	25 mg	0.5 mL
5'6"-6"	63 kg	32 mg	0.6 mL
> 6'	75 kg	38 mg	0.8 mL

ADULT FEMALE:			
Initial Induction Dose			
height	Kg	dose	vol.
< 5'	45 kg	90 mg	1.8 mL
5'-5'6"	50 kg	100 mg	2 mL
> 5'6"	63 kg	126 mg	2.5 mL

Second Induction Dose			
height	Kg	dose	vol.
< 5'	45 kg	45 mg	0.9 mL
5'-5'6"	50 kg	50 mg	1 mL
> 5'6"	63 kg	63 mg	1.3 mL

Post-Intubation Sedation			
height	Kg	dose	vol.
< 5'	45 kg	23 mg	0.5 mL
5'-5'6"	50 kg	25 mg	0.5 mL
> 5'6"	63 kg	32 mg	0.6 mL

Richmond Agitation Sedation Scale (RASS)		
4	Combative	Overtly combative, violent, immediate danger to staff
3	Very agitated	Pulls/removes tubes, aggressive
2	Agitated	Frequent non-purposeful movement
1	Restless	Anxious, non-aggressive/violent movement
0	Alert and calm	
-1	Drowsy	Sustained awakening to voice (>10 sec)
-2	Light sedation	Briefly awakens with eye contact (< 10 sec)
-3	Moderate sedation	Movement/eye movement to voice, no eye contact
-4	Deep sedation	Movement/eye opening to physical stimulation
-5	Cannot be aroused	No response to voice/physical stimulation

ADULT ALTERED MENTAL STATUS
SYNCOPE / PRE-SYNCOPE

HPI	Signs & Symptoms	Differential Diagnosis
- PMH: cardiac, stroke, seizures, diabetes - GI bleed - Ruptured ectopic - Fluid loss: vomiting, diarrhea, dehydration - Recent trauma	- LOC with/without recovery - Lightheadedness, dizziness - Pulse irregularity, rate - Hypotension - Hallucination, delusion	- Vasovagal - Orthostatic hypotension - Dysrhythmia - Alcohol/Overdose Hypo/er-glycemia Hypo/er-thermia - Shock Stroke/TIA Sepsis Behavioral emergency

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and 12 lead, monitor Capnography.		•	•
3. If respiratory depression or narcotic overdose suspected, see Depressant Overdose protocol	•	•	•
4. Obtain blood glucose level. If blood glucose is < 60 or > 250, see Diabetic Emergencies protocol	•	•	•
5. Perform BEFAST Stroke Scale. If positive, see Stroke protocol	•	•	•

NOTE:

- Bring container of drug or substance providing that the transport of the item does not pose a safety risk.

ADULT ALLERGIC REACTION
ANAPHYLACTIC SHOCK

HPI	Signs & Symptoms	Differential Diagnosis
- Onset and location - Insect sting/bite - Food allergy/exposure - New clothing, soap, detergent - PMH of reactions - Medication allergy	- Itching, hives - Coughing, wheezing - Respiratory distress - Difficulty swallowing - Vomiting, diarrhea - Hypotension - Edema	- Urticaria (rash only) - Anaphylaxis - Angioedema - Aspiration/obstruction - Asthma/COPD Heart - Failure

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

UNSTABLE/ ANAPHYLACTIC SHOCK: Altered Mental Status, MAP<65 or SBP<90

3. Secure airway		•
4. EPINEPHRINE 1 mg/mL: 0.3 mg IM in anterolateral thigh, may repeat 5 min., no max.	•	•
5. DIPHENHYDRAMINE : 50 mg IVP/IO slowly over 2 minutes		•
or : 50mg IM	•	•
6. NORMAL SALINE in 500 mL increments, titrate to MAP ≥ 65		•
7. DuoNeb: ALBUTEROL: 2.5 mg /IPRATROPIUM BROMIDE: 0.5 mg , may repeat x1	•	•
8. ALBUTEROL : 2.5 mg NEB, may repeat 5 min., no max. dosage	•	•
9. If condition continues to deteriorate and MAP <65 despite NORMAL SALINE contact OLMC : PUSH DOSE EPINEPHRINE : 50 mcg (5mL) IVP/IO, repeat 5 min., max. 100mcg, target MAP ≥ 65		•

STABLE: GCS 15, MAP >65, SBP >90, skin warm/dry
WITH AIRWAY INVOLVEMENT:

3. EPINEPHRINE 1 mg/mL: 0.3mg IM in anterolateral thigh, may repeat 5 min., no max.	•	•
4. DIPHENHYDRAMINE : 50 mg IVP slowly over 2 minutes		•
or : 50mg IM	•	•
5. DuoNeb: ALBUTEROL: 2.5 mg /IPRATROPIUM BROMIDE: 0.5 mg , may repeat x1	•	•
6. ALBUTEROL : 2.5 mg NEB, may repeat 5 min., no max. dosage	•	•

STABLE: GCS 15, MAP >65, SBP >90, skin warm/dry
WITHOUT AIRWAY INVOLVEMENT:

3. Apply cold pack to site.	•	•	•
4. DIPHENHYDRAMINE : 25 mg IVP slowly over 2 minutes			•
or : 25mg IM	•	•	

NOTE:

1. DuoNeb **ALBUTEROL/IPRATROPIUM BROMIDE & ALBUTEROL** NEB treatment may be administered in-line for those patients requiring CPAP or intubation.

PEARLS:

1. Patients with asthma, atopic dermatitis (eczema), prior anaphylactic history, and those who delay treatment can be at greater risk for a fatal reaction.
2. Gastrointestinal symptoms occur most commonly in food-induced anaphylaxis, and oral pruritus is often the first symptom observed.
3. New onset hoarseness or underpowered voice indicates vocal cord edema and should prompt concern for airway compromise

ADULT DIABETIC EMERGENCIES

HPI	Signs & Symptoms	
- PMH diabetes - Onset of symptoms - Last meal	<u>Hypoglycemia</u> - Altered mental status - Cool, clammy skin - Diaphoresis Seizure - Decreased visual acuity, blindness	<u>Hyperglycemia</u> - Altered mental status - Dry, red, warm skin - Acetone (fruity) smell on breath - Kussmaul respirations - Dry mouth, intensive thirst - Nausea/vomiting and abdominal pain

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography		•	•
3 Obtain blood glucose level	•	•	•

HYPOGLYCEMIA: Altered Mental Status, BG < 60 or cool/clammy skin

4. ORAL GLUCOSE GEL : 15 gm (If patient is able to tolerate oral preparation, has gag reflex and able to protect own airway)	•	•	•
5. DEXTROSE 10% : 25 gm/250 mL IV, may repeat, max. 50 gm			•
6. or if no IV, GLUCAGON 1 mg IM/IN		•	•

HYPERGLYCEMIA with DEHYDRATION:

BG >250 with dehydration, vomiting, abdominal pain, AMS
or unable to obtain BG with warm, flushed skin and deep, rapid respirations

4. NORMAL SALINE in 500 mL increments, titrate to desired patient response	•
--	---

PEARLS:

1. Asymptomatic hyperglycemia poses no risk to the patient while inappropriately aggressive interventions to manage blood sugar can harm patients.
2. Glucagon causes a breakdown of stored glycogen to glucose. Glucagon may not work if glycogen stores are previously depleted due to liver dysfunction, alcoholism, or malnutrition. Effects of Glucagon may take up to 30 minutes.

SEVERE FEBRILE RESPIRATORY ILLNESS

HPI	Signs & Symptoms	Differential Diagnosis
- Cough - Fever - Contact - Associated high risk	- Fever > 100.4 F - Cough, shortness of breath or hypoxia - Contact with confirmed/suspected ≤10 days - Occupation with risk - Travel history - Atypical pneumonia	Asthma/COPD

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Patient to don surgical mask, hand hygiene with waterless soap, instruct on tissue use	•	•	•
3. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
4. Notify receiving facility of precautions in place	•	•	•
5. Consider initial facility evaluation of patient in back of ambulance to determine isolation needs			•
6. Leave equipment in patient room until appropriately cleaned	•	•	•
7. Document exposure to possible communicable disease	•	•	•

PEARLS:

1. Refer to agency policy on decontamination of ambulance and equipment.
2. Limit number of personnel exposed. Provider to don N95 mask, gloves, gown, and eye protection. Avoid touching outside of N95 mask.

ADULT NAUSEA/VOMITING

HPI	Signs & Symptoms	Differential Diagnosis
- Time of last meal - Last bowel movement - Last emesis - Relation to food - Menstrual cycle - Trauma	- Pain, character - Distention - Constipation - Diarrhea - Radiation	Cardiac - Drugs (NSAID's, antibiotics, narcotics) - GI or renal disorders - Food or toxin induced - Pregnancy, GYN

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
2. Obtain ECG and/or 12 lead, monitor Capnography.		●	●
3. ONDANSETRON : 4 mg IVP over 30 seconds			●
or 4 mg ORAL, May repeat 10 min., max. 8 mg		●	●

NOTE:

- 1. Do not administer to patients who are pregnant

ACETYLCHOLINESTERASE INHIBITOR EXPOSURE

HPI	Mild	Moderate	Severe
- Nerve agent, carbamate, organophosphate, insecticide (tabun, sarin, soman, VX) - Agricultural pesticides - Industrial manufacturing products	- Pupil constriction (primary sign in vapor exposure, may not be present in all exposures) - Severe rhinorrhea	- Nausea/vomiting - Weakness - Muscle fasciculations - Localized swelling - Shortness of breath	- Unconscious - Convulsions - Apnea or severe respiratory distress requiring assisted ventilation - Flaccid paralysis

	EMR	EMT	PM
1. Ensure patient has been thoroughly decontaminated.	●	●	●
2. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
3. Administer ATROPINE IM or 2-PAM per tables below, based on patient presentation		●	●
4. Obtain ECG and/or 12 lead, monitor Capnography.		●	●

MILD:	ATROPINE auto-injector	OR ATROPINE	600MG auto-injector	OR 2-PAM
0-2yrs (<13kg)	0.5 mg	0.05 mg/kg		
3-7yrs (13-25 kg)	1 mg	1 mg		
8-14yrs (26-50 kg)	2 mg	2 mg		
>14 yrs	2 mg	2 mg		
Pregnant	2 mg	2 mg		
Geriatric	1 mg	1 mg		

MODERATE:	ATROPINE auto-injector	OR ATROPINE	600MG auto-injector	OR 2-PAM
0-2yrs (<13kg)	0.5 mg	0.05 mg/kg		15 mg/kg
3-7yrs (13-25 kg)	1 mg	1 mg	1	15 mg/kg
8-14yrs (26-50 kg)	2 mg	2 mg	1	15 mg/kg
>14 yrs	2-4 mg	2-4 mg	1	600 mg
Pregnant	2-4 mg	2-4 mg	1	600 mg
Geriatric	2 mg	2 mg	1	10 mg/kg

SEVERE:	ATROPINE auto-injector	OR ATROPINE	600MG auto-injector	OR 2-PAM
0-2yrs (<13kg)	0.5 mg	0.1 mg/kg		45 mg/kg
3-7yrs (13-25 kg)	2 mg	0.1 mg/KG	1	45 mg/kg
8-14yrs (26-50 kg)	4 mg	4 mg	2	45 mg/kg
>14 yrs	6 mg	6 mg	3	1800 mg
Pregnant	6 mg	6 mg	3	1800 mg
Geriatric	2-4 mg	2-4 mg	2-3	25 mg/kg

PEARLS:

1. Clinical improvement based on drying of secretions and respiratory effort
2. Atropine is primary antidote; administer repeated doses liberally to patients who exhibit signs/symptoms until excessive secretions resolve.
3. Treat [Seizures](#) with benzodiazepines per SOP
4. Can give meds IV/IO; however, IM recommended eliminating delay establishing IV/IO.

NOTE:

1. If activating the Multiple Patient Management Plan - consider activation of [CHEMPACK Resources](#).

ADULT BETA BLOCKER OVERDOSE

HPI	Signs & Symptoms	Differential Diagnosis
- Available medications - Ingestion history - Route, and quantity used - Reason (suicidal, accidental, criminal) - PMH	AMS - Hypotension Bradycardia - Slurred speech Nausea/vomiting	- Co-ingestions - Conduction delays

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90

3. NORMAL SALINE in 500 mL increments, titrate to MAP ≥ 65	•
4. If condition deteriorates and MAP <65 despite NORMAL SALINE: ATROPINE : 1 mg IVP/IO, rapid push, may repeat 5 min., max. 3 mg	•
5. GLUCAGON : 1 mg IVP/IO, repeat 5 min., max 2 mg	•
6. If widened QRS, consider SODIUM BICARBONATE 50 mEq IV/IO	•
7. If condition continues to deteriorate, consider Transcutaneous Pacing	•

PEARLS:

1. Glucagon increases heart rate and myocardial contractility, and improves atrioventricular conduction.
2. Cardiac pacing may be effective in increasing the rate of myocardial contraction. Electrical capture is not always successful and, if capture does occur, blood pressure is not always restored. Reserve cardiac pacing for patients unresponsive to pharmacological therapy.
3. Do not give patient anything to eat or drink by mouth.
4. Anticipate vomiting, respiratory arrest, seizure, dysrhythmias and refer to indicated protocols.
5. Do not induce vomiting, especially in cases where caustic substance ingestion is suspected.
6. Call Illinois Poison Center (800) 222-1222 for additional information.

ADULT CALCIUM CHANNEL BLOCKER OVERDOSE

HPI	Signs & Symptoms	Differential Diagnosis
- Available medications - Ingestion history - Route, and quantity used - Reason (suicidal, accidental, criminal) - PMH	AMS - Hypotension Bradycardia - Slurred speech Nausea/vomiting	- Co-ingestions - Conduction delays

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90

3. NORMAL SALINE in 500 mL increments, titrate to MAP ≥ 65	•
4. If condition deteriorates and MAP <65 despite NORMAL SALINE: ATROPINE : 1 mg IVP/IO, rapid push, may repeat 5 min., max. 3 mg	•
5. If condition continues to deteriorate, consider Transcutaneous Pacing	•

PEARLS:

- Cardiac pacing may be effective in increasing the rate of myocardial contraction. Electrical capture is not always successful and, if capture does occur, blood pressure is not always restored. Reserve cardiac pacing for patients unresponsive to pharmacological therapy.
- Do not give patient anything to eat or drink by mouth.
- Anticipate vomiting, respiratory arrest, seizure, dysrhythmias and refer to indicated protocols.
- Do not induce vomiting, especially in cases where caustic substance ingestion is suspected.
- Call Illinois Poison Center (800) 222-1222 for additional information.

ADULT POISONING / OVERDOSE
OPIATES / DEPRESSANTS

HPI	Signs & Symptoms	Differential Diagnosis
- Available medications - Ingestion history - Route, and quantity used - Reason (suicidal, accidental, criminal)	- AMS - Decreased respiratory rate - Dysrhythmias - Hypo/er-tension - Hypo/er-thermia	- Benzodiazepine OD - Cardiac medication OD - Tricyclic antidepressants OD - Mix of stimulants/opiates

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, respiratory depression

3. Consider physical restraints before administering medication	•	•
4. NALOXONE : 2 mg IN/IM, may repeat 3 min., max. 10 mg	•	•
or : 2 mg IV/IO, may repeat 3 min., max. 10 mg		•

NOTE:

- Bring container of drug or substance providing that the transport of the item does not pose a safety risk.
- Consider the use of restraints prior to the administration of
- If NALOXONE was administered prior to the arrival of EMS, obtain dose/route information for reporting and documentation.
- Do not give patient anything to eat or drink by mouth.
- Anticipate vomiting, respiratory arrest, seizure, dysrhythmias and refer to indicated protocols.
- Do not induce vomiting, especially in cases where caustic substance ingestion is suspected.
- Call Illinois Poison Center (800) 222-1222 for additional information.

ADULT STIMULANT OVERDOSE

HPI	Signs & Symptoms	Differential Diagnosis
- Available medications - Ingestion history - Route, and quantity used - Reason (suicidal, accidental, criminal)	- AMS - Hyperthermia - Dysrhythmias - Hypo/er-tension - Tachypnea - Seizures	- Diabetic Ketoacidosis - Anticholinergic OD - Solvents, cleaning agents

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
3. Consider physical restraints before administering medication		•	•
4. If signs of hyperthermia, initiate rapid cooling		•	•

UNSTABLE: SEVERE agitation, aggression, or violent behavior

5. > 5 years old and ≤ 65 years old: KETAMINE : IM dosing, use ideal body weight chart below, max. single dose 300mg If severe agitation, aggression, or violent behavior continues after 5 minutes: KETAMINE : IM dosing, use ideal body weight chart below, max. single dose 450mg >65 years old: MIDAZOLAM : 2mg IN/IV/IO/IM, may repeat as needed every 2 min., max total dose 10mg	•
6. If emergence reaction, contact OLMC to consider MIDAZOLAM for agitation/hallucinations	•
7. ONDANSETRON : 4 mg IVP over 30 seconds	•
or 4 mg ORAL, May repeat 10 min., max. 8 mg	•

NOTE:

- Four point restraints for patient and provider safety should be the first line. Sedatives including **KETAMINE** should only be used for the treatment of severe agitation and never to facilitate the transport of an involuntary patient.
- KETAMINE** should NOT be a first-line medication for NON-behavioral or NON-drug-induced psychosis (e.g., postictal confusion). Instead **MIDAZOLAM** should be considered. Contact **OLMC** for questions.
- Do NOT cover patient's face after administering sedation.
- NEVER transport patient prone.

IDEAL BODY WEIGHT		
ADULT MALE:		
	Initial Dose:	
height	dose	vol.
< 5'6"	200 mg	4 mL
5'6"-6"	250 mg	5 mL
> 6'	300 mg	6 mL
	Repeat Dose:	
height	dose	vol.
< 5'6"	100 mg	2 mL
5'6"-6"	125 mg	2.5 mL
> 6'	150 mg	3 mL

ADMINISTRATION LOCATION:
anterolateral thigh: max 5mL
shoulder: max 2mL

ADULT FEMALE:		
	Initial Dose:	
height	dose	vol.
< 5'	180 mg	3.6 mL
5'-5'6"	200 mg	4 mL
> 5'6"	250 mg	5 mL
	Repeat Dose:	
height	dose	vol.
< 5'	90 mg	1.8 mL
5'-5'6"	100 mg	2 mL
> 5'6"	125 mg	2.5 mL

ADULT PAIN MANAGEMENT

HPI	Signs & Symptoms	Differential Diagnosis
• Age • Location • Duration • Severity • PMH • Medications • Allergies to medications	• Quality • Relation to movement • Relation to breathing • Increase with palpation	• Musculoskeletal • Visceral (abdominal) • Cardiac • Pleural, respiratory • Neurogenic • Renal (colic)

	EMR	EMT	PM
1. Provide Adult Routine Medical/Trauma Care - treat life threatening problems with airway, breathing, circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
3. FENTANYL : 1 mcg/kg IVP/IN/IO/IM (100 mcg max single dose), may repeat 10 min., max total dose 200mcg			•
4. If respiratory depression occurs NALOXONE : 2 mg IN/IM, may repeat 3 min., max. 10 mg		•	•
or 2 mg IV/IO, may repeat 3 min., max. 10 mg			•
5. Consider prophylactic nausea/vomiting : ONDANSETRON : 4 mg IV/IO, over 30 sec.			•
or 4 mg ORAL, may repeat 10 min., max. 8 mg		•	•

ADULT SEIZURES

STATUS EPILEPTICUS

HPI	Signs & Symptoms	Differential Diagnosis
- Seizure activity description - Previous seizure history - History of trauma - History of diabetes mellitus - History of pregnancy	- Decreased mental status - Sleepiness - Incontinence - Observed seizure activity - Evidence of trauma - Evidence of mouth trauma	TBI, increased ICP Diabetes Eclampsia Poisoning/Overdose - Fever Stroke

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
3. Nausea/vomiting precautions. Protect patient from injury.		•	•

ACTIVE SEIZURE: Altered Mental Status

4. MIDAZOLAM : 10 mg IM or 2 mg IVP/IO/IN, may repeat 2 min., max. 10 mg			•
5. Obtain blood glucose level. If blood glucose is < 60 or > 250, see Diabetic Emergencies protocol	•	•	•
6. If seizure activity continues or recurs, contact OLMC to administer further doses of MIDAZOLAM .			•

PEARLS:

- Status epilepticus is defined as two or more consecutive seizures without a period of consciousness or recovery. This is a true emergency requiring rapid airway control, treatment, and transport.
- Do NOT place anything in mouth if actively seizing.

ADULT SEPSIS

HPI	Criteria (≥ 2)	Differential Diagnosis
- Fever, chills, sweats - Recent antibiotic use - Suspected, documented infection	- SBP < 100 - MAP < 65 - Temp. > 38° C (100.4° F) or < 36° C (96.8° F) - Respirations > 20/min - Heart Rate > 90 BPM - EtCO2 ≤ 25 mmHg <u>AMS</u>	- Hypovolemic shock Cardiogenic shock

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
3. Evaluate for suspected or documented infection		•	•

UNSTABLE: Any 2 or more of the above listed criteria

4. If ≥ 2 criteria met, call OLMC for SEPSIS ALERT	•	•
5. NORMAL SALINE in 500 mL increments, target 30mL/kg		•
6. If condition continues to deteriorate and MAP <65 despite NORMAL SALINE: PUSH DOSE EPINEPHRINE: 50 mcg (5mL) IVP/IO, repeat 5 min., max. 100mcg, target MAP ≥ 65		•

PEARLS:

1. Bedside report to hospital to include total amount of IVF infused
2. Early, aggressive IV fluid administration is essential in the treatment of suspected sepsis or septic shock.
3. Sometimes patients may present with complaints of weakness, malaise, altered mental status, or simply "not eating." The source of infection may be readily apparent (cellulitis), may require extensive testing, or may be completely obscure.

STROKE

TIA

HPI	Signs & Symptoms	Differential Diagnosis
• Previous CVA/ TIA's • Associated diseases; diabetes, hypertension, CAD, atrial fibrillation • Medications: Coumadin, Eliquis, Xarelto • Recent trauma	• Altered mental status • Weakness, paralysis • Blindness/sensory loss • Aphasia, dysarthria • Headache, Hypertension • Respiratory pattern change • Seizures	• Hypo/er-glycemia • Seizure • Brain tumor • TBI

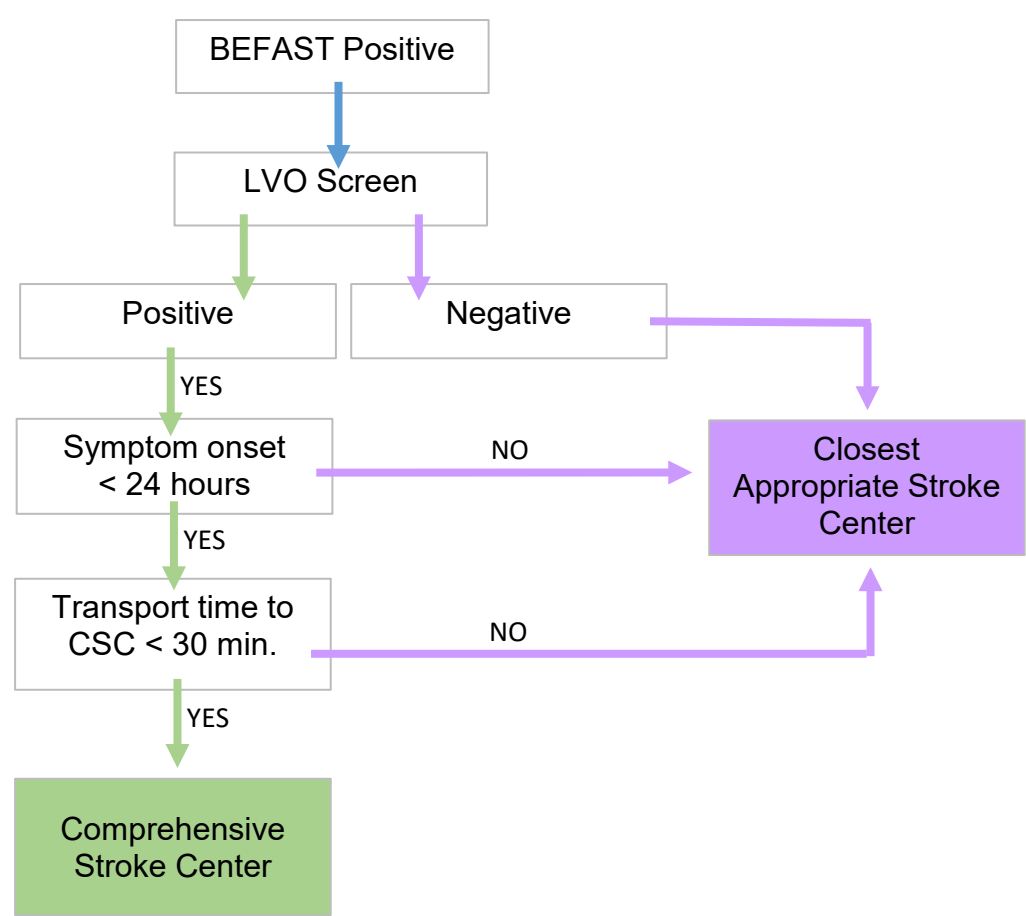
	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing, circulation and disability.	•	•	•
2. Determine last known well time	•	•	•
3. Obtain blood glucose level. If blood glucose is < 60 or > 250, see Diabetic Emergencies protocol	•	•	•
4. Perform BEFAST Stroke Scale and if positive, perform Large Vessel Occlusion (LVO) Screen If negative LVO Screen, transport to closest appropriate Stroke Center. Contact OLMC with Stroke Alert. If positive LVO Screen, see chart below for appropriate destination.	•	•	•
5. Obtain ECG and 12 lead, monitor Capnography.		•	•

	Assess	Abnormal (acute findings)
B Balance	performance of finger to nose and heel to shin test	loss of balance, dizziness, or trouble walking
E Eyes	conjugate gaze	vision loss, diplopia, blurred vision, or fixed gaze deviation
F Face	smile	facial numbness or weakness
A Arm	arms extended with palms up and leg raise for drift	unilateral weakness or numbness to arms or legs
S Speech	repeating a phrase or naming a common object	difficulty speaking or comprehension
T Time	Established last known well time	Activate Stroke Alert if symptoms within 24 hours

LARGE VESSEL OCCLUSION SCREEN: Any one of the following

- 1. Dense/profound deficit
 - new sudden onset speech deficit
 - sudden significant gait disturbance
 - significant extremity weakness
- 2. Fixed eye gaze deviation
- 3. Hemineglect (ignoring one side of the body)

DESTINATION DETERMINATION:



NOTE:

- 1. Rapid neurologic deterioration (unequal pupils, decerebrate posturing, lateralizing signs) often seen secondary to increased ICP should be treated with ventilations guided by Capnography. Ventilate patient 1 breath/3-6 seconds to Capnography of 35 mmHg.
- 2. Transport to Comprehensive Stroke Center (CSC) if transport time is less than 30 minutes and within agency's normal transport destination.
- 3. If positive BEFAST, notify **OLMC** of Stroke Alert. If positive LVO screen, notify **OLMC** of Stroke Alert with positive LVO.

REGION X FIELD TRIAGE AND TRANSPORT CRITERIA

- [Traumatic arrest](#) - rapid transport to closest Trauma Center
- No airway - rapid transport to closest Comprehensive Emergency Center

CATEGORY I: Unstable Vitals and/or Anatomic Criteria

Rapid transport to **highest level** Trauma Center ≤ 25 min. transport time

Unstable Vital Signs

- SBP: Adult ≤ 90 (2 consecutive readings)
Pediatric ≤ 80 (2 consecutive readings)
- GCS: < 14 with associated head trauma
- RR: 0-12mo.: < 20/min.
> 1yr.: <10 or >29/min.
need for ventilatory support
- Adult Shock Index: HR > SBP

Anatomic Criteria

- Penetrating injuries to head, neck, torso, or extremities proximal to elbow/knee
- ≥ 2 proximal long bone fractures
- Unstable pelvis
- Chest wall instability or deformity (e.g. flail chest)
- Crushed, degloved, mangled or pulseless extremity
- Open or depressed skull fractures
- Amputation proximal to wrist or ankle
- Paralysis

CATEGORY II: Mechanism of Injury

If patient does not meet Category I, rapid transport to **closest** Trauma Center

High Risk Crash

- Ejection from vehicle (partial or complete)
- Death in same passenger compartment
- Intrusion into passenger compartment > 12 in.
- Intrusion into non-passenger compartment > 18 in.
- Vehicle telemetry data consistent with a high risk for injury
- Unrestrained rollover
- Motorcycle crash > 20 mph
- Auto vs. Pedestrian thrown, run over, or with > 20 mph impact
- Auto vs. Bicyclist thrown, run over, or with > 20 mph impact

Falls Adult: ≥ 20 feet (1 story = 10 feet)
Pediatric: ≥ 10 feet or 2x height of the child

Other Burns with or without associated trauma

SPECIAL CONSIDERATIONS:

If patient does not meet Category I or Category II, transport to **closest appropriate** facility. If this is not a Trauma Center, contact **OLMC**

- Adults > 55 (increased risk for injury/death)
- Adults > 65 with SBP < 110 may indicate shock
- Adults > 65 with low MOI/standing falls may cause severe injury
- Anticoagulants/bleeding disorders w/ head injury may rapidly deteriorate
- EMS provider judgement

All other transport to closest appropriate Comprehensive Emergency Center

NOTE:

1. Pediatrics should be preferentially transported to appropriate level pediatric-capable Trauma Center
2. Pregnancy > 20 weeks should be preferentially transported to facility with emergency obstetrics capabilities

TRAUMATIC ARREST

	EMR	EMT	PM
1. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●

WITHHOLDING RESUSCITATION

2. Patients in asystole or Patients with an injury incompatible with life: decapitation thoracic transection incineration (90% surface area burn)	●
3. Contact OLMC for confirmation and time of death	●

INITIATE RESUSCITATION

2. Patients presenting in the following conditions: drowning hypothermia visible pregnancy pediatric witnessed loss of pulse for penetrating trauma MOI does not fit a traumatic cause for arrest	●	●
if any electrical activity is present		●
3. Minimize on scene time. Initiate rapid transport to closest Trauma Center	●	●
4. Treatment should focus on reversible causes of PEA/traumatic arrest		●
5. DO NOT administer EPINEPHRINE 1 mg/10mL for traumatic arrest		●

PEARLS:

1. When the mechanism of injury does not correlate with the clinical condition, suggesting a nontraumatic cause of cardiac arrest, standard resuscitative measures should be followed. In such cases, refer to appropriate protocol
2. Survival from traumatic cardiac arrest requires careful coordination between rapid prehospital assessment, EMS clinician treatment of reversible causes of traumatic cardiac arrest and transport that is rapid
3. Treatment priorities should initially focus on control of massive hemorrhage (including management of pelvis fractures), airway management, and consideration of bilateral [needle decompression](#). If CPR is performed, it should be performed en route to the hospital

SUSPECTED ADULT / ELDER ABUSE

HPI	Signs & Symptoms	Considerations
- Type of injury - MOI - Timeline of incidents	- Dehydration - Fractures - Recurring injuries - Decubitus	- Emotional trauma - Behavioral emergency - Traumatic injury - Failure to thrive

	EMR	EMT	PM
1. Provide Adult Routine Medical/Trauma Care - treat life threatening problems with airway, breathing, circulation and disability	•	•	•

Adult Abuse:	the mistreatment to any resident age 18-59 living with a disability and any adult 60 years of age or older who live in a domestic setting.
Abuse:	physical, sexual or emotional maltreatment or willful confinement
Neglect:	the failure of a caregiver to provide an adult with the necessities of life, including, but not limited to food, clothing, shelter or medical care. Neglect may be either passive (non-malicious) or willful.

NOTE:

- Abuse and/or neglect of elderly patients may occur in the non-institutional or nursing home setting.
- It is mandated by the State of Illinois to report suspected abuse cases to the Abuse Hot Line:
Elder Abuse 24-hour Hotline: 866-800-1409
- The prehospital provider must accurately and completely document any physical findings on the PCR and relay such findings to the Emergency Department Staff upon transfer to the hospital.

ADULT BLAST INJURY

HPI	Signs & Symptoms	Considerations
- Cause of blast - Time of exposure - Chemicals involved - Location to blast	- AMS, unresponsive - Hypotension - Severe hemorrhage - Deafness - Respiratory compromise, failure	- Management of HazMat - Management of MCI - Coordination of resources

	EMR	EMT	PM
1. Ensure scene safety. Remove patient as soon as is practical and safe	•	•	•
2. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
3. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
4. Refer to Airway Management as needed	•	•	•
5. Refer to Musculoskeletal Trauma as needed	•	•	•
6. Evaluate for: burns - thermal/chemical/electrical blunt/penetrating trauma crush injury barotrauma (pressure related injury) chemical contamination radiation injury	•	•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90

7. NORMAL SALINE in 500mL increments, titrate to MAP ≥ 65, max. 1L	•
8. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg	•

PEARLS:

1. Ensuring scene safety is especially important at the scene of an explosion. Always consider the possibility of subsequent explosions.
2. Patients sustaining blast injury may sustain complex, multi-system injuries, including blunt and penetrating trauma, shrapnel, barotrauma, burns, and toxic chemical exposure

ADULT BURNS

HPI	Signs & Symptoms	Differential Diagnosis
- Type of exposure - Need for decontamination - Percent surface area - Inhalation injury - PMH - Other trauma	- Pain, swelling - Hypotension - Hoarseness, wheezing - Singed facial, nasal hair - Airway compromise, distress - Carbonaceous sputum	- Chemical - Thermal - Radiation - Electrical

	EMR	EMT	PM
1. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
3. Consider Airway Management	•	•	•
4. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg			•
5. Estimate total burn surface area using Rule of 9's	•	•	•
6. NORMAL SALINE in 500mL increments, titrate to MAP ≥ 65			•

THERMAL:

Superficial			
7. Cool burned area with water or NORMAL SALINE	•	•	•
8. If < 20% total burn surface area, apply sterile NORMAL SALINE soaked dressings	•	•	•
Partial or Full Thickness			
7. Apply dry sterile dressing	•	•	•

ELECTRICAL or LIGHTNING:

7. Treat dysrhythmias per protocol			•
8. Identify and document entry/exit wounds	•	•	•
9. Assess neurovascular status (PMS) of affected area	•	•	•
10. Immobilize affected area	•	•	•
11. Apply dry sterile dressing	•	•	•

CHEMICAL:

7. Refer to department HazMat protocol for decontamination	•	•	•
8. If powder: remove affected clothing, brush away excess	•	•	•
9. Flush burn area with sterile water or NORMAL SALINE	•	•	•
10. If eye involvement: Assist patient in removing contact lens Flush burned eye with sterile water or NORMAL SALINE continuously	•	•	•

PEARLS:

- Onset of stridor and change in voice are sentinel signs of potentially significant airway burns, which may rapidly lead to airway obstruction or respiratory failure.

ADULT CRUSH INJURY

HPI	Signs & Symptoms	Differential Diagnosis
- Traumatic crush mechanism of injury - Entrapment of extremity or trunk - Time entrapped	- Pain out of proportion to clinical exam - Tense swelling - Absent pulses - Paresthesias - Peaked T waves on ECG - Widened QRS on ECG	- Traumatic asphyxia - Hyper-kalemia - Extremity fracture - Spinal cord injury

	EMR	EMT	PM
1. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Manage severe hemorrhage. Adult Musculoskeletal Trauma	•	•	•
3. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

UNSTABLE: Prolonged entrapment, Peaked T, Widened QRS

4. Coordinate extrication with treatment	•
5. Contact OLMC SODIUM BICARBONATE : 50mEq IVP/IO	•
6. and/or ALBUTEROL : 2.5 mg NEB, may repeat 5 min., max. 5mg	•
7. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg	•

PEARLS:

- Carefully monitor for dysrhythmias or signs of hyperkalemia before and immediately after release of pressure and during transport (e.g., peaked T waves, wide QRS, lengthening QT interval, loss of P wave)
- IV fluids should be administered prior to releasing the crushed body part. Crush injury without adequate fluid resuscitation develops into crush syndrome
- A patient with a crush injury may initially present with very few signs and symptoms. Maintain a high index of suspicion for any patient with a compressive mechanism of injury

DOMESTIC VIOLENCE

HPI	Signs & Symptoms	Considerations
- Type of injury - MOI - Timeline of incidents	- Physical injuries - Emotional injuries - Recurring injuries - Withdrawal, hostility	- Emotional trauma - Behavioral emergency - Traumatic injury - Evidence preservation

	EMR	EMT	PM
1. Provide Adult Routine Medical/Trauma Care - treat life threatening problems with airway, breathing, circulation and disability	•	•	•
2. BE NON JUDGMENTAL AND NON THREATENING Respect and take the patient seriously Maintain privacy. The patient should be interviewed and examined alone Questions should be asked when household members are not within hearing distance The patient must be asked directly if their injuries are a result of physical attack Have a high index of suspicion; battered patients rarely admit the source of their injury Aside from the typical injuries look for: Suicide attempts Depression Substance abuse Hysterics Multiple vague somatic complaints Anxiety Miscarriage Maintain a helping approach and be as non-threatening as possible	•	•	•

PEARLS:

1. Domestic Violence is the MOST common form of violence and the least reported. Domestic Violence is the act of attacking, threatening, harassing or interfering with the personal liberty of any family or household member by any other family or household member, excluding any reasonable discipline of a minor child by a parent or guardian of such minor child.
2. If the victim signs a refusal, inform the patient that EMS personnel are mandated by the State to report all cases of domestic violence to the local police. Document this conversation on the PCR. Also, offer immediate and adequate information regarding services available to victims of abuse, for any person suspected to be a victim of domestic abuse.
3. National Domestic Violence Hotline: 800-799-7233

ELECTRICAL DEVICE WEAPON EXPOSURE

HPI	Signs & Symptoms	Considerations
- Events leading up to - Location of darts - Last tetanus	- Local injury - AMS - Chest pain - Respiratory distress, failure	- Drug/alcohol use - Cardiac dysrhythmia STEMI - Respiratory arrest - Cardiac arrest

	EMR	EMT	PM
1. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Manage the patient’s condition that triggered the application of the electrical weapon with attention to scene safety. See Behavioral Emergencies		•	•
3. Ensure patient is appropriately secured/ restrained with assistance of law enforcement to protect the patient and clinicians. Ensure the Taser cartridge is removed from the gun or the wires are cut		•	•
4. Perform comprehensive trauma and medical assessment for injuries (e.g., from falls or altercations)		•	•
5. Obtain ECG and/or 12 lead, monitor Capnography as appropriate		•	•
6. Treat all imbedded barbed darts as a foreign body and leave them for physician removal, stabilize in place and transport the patient		•	•

NOTE:

1. Patients who have been subject to a conducted electrical weapon are not eligible for medical clearance and release to law enforcement. Call **OLMC** if this is requested.

ADULT HEAD SPINAL FACIAL INJURIES

HPI	Signs & Symptoms	Differential Diagnosis
- Time of injury - MOI - blunt vs. penetrating - Loss of consciousness - Evidence of multi-system trauma	- AMS - Pain, swelling, bleeding - Cushing's Triad - Unequal, sluggish pupils - Respiratory distress or failure - Vomiting	Stroke Diabetic Emergency - Hypovolemic Shock

	EMR	EMT	PM
1. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Utilize Spinal Motion Restriction as indicated	•	•	•
3. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90

4. Perform Rapid Trauma Assessment and manage airway as indicated	•	•	•
5. Obtain blood glucose level. If blood glucose is < 60 or > 250, see Diabetic Emergencies protocol	•	•	•
6. NORMAL SALINE in 500 mL increments, titrate to MAP ≥ 65 or SBP ≥ 110			•
7. If signs of increased ICP (see PEARL 1), ventilate 1 breath/3-6 seconds to Capnography of 35 mmHg.		•	•
8. If condition continues to deteriorate and failure to maintain airway/ventilate/oxygenate, consider Drug Assisted Intubation			•
9. For seizures MIDAZOLAM : 10 mg IM or 2 mg IVP/IO/IN, may repeat 2 min., max. 10 mg			•

STABLE: GCS 15, MAP >65, SBP >90, skin warm/dry

4. Perform Focused Exam		•	•
5. Obtain blood glucose level. If blood glucose is < 60 or > 250, see Diabetic Emergencies protocol	•	•	•
6. NORMAL SALINE in 500 mL increments, titrate to MAP ≥ 65 or SBP ≥ 110			•

PEARLS:

- 1. Rapid neurologic deterioration (unequal pupils, decerebrate posturing, lateralizing signs) often seen secondary to increased ICP should be treated with ventilations guided by Capnography. Ventilate patient 1 breath/3-6 seconds to Capnography of 35 mmHg.
- 2. If seizure activity continues or recurs, contact **OLMC** to administer further doses of **MIDAZOLAM**.
- 3. A nasal airway should not be used with significant facial injury or possible basilar skull fracture

ADULT MUSCULOSKELETAL TRAUMA

EXTREMITY TRAUMA

HPI	Signs & Symptoms	Differential Diagnosis
- MOI consistent with extremity injury - Pain scale - Time of injury - Anticoagulant use - Age	- Deformity, instability - shortening, lengthening - New onset PMS changes	- Crush injury/syndrome - Hypovolemic shock

	EMR	EMT	PM
1. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
3. Evaluate for deformity, shortening, rotation, instability	•	•	•
4. Evaluate the neurologic status of extremity (sensation and movement)	•	•	•
5. Evaluate the vascular status of extremity (pulse, pallor, cap. refill, degree of blood loss)	•	•	•
6. Manage life-threatening external bleeding: Expose the wound and apply direct pressure to bleeding site, followed by a pressure dressing If direct pressure/pressure dressing is ineffective or impractical, apply a commercial tourniquet to extremity Junctional injury or any limb wound with persistent bleeding, wound pack tightly to the depth of the wound until bleeding stops. Then apply direct pressure (≥ 3 min.) and/or pressure dressing	•	•	•
7. Manage life-threatening internal bleeding: Stabilize suspected fractures/dislocations	•	•	•
8. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg			•

UNSTABLE: SBP < 90, (< 110 patients ≥ 65), tachycardia > 120 BPM

9. NORMAL SALINE in 500 mL increments, titrate to MAP 65 or SBP 90 (SBP 110 for ≥ 65 years)	•
10. Contact OLMC : TRANEXAMIC ACID (TXA): 1g diluted in 100mL D5W IVPB, over 10 min.	•

PEARLS:

1.

Survival is markedly improved when a tourniquet is placed before shock develops
2.

Properly-applied tourniquets in conscious patients are painful – treat pain with analgesics, but do not loosen a tourniquet to relieve discomfort
3.

[Amputated](#) body parts should be transported with patient for possible re-implantation

SEXUAL ASSAULT

HPI	Signs & Symptoms	Considerations
- Type of injury - MOI - Timeline of incidents	- Physical injuries - Emotional injuries - Recurring injuries - Withdrawal, hostility	- Emotional trauma - Behavioral emergency - Traumatic injury - Evidence preservation

	EMR	EMT	PM
1. Provide Adult Routine Medical/Trauma Care - treat life threatening problems with airway, breathing, circulation and disability	●	●	●
2. Approach the patient calmly and professionally	●	●	●
3. EMS should limit questioning concerning the incident to the minimum necessary to provide appropriate patient care	●	●	●
4. Respect the patient's autonomy. Explain all procedures before beginning the procedures	●	●	●
5. Avoid touching the patient other than taking vital signs or examining physical injuries. (Do not examine the genitalia unless there is a life threatening hemorrhage)	●	●	●
6. Attempt to preserve physical evidence	●	●	●
7. Provide emotional support with a non-judgmental attitude	●	●	●

PEARLS:

1. If the patient signs a refusal, inform the patient that EMS personnel are mandated by the State to report all cases of sexual assault to the local police. Document this conversation on the PCR. Also, offer immediate and adequate information regarding services available to victims of abuse, for any person suspected to be a victim of domestic abuse.
2. National Sexual Assault Hotline: 800-656-4673

BITES AND ENVENOMATIONS

HPI	Signs & Symptoms	Differential Diagnosis
- Type of bite/sting - Description of creature - Time, location, size - Previous reaction - Domestic vs. wild - Tetanus or rabies risk	- Rash, skin break - Pain, soft tissue swelling - Evidence of infection - Shortness of breath - Wheezing - Hypotension	- Allergic reaction Anaphylaxis - Infection

	EMR	EMT	PM
1. Provide Routine Medical/Trauma Care - treat life threatening problems with airway, breathing, circulation and disability	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•
3. Locate the puncture wounds	•	•	•
4. Keep the patient supine with minimal movement, and keep the bitten extremity at the level of the heart	•	•	•
5. Remove all constricting jewelry, rings, watches, bracelets from affected extremity	•	•	•
6. Mark the border of any swelling and/or ecchymosis and note time	•	•	•
7. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg			•

NOTE:

- Do not remove clothing unless causing constriction (minimal movement of patient)
- Do not apply a tourniquet, compression wrap, or ice pack
- Do not place IV in the same extremity as the bite
- If snakebite, include the bite time and species (if known) in verbal and written report. Transfer any personal anti-venom supply with the patient

ADULT DROWNING/SUBMERSION

SCUBA INJURY

HPI	Signs & Symptoms	Differential Diagnosis
- Submersion in water - Possibility of trauma - Duration of submersion - Temperature of water - Depth achieved - Type of water	- AMS, unresponsive - Vomiting - Coughing	- Pre-existing medical - Trauma - Pressure injury

	EMR	EMT	PM
1. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
2. Obtain ECG and/or 12 lead, monitor Capnography.		●	●

SCUBA

3. Maintain in supine position	●	●	●
4. NORMAL SALINE in 500mL increments, titrate to MAP ≥ 65			●
5. Consider needle decompression if tension pneumothorax is suspected			●
6. Keep warm but not hyperthermic. Consider Hypothermia	●	●	●

DROWNING:

3. Consider Hypothermia	●	●	●
4. Contact OLMC to consider CPAP		●	●

NOTE:

1.

Diver’s Alert Network (DAN) 24-hour emergency hotline:
919- 684-9111
2.

If no pulse present follow appropriate [cardiac protocol](#)

ADULT DROWNING/SUBMERSION
SCUBA INJURY

HPI	Signs & Symptoms	Differential Diagnosis
- Submersion in water - Possibility of trauma - Duration of submersion - Temperature of water - Depth achieved - Type of water	- AMS, unresponsive - Vomiting - Coughing	- Pre-existing medical - Trauma - Pressure injury

	EMR	EMT	PM
1. Provide Adult Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

SCUBA

3. Maintain in supine position	•	•	•
4. NORMAL SALINE in 500mL increments, titrate to MAP ≥ 65			•
5. Consider needle decompression if tension pneumothorax is suspected			•
6. Keep warm but not hyperthermic. Consider Hypothermia	•	•	•

DROWNING:

3. Consider Hypothermia	•	•	•
4. Contact OLMC to consider CPAP		•	•

NOTE:

- 1. Diver’s Alert Network (DAN) 24-hour emergency hotline: 919- 684-9111
- 2. If no pulse present follow appropriate [cardiac protocol](#)

EXERTIONAL HEAT STROKE

HPI	Signs & Symptoms	Differential Diagnosis
- Exposure time to heat - Extremities of age - Strenuous exercise in hot weather	- AMS - Rectal temperature >104F	- Heat cramps - Heat exhaustion - Fever - Dehydration

	EMR	EMT	PM
1. Provide Adult/Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
2. Move patient to cool environment, remove clothing to facilitate cooling	●	●	●
3. Obtain ECG and/or 12 lead, monitor Capnography.		●	●

EXERTIONAL HEAT STROKE: AMS, rectal temperature > 104F°

4. Cool on scene and then transport patient	●
5. <u>May</u> assist with rectal temperature measurement if performed by athletic staff on scene with event (probe inserted 6 inches)	●
6. Initiate immersion cooling with or without rectal temperature: a. If tank available, perform whole-body cold-water immersion (assist athletic staff if patient is already in a tank) b. If no tank available, perform Tarp Assisted Cooling with Oscillation (TACO method): Place patient on a large tarp/body bag with ice water and tilt back and forth continuously (assist athletic staff if patient is already in a tarp with oscillating ice water). c. Notify OLMC of onsite immersion cooling for an exertional heat stroke patient	●
7. Stop cooling and transport when rectal temperature < 102.2F° or if mental status improves (may assist staff with rectal temp. if available)	●
8. ALL heat stroke patients require transport	●

NOTE:

- 1. Oral and axillary temperature measurements are unreliable
- 2. Stop immersive cooling if patient begins to shiver.
- 3. If patient vomits, refer to [Nausea/Vomiting Protocol](#)
- 4. If patient seizes, refer to [Seizure Protocol](#)

CONSIDERATIONS FOR COLLAPSED ATHLETE

- 1. Arrhythmia and [cardiac arrest](#)
- 2. [Traumatic brain injury](#)
- 3. [Hypoglycemia](#)
- 4. Hyponatremia - evident by abdominal cramps, vomiting, widened QRS - IV fluid bolus CONTRAINDICATED
- 5. [Anaphylaxis](#)

PEARLS:

- 1. The care goals include: rapid recognition, rapid assessment, and rapid cooling. Rapid recognition of exertional heat stroke and immersion cooling are essential to survival. Critical cell damage occurs in 30 minutes. Cool first/Transport second.

ADULT HYPERTHERMIA
HEAT EMERGENCIES

HPI	Signs & Symptoms	Differential Diagnosis
- Exposure time to heat - Extremities of age - PMH - Poor PO intake	- AMS - Hot, dry, sweaty skin - Pale, clammy skin - Fatigue, dizziness - Nausea/vomiting - Frontal headache - Seizures	- Heat cramps - Heat exhaustion - Heat stroke - Dehydration - Cocaine, amphetamines - Fever

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Move patient to cool environment, remove clothing to facilitate cooling	•	•	•
3. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

STROKE: Altered Mental Status, flushing, hot skin (dry or moist), temperature ≥104F°

4	Establish IV/IO and administer NORMAL SALINE only if patient is hypotensive		•
5	Initiate active rapid cooling: Douse towels or sheets with cool water, place on patient, and fan body Apply cold packs to neck, axilla, and groin	•	•

EXHAUSTION: AMS, sweating, weakness, nausea/vomiting, syncope

4.	NORMAL SALINE in 500mL increments, titrate to desired effect	•
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NOTE:

- Stop active cooling if patient begins to shiver.
- Heat cramps are indicated by normal level of consciousness and muscle cramps or spasms. There is no treatment for these patients.
- If patient seizes, refer to [Seizure Protocol](#)

PEARLS:

- Geriatrics are more sensitive to extreme temperatures.
- Sweating generally stops as core temperature rises above 104F° which is indicative of heat stroke

ADULT HYPOTHERMIA
COLD EMERGENCIES

HPI	Signs & Symptoms	Differential Diagnosis
- Exposure to environment - Extremes of age - Length of exposure - Water involvement - Alcohol, barbiturates use	- Cold, clammy skin - Shivering - AMS - Extremity pain - Sensory abnormality - Bradycardia - Hypotension	- Environmental exposure - Sepsis - Hypoglycemia

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - support life threatening problems with airway, breathing, and circulation.	•	•	•
2. Move patient to warm environment. Remove wet clothing.	•	•	•
3. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

SYSTEMIC HYPOTHERMIA: Whole body involvement

4. Apply heat packs to neck, axilla, groin and thorax	•	•	•
5. Assess for pulse. If no pulse present follow appropriate cardiac protocol a. <u>extremities flex</u>: extend time between medication, repeat defibrillation as core temp. rises b. <u>extremities DO NOT flex</u>: withhold medications, only defibrillate x1			•
			•

FROSTBITE: Isolated body areas of involvement

4. Rapidly re-warm frozen areas with warm water or hot packs wrapped in a towel	•	•	•
5. Apply dry sterile dressing (treat like a burn)	•	•	•
6. Immobilize and elevated frozen areas	•	•	•
7. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg			•

NOTE:

1. [Termination of Resuscitative Efforts](#) does not apply to these patients.

PEARLS:

1. In patients suffering from moderate to severe hypothermia, it is critical to not allow these patients to stand or exercise as this may cause circulatory collapse
2. Shivering typically ceases when core temperature is below 32°C / 90°F

TOXIC EXPOSURE

INGESTIONS

HPI	Signs & Symptoms	Differential Diagnosis
- Availability of substance - Time of exposure - Suspected substance	Toxidromes for specific exposure	- Radiation exposure Nerve agent exposure - Drug lab exposure - Carbon monoxide - Illicit drug cutting agents

	EMR	EMT	PM
1. Assess for scene safety	●	●	●
2. Remove patient from hazardous environment. Decontaminate to remove continued sources of absorption, ingestion, inhalation, or injection	●	●	●
3. Provide Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
4. Obtain ECG and/or 12 lead, monitor Capnography.		●	●
5. Identify intoxicating agent by toxidrome or appropriate environmental clues	●	●	●
6. Contact OLMC for interventions as indicated by identified exposure		●	●

NOTE:

- Bring container of drug or substance providing that the transport of the item does not pose a safety risk.
- Do not give patient anything to eat or drink by mouth.
- Anticipate vomiting, respiratory arrest, seizure, dysrhythmias and refer to indicated protocols.
- Do not induce vomiting, especially in cases where caustic substance ingestion is suspected.
- Call Illinois Poison Center (800) 222-1222 for additional information.

EMERGENCY CHILDBIRTH

HPI	Signs & Symptoms	Differential Diagnosis
- Gravida (# of pregnancies) - Para (# of live births) - Due date - High risk concerns - Previous labor time - State of amniotic sac - Duration, frequency of contractions	- Signs of imminent birth - Crowning - Bulging Perineum - Involuntary Pushing - Signs of complications Prolapsed cord - Profuse bleeding - Meconium staining	- Braxton Hicks contractions - Premature rupture of membranes

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - support life threatening problems with airway, breathing, and circulation.	•	•	•
2. If birth is not imminent, transport patient on her left side	•	•	•

DELIVERY:

3. If contractions are 2 minutes apart, or signs of imminent birth are present, prepare for delivery Don standard precautions and sterile gloves from OB kit Drape mother's abdomen and perineum	•	•	•
4. Protect perineum with gentle hand pressure while supporting the neonate's head as it emerges from the vagina. Tear amniotic membrane if it is still intact at this point.	•	•	•
5. Check for nuchal cord (cord wrapped around the neck)		•	•
6. Facilitate delivery of the upper shoulder by gently guiding the head downwards	•	•	•
7. Facilitate delivery of the lower shoulder by lifting the head and neck slightly	•	•	•
8. Get a firm grasp on neonate. The rest of the neonate should deliver with minimal assistance	•	•	•
9. Note time of delivery and record on neonate's PCR.	•	•	•

NEWBORN and POST PARTUM CARE:

10. Stimulate neonate by drying, rubbing back or flicking the soles of the feet. Do not shake neonate	•	•	•
11. Perform rapid assessment: term gestation, crying or breathing and good muscle tone. Obtain 1 minute APGAR Score . If needed see Neonate Resuscitation		•	•
12. Keep neonate level with the vagina. After 1-3 min., double clamp the cord 8 in. from the newborn and 2 in. apart. Cut between the two clamps	•	•	•
13. Maintain neonate warmth - wrap in blanket, keep head covered	•	•	•
14. Obtain 5 min. APGAR Score. If < 7, repeat score every 5 min. for 20 min.	•	•	•

15. Allow placenta to deliver spontaneously. Do not delay transport while waiting for placenta to deliver. When delivered, collect placenta in plastic bag, bring to hospital and document time of placental delivery.	•	•	•
16. Check perineum for tears. If torn and bleeding, apply direct pressure with sanitary pads, and have patient bring legs together.		•	•
17. Monitor for vaginal bleeding > 500mL and hypotension		•	•
<u>NORMAL SALINE</u> in 500mL increments			•
Once placenta is delivered, massage fundus to stop bleeding. Check every 5 min. for firmness. Massage as necessary		•	•
Contact OLMC <u>TRANEXAMIC ACID</u> (TXA): 1g diluted in 100mL D5W IVPB, over 10 min.			•
18. Utilize identification tags for mother and neonate, must include mothers name, gender of newborn, time of delivery.	•	•	•
19. Transport neonate in a secured seat/device unless resuscitation is needed.	•	•	•

MATERNAL RESUSCITATION MODIFICATIONS

3. Perform left uterine displacement while the patient is in the supine position.	•	•	•
4. Chest compressions should be performed slightly higher on the sternum than normal		•	•
5. Defibrillation should be performed following standard guidelines.			•

PEARLS:

- 1. After delivery of neonate, suctioning (with a bulb syringe) should be reserved for infants who have obvious obstruction to the airway or require positive pressure ventilation
- 2. If neonate presents with [shoulder dystocia](#) (as evident by "Turtle sign"), hyperflex mother's hips to severe supine knee-chest position (McRoberts's). Apply firm suprapubic pressure to attempt to dislodge shoulder.
- 3. If strong cry, regular respiratory effort, good tone, and term gestation, infant should be placed skin-to-skin with mother and covered with dry linen

NEONATE PCR DOCUMENTATION:

- 1. Time of delivery, gender and mother's name.
- 2. If nuchal cord was present.
- 3. If meconium flecks were noted in amniotic fluid.
- 4. APGAR scores at 1 minute and 5 minutes.
- 5. Any infant resuscitation initiated and response.

NEONATE RESUSCITATION

	EMR	EMT	PM
1. Stimulate neonate by drying, rubbing back or flicking the soles of the feet. Keep warm. Lay supine with neutral neck	•	•	•
2. Assess airway and pulse	•	•	•
3. If obvious airway obstruction, suction mouth then nose with bulb syringe		•	•

HR < 60:

4. Ventilate via BVM at 40-60/min. on room air	•	•
5. Reassess every 30 sec., if pulse remains < 60, begin chest compressions at ratio of 3:1 (compressions:ventilations)	•	•
6. Contact OLMC for further instructions	•	•
7. Reassess every 30 sec., if pulse remains < 60, ventilate via BVM on 100% OXYGEN	•	•

APNEA or HR < 100:

4. Ventilate via BVM at 40-60/min. on room air	•	•
5. Reassess every 30 sec., if pulse remains 60 - 100 bpm, continue ventilation via BVM at 40-60/min. on room air	•	•
6. Contact OLMC for further instructions	•	•

NOTE:

- 1. Obtain 5 min. [APGAR Score](#). If < 7, repeat score every 5 min. for 20 min.

PEARLS:

- 1. Initiate resuscitative efforts for neonate documented gestational age > 20 weeks (usually calculated by date of last menstrual period). If any doubt about accuracy of gestational age, initiate resuscitation
- 2. If immediate resuscitation is required and the newborn is still attached to the mother, clamp the cord in two places and cut between the clamps. If no resuscitation is required, warm/dry/stimulate the newborn, and then cut/clamp the cord after 60 seconds or WHEN the cord stops pulsating
- 3. Hypothermia is common in neonates and worsens outcomes of nearly all post-natal complications. Wrap neonate in dry towel or thermal blanket to keep as warm as possible during resuscitation; keep head covered if possible

DELIVERY COMPLICATIONS
BREECH BIRTH / SHOULDER DYSTOCIA

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - support life threatening problems with airway, breathing, and circulation.	●	●	●
2. If birth is not imminent, transport patient on her left side	●	●	●

BREECH BIRTH:

3. Prepare to transport with care enroute if only the buttocks or lower extremities are delivered	●	●
4. If delivery is in progress, contact OLMC	●	●
5. Once legs are delivered, support the baby's body wrapped in a towel/chux	●	●
6. If the cord is accessible, palpate frequently for pulsations and attempt to loosen to create slack for delivery of the head.	●	●
7. After the torso/shoulders are delivered, sweep down the arms.	●	●
8. To facilitate the delivery of the head: If face down, elevate the legs/trunk Apply firm pressure over the mother's fundus Allow the entire body to be delivered with contractions while supporting the newborn's body The head should deliver in ≤ 30 sec. Once head delivers, anticipate need for neonatal resuscitation	●	●
9. If head does NOT deliver ≤ 30 sec. Reach 2 gloved fingers in the shape of a "V" into the vagina with the palm facing the neonate's face to locate mouth/nose Push vaginal wall away from the neonate's face to maintain airway Keep your fingers in place during transport. Alert receiving hospital Keep the delivered portion of the body warm and dry	●	●
10. Anticipate maternal hemorrhage after the birth of the infant	●	●

SHOULDER DYSTOCIA

3. Hyperflex the mother's hips to severe supine knee-chest position. (McRobert's position).	●	●
4. Apply firm suprapubic pressure to attempt to dislodge shoulder.	●	●

PEARLS:

- 1. Take care not to hyperextend the neck. Never attempt to pull the infant by the legs or trunk.
- 2. Do not attempt delivery or delay transport of any single limb or other abnormal presentation
- 3. If needed, put the mother in a kneeling position which may assist in the delivery of the newborn

DELIVERY COMPLICATIONS
PROLAPSED CORD or NUCHAL CORD

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - support life threatening problems with airway, breathing, and circulation.	•	•	•
2. If birth is not imminent, transport patient on her left side		•	•

PROLAPSED CORD: umbilical cord is visualized prior to delivery

3. Elevate mother’s hips. Instruct patient to pant during contractions.	•	•
4. If the umbilical cord presents externally or can be visualized in the vagina, use two fingers of a gloved hand to prevent any presenting part of the delivering fetus from occluding/compressing the cord.	•	•
5. Pressure relieving maneuvers must be maintained throughout transport.	•	•
6. Monitor cord pulsations	•	•

NUCHAL CORD: cord wrapped around neck

3. Slip two fingers around the cord and lift over neonate’s head, proceed with delivery.	•	•
4. If the loop is too tight to slip over the head, attempt to slip the cord over the shoulders and deliver the body through the loop	•	•
5. If unsuccessful, double-clamp cord, cut cord between clamps with sterile scissors to allow for release of cord from neck.	•	•
6. Proceed with delivery.	•	•

PEARLS:

- 1. Cover exposed cord with moist dressing and keep warm

OBSTETRICAL COMPLICATIONS

BLEEDING DURING PREGNANCY

HPI	Signs & Symptoms	Differential Diagnosis
- Due date - Time/amount of bleeding - Trauma - Drug/tobacco use - Prenatal care - Sensation of fetal activity	- Abdominal pain - Vaginal bleeding - Rigid abdomen - Lack of abdominal pain with bright red vaginal bleeding	- Ectopic pregnancy - Placenta previa - Placenta abruptio - Spontaneous abortion

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - support life threatening problems with airway, breathing, and circulation.	•	•	•
2. Position mother on her left side	•	•	•
3. Obtain ECG and/or 12 lead, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or SBP < 90, bleeding > 500mL

4. <u>NORMAL SALINE</u> in 500mL increments, titrate to MAP ≥ 65	•
5. Contact OLMC : <u>TRANEXAMIC ACID</u> (TXA): 1g diluted in 100mL D5W IVPB, over 10 min. ***Contraindicated if presumed or confirmed fetus still in uterus	•

NOTE:

- Type, color and amount of bleeding and/or discharge. If tissue passes, collect and transport to hospital with the patient.

PEARLS:

- Ectopic pregnancy: implantation outside the uterus, first trimester. Presenting with abdominal/pelvic pain with or without minimal bleeding. Shock is possible even with minimal or no vaginal bleeding
- Spontaneous abortion: generally, first trimester presenting with intermittent pelvic pain (uterine contractions) with vaginal bleeding/passage of clots or tissue
- Placenta previa: placenta covers part or all of the cervical opening generally, late second or third trimester. Presents with painless vaginal bleeding
- Abruptio placenta: Most frequently occurs in third trimester of pregnancy; placenta prematurely separates from the uterus causing intrauterine bleeding

HYPERTENSIVE DISORDERS
PRE-ECLAMPSIA and ECLAMPSIA

HPI	Signs & Symptoms	Differential Diagnosis
- Due date - PMH - Recent infection - Prenatal care - Trauma - pregnancy > 20 weeks or postpartum ≤ 4 weeks	- Hypertension - Seizures - Headache - Edema - Visual disturbances	- Pre-existing seizure - Drug/alcohol withdrawal

	EMR	EMT	PM
1. Provide Adult Routine Medical Care - support life threatening problems with airway, breathing, and circulation.	●	●	●
2. Position mother on her left side	●	●	●
3. Obtain ECG and/or 12 lead, monitor Capnography.		●	●
4. Seizure precautions: gentle handling, low CNS stimulation		●	●

UNSTABLE: SBP >160 or DBP >110

5. MAGNESIUM SULFATE : 4g diluted in 100 mL D5W IVPB, over 15	●
6. If hypertension persists, HYDRALAZINE 5 mg IVP over 1 min.	●
7. If hypertension persists after 10 min., HYDRALAZINE 10 mg IVP over 1 min.	●
8. For seizures MIDAZOLAM : 10 mg IM or 2 mg IVP/IO/IN, may repeat 2 min., max. 10 mg	●
9. If seizure activity continues/recurs contact OLMC to repeat MIDAZOLAM	●

PEARLS:

- Respirations during an active seizure should be considered ineffective and airway maintenance should occur per assessment.
- The predominant during pregnancy risk factors for development of preeclampsia include: age extremes (< 20 years or > 35 years), primigravida, and multiple gestation
- Definitive treatment can only be accomplished through delivery of the placenta.
- Although it occurs primarily in first pregnancies, women with pre-eclampsia in a previous pregnancy are 7x more likely to develop again in later pregnancy

PEDIATRIC BRADYCARDIA

HPI	Signs & Symptoms	Possible Causes	
- Foreign body obstruction - Respiratory distress - Toxic exposure - Congenital disease - PMH	- Heart rate < 60 bpm - Cyanosis, mottling - Delayed capillary refill > 2 sec. - Hypotension - AMS	- Hypovolemia - Hypoxia - Acidosis - Hyper/o-kalemia - Hypothermia - Hypoglycemia	- Toxins/OD - Cardiac tamponade - Acute coronary syndrome - Pulmonary embolism - Trauma - Tension Pneumo

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing, circulation, and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•
3. Ventilate with BVM on 100% OXYGEN with airway adjuncts, 1 breath every 2-3 sec.	•	•	•

UNSTABLE: Weak/absent pulses, hypotension, pallor/cyanosis, altered mental status

4. Begin 2-rescuer CPR 15:2 compressions to ventilations for a 2 min. cycle	•	•	•
5. Obtain vascular access IV/IO			•
6. EPINEPHRINE 1 mg/10mL: 0.01 mg/kg IVP/IO, max. single dose 1 mg, repeat 3-5min., no max.			•
7. NORMAL SALINE in 20 mL/kg, then TKO			•
8. Contact OLMC to consider ATROPINE : 0.02 mg/kg IVP/IO, max. single dose 0.5 mg, may repeat 3 min., max. total dose 1 mg			•
9. Contact OLMC to consider transcutaneous pacing			•

STABLE: GCS 15, MAP >65, skin warm/dry

4. Observe for deterioration, need to continue bagging or UNSTABLE protocol	•	•	•
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PEARLS:

1. Although noninvasive pacing may be attempted with OLMC, typically bradycardias of hypoxic etiology do not respond. First line therapy is prompt airway support, ventilation and oxygenation.
2. Consider potential accidental/intentional overdose on [beta-blockers](#) and [calcium channel blockers](#). Refer to specific protocols.

PEDIATRIC CARDIOGENIC SHOCK

HPI	Signs & Symptoms	Differential Diagnosis
• Congenital heart disease • Cardiac surgery • Arrhythmia • Post cardiac arrest	• Restlessness, confusion • Weakness, dizziness • Weak, rapid pulse • Pale, cool, diaphoretic • > 2 sec. capillary refill • mottled, cyanotic	• Hypovolemic shock • Septic shock • Medication reaction • Arrhythmia

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or Capillary refill >2 sec.

3. Verify and check often that lung sounds are clear	•	•
4. NORMAL SALINE in 10 mL/kg, titrate to MAP ≥ 65		•
5. NORMAL SALINE in 20 mL/hour, titrate to MAP ≥ 65		•
6. If condition continues to deteriorate and MAP <65 despite NORMAL SALINE contact OLMC: to consider additional NORMAL SALINE and/or PUSH DOSE EPINEPHRINE : 1mcg/kg IVP/IO, max single dose 50mcg, repeat 5 min, max total 100mcg, target MAP 65		•

Use calculator for dosing

STABLE PEDIATRIC TACHYCARDIA
ADEQUATE PERFUSION

Signs & Symptoms			Possible Causes	
<u>Sinus Tachycardia</u> •P wave normal activity Infant <220 beats/min Child <180 beats/min	<u>Supraventricular Tachycardia</u> •P wave absent •HR not variable •Abrupt rate change Infants ≥220 beats/min Child ≥180 beats/min	<u>Ventricular Tachycardia</u> •P wave absent •HR not variable •Wide QRS complex	• Hypovolemia • Hypoxia • Acidosis • Hyper/o-kalemia • Hypoglycemia • Hypothermia	• Toxins/OD • Cardiac tamponade • Trauma • PE • Tension pneumo

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•

STABLE: GCS 15, MAP >65, skin warm/dry

3. Instruct patient to perform vagal maneuver	•	•
4. <u>ADENOSINE</u> : 0.1 mg/kg, max. dose 6 mg, rapid IVP, followed by rapid flush NORMAL SALINE : 5 mL		•

SUPRAVENTRICULAR TACHYCARDIA

5. If no response in 2 min. - ADENOSINE : 0.2 mg/kg, max. dose 12 mg, rapid IVP, followed by rapid flush NORMAL SALINE : 5 mL	•
6. If no response in 2 min., consider sedation for conscious patients <u>MIDAZOLAM</u> : 0.2 mg/kg IVP/IO, max. single dose 2 mg, may repeat 2 min., max. 10 mg	•
7. <u>Synchronized cardioversion</u> 1 J/kg	•
8. If no immediate response, synchronized cardioversion 2 J/kg	•
9. For Pain Management : <u>FENTANYL</u> : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200	•

VENTRICULAR TACHYCARDIA

5. If no response in 2 min. - <u>AMIODARONE</u> : 5 mg/kg, max. dose 150 mg, diluted in 100 mL D5W IVPB, over 10 min.	•
6. If no response in 2 min., consider sedation for conscious patients MIDAZOLAM : 0.2 mg/kg IVP/IO, max. single dose 2 mg, may repeat 2 min., max. 10 mg	•
7. Synchronized cardioversion 1 J/kg	•
8. If no immediate response, synchronized cardioversion 2 J/kg	•
9. For Pain Management: FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg	•

NOTE:

- 1. If receiving sedation or pain medication, monitor patient with continuous Capnography
- 2. Do not delay cardioversion for sedation

Use Calculator for dosing

UNSTABLE PEDIATRIC TACHYCARDIA

INADEQUATE PERFUSION

Signs & Symptoms			Possible Causes	
<u>Sinus Tachycardia</u> •P wave normal •HR varies with activity Infant <220 beats/min Child <180 beats/min	<u>Supraventricular Tachycardia</u> •P wave absent •HR not variable •Abrupt rate change Infants ≥220 beats/min Child ≥180 beats/min	<u>Ventricular Tachycardia</u> •P wave absent •HR not variable •Wide QRS complex	• Hypovolemia • Hypoxia • Acidosis • Hyper/o-kalemia • Hypoglycemia • Hypothermia	• Toxins/OD • Cardiac tamponade • Trauma • PE • Tension pneumo

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or Capillary refill >2 sec.

SUPRAVENTRICULAR TACHYCARDIA

3. Instruct patient to perform vagal maneuver	•	•
4. ADENOSINE : 0.1 mg/kg, max. dose 6 mg, rapid IVP, followed by rapid flush NORMAL SALINE : 5 mL		•
5. If no response in 2 min. - ADENOSINE : 0.2 mg/kg, max. dose 12 mg, rapid IVP, followed by rapid flush NORMAL SALINE : 5 mL		•
6. If no response in 2 min., consider sedation for conscious patients MIDAZOLAM : 0.2 mg/kg IVP/IO, max single dose 2mg, may repeat 2 min., max. 10 mg		•
7. Synchronized cardioversion 1 J/kg		•
8. If no immediate response, synchronized cardioversion 2 J/kg		•
9. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max. single dose 100 mcg, repeat every 10 min., max. total dose 200 mcg		•

VENTRICULAR TACHYCARDIA

3. Consider sedation for conscious patients MIDAZOLAM: 0.2 mg/kg IVP/IO, max single dose 2mg, may repeat 2 min., max. 10 mg	•
4. Synchronized cardioversion 1 J/kg	•
5. If no immediate response, synchronized cardioversion 2 J/kg	•
6. Contact OLMC to consider AMIODARONE	•
7. For Pain Management: FENTANYL: 1 mcg/kg IVP/IN/IO/IM, max. single dose 100 mcg, repeat every 10 min., max. total dose 200mcg	•

NOTE:

1. If receiving sedation or pain medication, monitor patient with continuous Capnography
2. Do not delay cardioversion for sedation

Use Calculator for dosing

PEDIATRIC ASYSTOLE

PULSELESS ELECTRICAL ACTIVITY

HPI	Signs & Symptoms	Possible Causes	
• Witnessed vs unwitnessed • Last known well • PTA AED use • Bystander CPR • Suspected etiology • PMH	• Pulseless • Apneic • No signs of rigor or lividity • No findings of signs incompatible with life	• Hypovolemia • Hypoxia • Acidosis • Hyper/o-kalemia • Hypothermia • Hypoglycemia	• Toxins/OD • Cardiac tamponade • Acute coronary syndrome • Pulmonary embolism • Trauma • Tension Pneumo

UNIVERSAL EMERGENCY CARDIAC CARE	EMR	EMT	PM
1. Assess responsiveness, pulse and breathing for no more than 10 seconds.	•	•	•
2. Consider and treat reversible causes of arrest	•	•	•
3. If no pulse and not breathing/gasping - Begin 2-rescuer CPR 15:2 compressions to ventilations for a 2 min. cycle	•	•	•
4. Ventilate with BVM on 100% OXYGEN with airway adjuncts, 1 breath every 2-3 sec.	•	•	•
5. Monitor Capnography.		•	•
6. Place monitor/defibrillator immediately as staffing allows	•	•	•
7. Analyze rhythm and continue CPR 2 min.			•
8. Obtain vascular access IV/IO			•
9. Consider advanced airway: ETT			•
or i-Gel.		•	•
Monitor placement with Capnography. Deliver 1 breath/6 sec.			

PERSISTENT NON-SHOCKABLE	
10. Analyze rhythm. Continue CPR 2 min.	•
11. EPINEPHRINE 1 mg/10mL: 0.01 mg/kg IVP/IO, max. single dose 1 mg, repeat 3-5min., no max.	•
12. NORMAL SALINE in 20 mL/kg, may repeat	•
13. If return of spontaneous circulation see ROSC	•

- NOTE:**
1. Utilize [High Performance CPR](#) for best patient outcome.
 2. The best airway is an effective airway with the least potential complications.

Use Calculator for dosing

PEDIATRIC VENTRICULAR FIBRILLATION
PULSELESS VENTRICULAR TACHYCARDIA

HPI	Signs & Symptoms	Possible Causes	
- Witnessed vs unwitnessed - Last known well - PTA AED use - Bystander CPR - Suspected etiology - PMH	- Pulseless - Apneic - No signs of rigor or lividity - No findings of signs incompatible with life	- Hypovolemia - Hypoxia - Acidosis - Hyper/o-kalemia - Hypothermia - Hypoglycemia	- Toxins/OD - Cardiac tamponade - Acute coronary syndrome - PE - Trauma

UNIVERSAL EMERGENCY CARDIAC CARE

	EMR	EMT	PM
1. Assess responsiveness, pulse and breathing for no more then 10 seconds.	•	•	•
2. Consider and treat reversible causes of arrest	•	•	•
3. If no pulse and not breathing/gasping - Begin 2-rescuer CPR 15:2 compressions to ventilations for a 2 min. cycle	•	•	•
4. Ventilate with BVM on 100% OXYGEN with airway adjuncts, 1 breath every 2-3 sec.	•	•	•
5. Monitor Capnography.		•	•
6. Place monitor/defibrillator immediately as staffing allows	•	•	•
7. Analyze rhythm and defibrillate 2 J/kg or equivalent biphasic. Continue CPR 2 min.			•
8. Obtain vascular access IV/IO			•

PERSISTENT V.FIB/PULSELESS V.TACH

9. Analyze rhythm and defibrillate 4 J/kg or equivalent biphasic. Continue CPR 2 min.	•
10. EPINEPHRINE 1 mg/10mL: 0.01 mg/kg IVP/IO, max. single dose 1 mg, repeat 3-5min., no max.	•
11. NORMAL SALINE in 20 mL/kg, may repeat	•
12. Analyze rhythm and defibrillate 6 J/kg or equivalent biphasic. Continue CPR 2 min.	•
13. AMIODARONE : 5 mg/kg IVP/IO, max. dose 300 mg	•
14. Analyze rhythm and defibrillate 8 J/kg or equivalent biphasic. Continue CPR 2 min.	•
15. If no response - AMIODARONE : 5 mg/kg IVP/IO, max. dose 150 mg	
16. Analyze rhythm and defibrillate 10 J/kg or equivalent biphasic. Continue CPR 2 min.	•
17. If return of spontaneous circulation see ROSC	•

NOTE:

- If rhythm appears to be Torsades de Points (polymorphic ventricular tachycardia), contact **OLMC** to consider [MAGNESIUM SULFATE](#) 25 mg/kg, max. 2 g, diluted in 100mL D5W IVPB, over 5 minutes.
- Utilize [High Performance CPR](#) for best patient outcome.
- The best airway is an effective airway with the least potential complications.

PEDIATRIC AIRWAY OBSTRUCTION

HPI	Signs & Symptoms	Differential Diagnosis
- Recent witnessed choking - Lack of teeth - Age	- Universal choking sign - Stridor - Inability to speak complete sentences - Increased respiratory rate/effort - Altered mental status	- Anaphylaxis - Aspiration/obstruction - Pulmonary embolism - Upper respiratory infection - Anxiety, hyperventilation

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•

UNRESPONSIVE:

3. Immediately begin CPR regardless of pulses.	•	•	•
4. Open the airway and check the mouth for foreign body	•	•	•
5. Assess for positive gag reflex. If present, utilize Pediatric Drug Assisted Airway Management protocol to proceed.			•
6. If FBO is not able to be removed, utilize direct visualization of the airway via video laryngoscopy and remove with Magill forceps.			•
7. If unable to remove and FBO is passed the vocal cords, intubate. Attempt to push the FBO into the right mainstem bronchus, then pull ETT back and ventilate left lung			•
8. If unable to remove and FBO is supraglottic (obstructing the view of the vocal cords), remove the videoscope and attempt to suction ≤ 10 seconds If suction does not remove FBO, utilize appropriate cricothyrotomy: < 10 years old: Needle Cricothyrotomy ≥ 10 years old: Surgical Cricothyrotomy			•

CONSCIOUS: CANNOT speak or cough

3.	0 - 1 year: Cycles of 5 back blows/ slaps, & 5 chest thrusts > 1 year: Perform repeated cycles of 5 back blows followed by 5 abdominal thrusts until the foreign body obstruction is relieved or the patient becomes unresponsive.	•	•	•
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CONSCIOUS: Able to speak or cough

3.	Allow patient to attempt to clear obstruction on their own.	•	•	•
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PEDIATRIC ASTHMA
RESPIRATORY DISTRESS WITH WHEEZING

HPI	Signs & Symptoms	Differential Diagnosis
- Time of onset - Possibility of foreign body - PMH and medications - Fever, infection	- Wheezing or stridor - Retractions - See-saw respirations - Diaphoresis - Tripod - AMS - Hypoxia	- Aspiration - Foreign body obstruction - Infection - Pneumonia, croup - Epiglottitis - Medication, toxin - Congenital heart disease

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing, circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•

SEVERE/IMPENDING FAILURE: Altered Mental Status, pale/diaphoretic, exhaustion

3. Consider ventilating with BVM on 100% OXYGEN with airway adjuncts, 1 breath every 2-3 sec.	•	•	•
4. DuoNeb: ALBUTEROL : 2.5mg / IPRATROPIUM BROMIDE : 0.5mg	•	•	
5. EPINEPHRINE 1 mg/mL: 0.01 mg/kg IM in anterolateral thigh, max. 0.3mg	•	•	
6. MAGNESIUM SULFATE : 25 mg/kg diluted in 100 mL D5W IVPB, over 15 min. , max. 2 g			•
7. If patient continues to deteriorate contact OLMC : EPINEPHRINE 1 mg/mL: 0.01 mg/kg IM in anterolateral thigh, max. 0.3mg	•	•	
8. DuoNeb: ALBUTEROL : 2.5mg / IPRATROPIUM BROMIDE : 0.5mg	•	•	
9. ALBUTEROL : 2.5 mg NEB, may repeat 5 min.	•	•	

MILD/MODERATE: GCS 15, MAP >65, skin warm/dry

3. Administer OXYGEN to maintain SPO2 94 - 99%.	•	•	•
4. DuoNeb: ALBUTEROL : 2.5mg / IPRATROPIUM BROMIDE : 0.5mg, may repeat once	•	•	
5. ALBUTEROL : 2.5 mg NEB, may repeat 5 min.	•	•	

PEARLS:

- 1. **MAGNESIUM SULFATE** may cause hypotension that will usually respond to a fluid bolus.

Use Calculator for dosing

PEDIATRIC CROUP

HPI	Signs & Symptoms	Differential Diagnosis
• Infant/toddler • Possibility of foreign body • Medical history • Fever, respiratory • Other sick siblings	• Low grade fever • Barking cough • Hoarseness • Stridor	• Asthma • Aspiration • Foreign body • Infection • Pneumonia

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Keep patient calm - DO NOT AGITATE	•	•	•
3. Provide emotional support and allow position of comfort	•	•	•

UNSTABLE: Resting stridor, respiratory distress

4. Ventilate with BVM on 100% OXYGEN with airway adjuncts, 1 breath every 2-3 sec.	•	•	•
5. Consider Pediatric Drug Assisted Advanced Airway			•
6. NEBULIZED EPINEPHRINE 1mg/mL: 0.5 mg/kg NEB, may repeat, max total dose 5 mg (5 mL)			•

STABLE: GCS 15, No cyanosis, good air exchange

4. Administer humidified OXYGEN by placing 6 mL NORMAL SALINE in nebulizer, deliver by mask or blow by	•	•	•
5. If no improvement, contact OLMC	•	•	•

PEARLS:

1. Foreign bodies can mimic croup, it is important to ask about a possible choking event.
2. Impending respiratory failure is indicated by change in mental status such as fatigue and listlessness, pallor, dusky appearance, decreased retractions, decreased breath sounds with decreasing stridor

PEDIATRIC DOSING CHART:

1:1 Epinephrine Normal Saline	3kg	4kg	5kg	6kg	7kg	8kg	9kg		≥ 10kg	
				dose	vol.	dose	vol.		dose	vol.
	Contact OLMC			3mg	3mL	4mg	4mL	5mg		5mL
					2mL		1mL	0mL		

PEDIATRIC AIRWAY MANAGEMENT
DRUG ASSISTED ADVANCED AIRWAY

Signs & Symptoms		Differential Diagnosis
<u>Respiratory Distress</u> - Increased work of breathing - Increased respiratory - Nasal flaring - Use of accessory muscles - Effectively compensating	<u>Respiratory Failure</u> - Exhausted energy reserves - Decreased respiratory rate - Decreased effort - Agitation or lethargy - Cyanosis - Fail oxygenate/ventilate	Anaphylaxis Asthma Airway obstruction OD/Poisoning - Pneumonia Croup, epiglottitis - Trauma

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing, circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•

UNSTABLE: Respiratory failure, failure to oxygenate/ventilate, expected clinical course

3. Ventilate with a BVM for 100% OXYGEN with airway adjuncts & padding under shoulders, 1 breath/2-3 sec., for ≥ 3 min.	•	•	•
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Indications for Drug Assisted Advanced Airway Placement:

- Cannot ventilate with BVM, potential impaired airway
- Increased WOB (retractions, nasal flaring, grunting) leading to failure
- Ventilatory failure (apnea, marked bradypnea or tachypnea, labored, hypoxia despite maximum oxygen delivery

4. Assess for positive gag reflex	•	•
5. If less invasive methods are ineffective or inappropriate: Call OLMC for orders to perform drug assisted advanced airway placement. KETAMINE : 1mg/kg IVP/IO, slowly over 1 min., max. 300mg	•	
6. After 1 min., reassess gag reflex for therapeutic effects of Ketamine. If inadequate sedation KETAMINE : 0.5 mg/kg IVP/IO, slowly over 1 min., max. total dose 450mg	•	
7. Place i-Gel. Verify placement with Capnography (continue to monitor throughout)	•	•
8. As needed, for post-intubation sedation, 15 min. following induction of sedation: KETAMINE : 0.5mg/kg IVP/IO, slowly over 1 min., NO REPEAT DOSING	•	

UNSTABLE: Respiratory failure due to **burns, inhalation injury, anaphylaxis**

3. Ventilate with a BVM for 100% OXYGEN with airway adjuncts & padding under shoulders, 1 breath/2-3 sec., for ≥ 3 min.	•	•	•
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Indications for Drug Assisted Advanced Intubation:

- Same as above and glottic or supraglottic swelling

4. Assess for positive gag reflex	•	•
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5. If less invasive methods are ineffective or inappropriate: Call OLMC for orders to perform drug assisted advanced airway placement. KETAMINE: 1mg/kg IVP/IO, slowly over 1 min., max. 300mg	•
6. After 1 min., reassess gag reflex for therapeutic effects of Ketamine. If inadequate sedation KETAMINE: 0.5 mg/kg IVP/IO, slowly over 1 min., max. total dose 450mg	•
7. Place endotracheal tube. Verify placement with Capnography (continue to monitor throughout)	•
8. As needed, for post-intubation sedation, 15 min. following induction of sedation: KETAMINE: 0.5mg/kg IVP/IO, slowly over 1 min., NO REPEAT DOSING	•

STABLE: Respiratory distress, effectively compensating

3. Administer supplemental OXYGEN	•	•	•
4. Support head in neutral position	•	•	•
5. Keep child calm, allow caregiver access to child	•	•	•

PEDIATRIC EPIGLOTTITIS

HPI	Signs & Symptoms	Differential Diagnosis
- Toddler - Possibility of foreign body - Medical history - Fever, respiratory - Other sick siblings	- High fever - Drooling - No cough - Stridor	- Asthma - Aspiration - Foreign body - Infection - Pneumonia

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Keep patient calm - DO NOT AGITATE	•	•	•
3. Provide emotional support and allow position of comfort	•	•	•

UNSTABLE: Resting stridor, respiratory distress

4. Administer humidified OXYGEN by placing 6 mL NORMAL SALINE in nebulizer, deliver by mask or blow by	•	•	•
5. NEBULIZED EPINEPHRINE 1mg/mL: 0.5 mg/kg NEB, may repeat, max total dose 5 mg (5 mL)			•
6. If patient continues to deteriorate, ventilate with BVM on 100% OXYGEN with airway adjuncts, 1 breath every 2-3 sec.	•	•	•
7. Consider Pediatric Drug Assisted Advanced Airway			•

PEARLS:

1. Impending respiratory failure is indicated by change in mental status such as fatigue and listlessness, pallor, dusky appearance, decreased retractions, decreased breath sounds with decreasing stridor

PEDIATRIC DOSING CHART:

1:1 Epinephrine Normal Saline	3kg 4kg 5kg	6kg 7kg	8kg 9kg		≥ 10kg	
		dose vol.	dose vol.		dose vol.	
	Contact OLMC	3mg	3mL	4mg	4mL	5mg 5mL
			2mL		1mL	0mL

PEDIATRIC ALTERED MENTAL STATUS
SYNCOPE/PRE-SYNCOPE

HPI	Signs & Symptoms	Differential Diagnosis
- PMH: cardiac, stroke, seizures, diabetes - GI bleed - Ruptured ectopic - Fluid loss: vomiting, diarrhea, dehydration - Recent trauma	- LOC with/out recovery - Lightheadedness, dizziness - Pulse irregularity, rate - Hypotension - Hallucination, delusion	- Vasovagal - Orthostatic hypotension - Dysrhythmia - Alcohol/Overdose Hypo/er-glycemia - Hypo/er-thermia - Shock Stroke/TIA Sepsis Behavioral emergency

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, MAP < 65 or Capillary refill >2 sec.

3. If respiratory depression or narcotic overdose suspected, see Depressant Overdose/Poisoning protocol	•	•	•
4. Obtain blood glucose level. If blood glucose is < 60 or > 250, see Diabetic Emergencies protocol	•	•	•

NOTE:

- 1. If age 2 years or less and signs/symptoms spontaneously resolve, refer to [BRUE/ALTE](#) protocol
- 2. Bring container of drug or substance providing that the transport of the item does not pose a safety risk.

PEDIATRIC ALLERGIC REACTION
ANAPHYLACTIC SHOCK

HPI	Signs & Symptoms	Differential Diagnosis
- Onset and location - Insect sting/bite - Food allergy/exposure - New clothing, soap, detergent - PMH of reactions - Medication allergy	- Itching, hives - Coughing, wheezing - Respiratory distress - Difficulty swallowing - Vomiting, diarrhea - Hypotension - Edema	- Urticaria (rash only) - Anaphylaxis - Angioedema - Aspiration/obstruction - Asthma/COPD Heart - Failure

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing, circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•

UNSTABLE/ ANAPHYLAXIS: Altered Mental Status, MAP<65, Capillary refill >2 sec.

3. Secure airway		•
4. EPINEPHRINE 1 mg/mL: 0.01mg/kg IM in anterolateral thigh, max single dose 0.3mg, may repeat 5 min., no max. or EpiPen <15kg contact OLMC 15-29kg: 0.15mg >30kg: 0.3mg	•	•
5. DIPHENHYDRAMINE : 1 mg/kg IVP/IO slowly over 2 min., max 50 mg		•
or : 1 mg/kg IM, max 50 mg	•	•
6. NORMAL SALINE : 20 mL/kg, titrate to MAP ≥ 65, max. 60mL/kg		•
7. DuoNeb : ALBUTEROL : 2.5 mg / IPRATROPIUM BROMIDE : 0.5 mg, may repeat x1	•	•
8. ALBUTEROL : 2.5 mg NEB, may repeat 5 min., no max. dosage	•	•
9. If condition continues to deteriorate and MAP <65 despite NORMAL SALINE contact OLMC : PUSH DOSE EPINEPHRINE : 1 mcg/kg IVP/IO, max. single dose 50 mcg, repeat 5 min., max. total 100mcg, titrate to MAP 65		•

STABLE: GCS 15, MAP >65, SBP >80, skin warm/dry
WITH AIRWAY INVOLVEMENT:

3. EPINEPHRINE 1 mg/mL: 0.01mg/kg IM in anterolateral thigh, max single dose 0.3mg, may repeat 5 min., no max. or EpiPen <15kg contact OLMC 15-29kg: 0.15mg >30kg: 0.3mg	•	•
4. DIPHENHYDRAMINE : 1 mg/kg IVP/IO over 2 min., max 50 mg		•
or : 1 mg/kg IM, max 50 mg	•	•
5. DuoNeb : ALBUTEROL : 2.5 mg / IPRATROPIUM BROMIDE : 0.5 mg, may repeat x1	•	•
6. ALBUTEROL : 2.5 mg NEB, may repeat 5 min., no max.	•	•

STABLE: GCS 15, MAP >65, SBP >80, skin warm/dry
WITHOUT AIRWAY INVOLVEMENT:

3.	Apply cold pack to site.	•	•	•
4.	DIPHENHYDRAMINE: 1 mg/kg IVP/IO over 2 min., max 25 mg			•
	or : 1 mg/kg IM, max 25 mg	•	•	

NOTE:

- 1. DuoNeb **ALBUTEROL/IPRATROPIUM BROMIDE & ALBUTEROL** NEB treatment may be administered in-line for those patients requiring CPAP or intubation.

PEARLS:

- 1. Patients with asthma, atopic dermatitis (eczema), prior anaphylactic history, and those who delay treatment can be at greater risk for a fatal reaction.
- 2. Gastrointestinal symptoms occur most commonly in food-induced anaphylaxis, and oral pruritus is often the first symptom observed.
- 3. New onset hoarseness or underpowered voice indicates vocal cord edema and should prompt concern for airway compromise

Use Calculator for dosing

BRIEF RESOLVED UNEXPLAINED EVENT
BRUE

Signs & Symptoms	Low Risk	High Risk
- Absent, decreased, or irregular breathing - Color change (Central cyanosis or pallor) - Marked change in muscle tone (hyper/-otonia) - AMS	- Age ≥ 60 days - Premature but ≥ 32 weeks - Postconceptional age ≥ 45 weeks 1 BRUE - Duration BRUE < 1 min.	- Age < 60 days - Premature < 32 weeks - Postconceptional age < 45 weeks >1 BRUE - Family PMH sudden cardiac death

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
2. Obtain ECG, monitor Capnography.		●	●
3. Transport for medical evaluation, even after resolved		●	●
4. Contact OLMC if transport is refused		●	●

PEARLS:

- Regardless of the patient’s well appearance, all infants with a history of signs or symptoms suggestive of BRUE should be transported for further evaluation
- The determination of a BRUE is made only after hospital evaluation, not in the field. A few of these infants will die even after hospital evaluation and treatment

PEDIATRIC DIABETIC EMERGENCIES

HPI	Signs & Symptoms	
• PMH diabetes • Onset of symptoms • Last meal	<u>Hypoglycemia</u>	<u>Hyperglycemia</u>
	• Altered mental status • Cool, clammy skin • Diaphoresis • Seizure • Decreased visual acuity, blindness	• Altered mental status • Dry, red, warm skin • Acetone (fruity) smell on breath • Kussmaul respirations • Dry mouth, intensive thirst • Nausea/vomiting and abdominal pain

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
2. Obtain ECG, monitor Capnography.		●	●
3 Obtain blood glucose level	●	●	●

HYPOGLYCEMIA: Altered Mental Status, BG < 60 or cool/clammy skin

4.	<u>ORAL GLUCOSE GEL</u> : 15 gm (If patient > 2 YO, is able to tolerate oral preparation, has gag reflex and able to protect own airway)	•	•	•
5.	<u>DEXTROSE 10%</u> : 0.5gm/kg IV/IO, may repeat, max. 50gm	•		
6.	or if no IV, <u>GLUCAGON</u> : < 20 kg: 0.5 mg IM/IN ≥ 20 kg: 1 mg IM/IN	•	•	

PEARLS:

1. Glucagon causes a breakdown of stored glycogen to glucose. Glucagon may not work if glycogen stores are previously depleted due to liver dysfunction, alcoholism, or malnutrition. Effects of Glucagon may take up to 30 minutes.

Use Calculator for dosing

PEDIATRIC NAUSEA/VOMITING

HPI	Signs & Symptoms	Differential Diagnosis
- Time of last meal - Last bowel movement - Last emesis - Relation to food - Menstrual cycle - Trauma	- Pain, character - Distention - Constipation - Diarrhea - Radiation	- Cardiac - Drugs (NSAID's, antibiotics, narcotics) - GI or renal disorders - Food or toxin induced - Pregnancy, GYN

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•
3. ONDANSETRON : ≥ 10kg and <40 0.1 mg/kg IVP/IO, over 30 sec., max. 4 mg ≥40 kg: 4 mg IVP/IO, over 30 sec.			•
or 4 mg ORAL, may repeat 10 min., max. 8 m		•	•

NOTE:

- 1. Do not administer to patients under 10kg
- 2. Do not administer to patients who are pregnant

Use Calculator for dosing

OPIATES / DEPRESSANTS

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
2. Obtain ECG, monitor Capnography.		●	●

3.	As age dictates, consider physical restraints before administering medication	•	•	•
4.	<u>NALOXONE</u> patients ≥ 5kg: : 0.1 mg/kg IN/IM, may repeat 3 min., max. 2 mg or : 0.1 mg IV/IO, may repeat 3 min., max. 2 mg		•	•
5.	If no response, contact OLMC to consider additional NALOXONE	•	•	•

1. Bring container of drug or substance providing that the transport of the item does not pose a safety risk.
2. If **NALOXONE** was administered prior to the arrival of EMS, obtain dose/route information for reporting and documentation.
3. Do not give patient anything to eat or drink by mouth.
4. Anticipate vomiting, respiratory arrest, seizure, dysrhythmias and refer to indicated protocols.
5. Do not induce vomiting, especially in cases where caustic substance ingestion is suspected.
6. Call Illinois Poison Center (800) 222-1222 for additional information.

1. Do not administer to patients under 5kg

Effective:5/25/2026
Region Reviewed/Revised: 1/13/2026

PEDIATRIC PAIN MANAGEMENT

HPI	Signs & Symptoms	Differential Diagnosis
• Age • Location, duration, severity • Allergies to medications	• Quality • Relation to movement • Relation to breathing • Increase with palpation	• Musculoskeletal • Visceral (abdominal) • Cardiac • Pleural, respiratory

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•
3. Assure SBP remains age appropriate	•	•	•
4. FENTANYL : 1 mcg/kg IVP/IN/IO/IM (100 mcg max single dose), may repeat 10 min., max total dose 200mcg			•
5. If respiratory depression occurs NALOXONE : ≥ 5kg: 0.1 mg/kg IN/IM, max. 2 mg		•	•
or 0.1 mg/kg IVP/IO, max. 2 mg			•
6. Consider prophylactic nausea/vomiting : ONDANSETRON : ≥ 10kg and <40 kg: 0.1 mg/kg IVP/IO, over 30 sec., max. 4 mg ≥40 kg: 4 mg IVP/IO, over 30 sec.			•
or 4 mg ORAL, may repeat 10 min., max. 8 mg		•	•

NOTE:

- 1. Do not administer NALOXONE to patients < 5kg
- 2. Do not administer ONDANSETRON to patients < 10kg

Use Calculator for dosing

PEDIATRIC SEIZURES

HPI	Signs & Symptoms	Differential Diagnosis
- Seizure activity - Previous seizure history - History of trauma - History of diabetes - History of pregnancy	- Decreased mental status - Sleepiness - Incontinence - Observed seizure activity - Evidence of trauma	- TBI, increased ICP - Diabetes Eclampsia - Poisoning/Overdose - Fever - Stroke

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•
3. Nausea/vomiting precautions. Protect patient from injury.		•	•

ACTIVE SEIZURE: Altered Mental Status

4. MIDAZOLAM : 0.2 mg/kg IM, max. 10 mg or 0.2 mg/kg IVP/IO/IN, max. single dose 2mg, may repeat 2 min., max. 10 mg			•
5. Obtain blood glucose level. If blood glucose is < 60 or > 250, see Diabetic Emergencies protocol	•	•	•
6. If seizure activity continues or recurs, contact OLMC to administer further doses of MIDAZOLAM .			•
7. Observe patient's sensorium and maintain airway	•	•	•

FEBRILE SEIZURE: ≥ 100.4F

8. Cool patient by removing clothing	•	•	•
9. Consider placing towels moistened in room temp. water over patient and fan	•	•	•
10 DO NOT induce shivering. DO NOT rub down with alcohol or place in ice-water bath	•	•	•

PEARLS:

- Status epilepticus is defined as two or more consecutive seizures without a period of consciousness or recovery. This is a true emergency requiring rapid airway control, treatment, and transport.
- Do NOT place anything in mouth if actively seizing.

CHECK VIAL CONCENTRATION PRIOR TO ADMINISTRATION

Use Calculator for dosing

PEDIATRIC SEPSIS

HPI	High Risk Condition	Differential Diagnosis
- Fever, chills, sweats - Recent antibiotic use - Suspected, documented infection	- Cancer - Asplenia - Sickle Cell - Bone marrow transplant - Solid organ transplant - Indwelling catheters - Immunosuppression - Immunodeficiency	- Hypovolemic shock - Cardiogenic shock

Pediatric Sepsis Inclusion Criteria

	< 28 days	1-12 mo.	1-2 yrs	2-4 yrs	5-12 yrs	13-15 yrs
Heart rate	> 205		> 190	> 140		> 100
Respiratory rate	> 60		> 40		> 34	> 25
Temperature	> 100.4 F					
Capillary refill	delayed > 3 sec., mottled					
SBP	< 60	< 70	< 70+2(age)			< 90
Mental status	Unresponsive, confused, inappropriate, lethargic					

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•
3. Evaluate for suspected or documented infection		•	•

UNSTABLE: Any 2 or more of the above listed criteria

4. If ≥ 2 criteria met, call OLMC for SEPSIS ALERT	•	•
5. NORMAL SALINE 20 mL/kg increments, may repeat if remain unstable, max. 60mL/kg, titrate to MAP ≥ 65		•
6. If condition continues to deteriorate and MAP <65 despite NORMAL SALINE contact OLMC : PUSH DOSE EPINEPHRINE : 1 mcg/kg IVP/IO, max. single dose 50 mcg, repeat 5 min., max. total 100mcg, target MAP 65		•

PEARLS:

1. Bedside report to hospital to include total amount of IVF infused
2. Monitor closely for signs of respiratory distress, rales or delayed capillary refill
3. Early, aggressive IV fluid administration is essential in the treatment of suspected sepsis or septic shock.

NOTE:

1. Volume sensitive children (neonates, congenital heart diseases, chronic lung disease, chronic renal failure) initial **NORMAL SALINE** of 10mL/kg (Max of 250mL)

PEDIATRIC SHOCK

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•
3. Establish IV/IO			•

Hypovolemic	Cardiogenic	Distributive
Hemorrhagic, Dehydration	Congenital heart disease, cardiac surgery, arrhythmia, post-ROSC	Sepsis, Anaphylaxis, Neurogenic shock
NORMAL SALINE 20 mL/kg NORMAL SALINE may repeat if remain unstable, max. 60mL/kg If condition continues to deteriorate and MAP <65 despite NORMAL SALINE contact OLMC: PUSH DOSE EPINEPHRINE : 1 mcg/kg IVP/IO, max. single dose 50 mcg, repeat 5 min., max. total 100mcg, titrate to MAP 65	NORMAL SALINE 10 mL/kg NORMAL SALINE 20 mL/hr If condition continues to deteriorate and MAP <65 despite NORMAL SALINE contact OLMC: Additional NORMAL SALINE and/or PUSH DOSE EPINEPHRINE : 1 mcg/kg IVP/IO, max. single dose 50 mcg, repeat 5 min., max. total 100mcg, titrate to MAP 65	NORMAL SALINE 20 mL/kg NORMAL SALINE may repeat if remain unstable, max. 60mL/kg If condition continues to deteriorate and MAP <65 despite NORMAL SALINE contact OLMC: PUSH DOSE EPINEPHRINE : 1 mcg/kg IVP/IO, max. single dose 50 mcg, repeat 5 min., max. total 100mcg, titrate to MAP 65

PEDIATRIC BURNS

HPI	Signs & Symptoms	Differential Diagnosis
- Type of exposure - Need for decontamination - Percent surface area - Inhalation injury - PMH - Other trauma	- Pain, swelling - Hypotension - Hoarseness, wheezing - Singed facial, nasal hair - Airway compromise, distress - Carbonaceous sputum	- Chemical - Thermal - Radiation - Electrical

	EMR	EMT	PM
1. Provide Pediatric Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Obtain ECG, monitor Capnography.		•	•
3. Consider Pediatric Airway Management	•	•	•
4. For Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg			•
5. Estimate total burn surface area using Rule of 9's	•	•	•
6. NORMAL SALINE for thermal/electrical burns, titrate to MAP ≥ 65 ≤ 5 YO: 125 mL/hr 6-13 YO: 250 mL/hr ≥ 14 YO: 500 mL/hr			•

THERMAL:

Superficial (1st degree)			
7. Cool burned area with water or NORMAL SALINE	•	•	•
8. If < 20% total burn surface area, apply sterile NORMAL SALINE soaked dressings	•	•	•
Partial (2nd) or Full (3rd) Thickness			
7. Apply dry sterile dressing	•	•	•

ELECTRICAL or LIGHTNING:

7. Treat dysrhythmias per protocol			•
8. Identify and document entry/exit wounds	•	•	•
9. Assess neurovascular status (PMS) of affected area	•	•	•
10. Immobilize affected area	•	•	•
11. Apply dry sterile dressing	•	•	•

CHEMICAL:

7. Refer to department HazMat protocol for decontamination	•	•	•
8. If powder: remove affected clothing, brush away excess	•	•	•
9. Flush burn area with sterile water or NORMAL SALINE	•	•	•
10. If eye involvement: Assist patient in removing contact lens Flush burned eye with sterile water or NORMAL SALINE continuously	•	•	•

PEARLS:

1. Onset of stridor and change in voice are impending signs of potentially significant airway burns, which may rapidly lead to airway obstruction or respiratory failure.
-

Use Calculator for dosing

SUSPECTED CHILD ABUSE / NEGLECT

HPI	Signs & Symptoms	Considerations
- Time of injury - Mechanism - LOC - Bleeding - PMH - Timeline of incidents	- Dehydration - Fractures - Recurring injuries - Varying age of healing - Decubitus	- Emotional trauma - Behavioral emergency - Traumatic injury

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Note environment, child’s interaction with parents, discrepancies in the history obtained from child and caregivers, and any signs of obvious injury	•	•	•
3. Treat obvious injuries	•	•	•
4. If parent/guardian refuses to let you transport the child, remain at the scene, contact police and request the child be placed in protective custody	•	•	•
5. Transport	•	•	•
6. EMS are mandated reporters: report your suspicions to the Emergency Department Physician and/or Nurse	•	•	•
7. Carefully document history and physical exam findings as well as environmental/circumstantial data on the EHR	•	•	•
8. Department of Children and Family Services must be notified at (800)-25-ABUSE (24-hour phone line) When contacting DCFS, identify self as a State Mandated Reporter to expedite the process Written confirmation of the verbal report must be filed with DCFS within 48 hours	•	•	•

NOTE:

1. Reporting information may be found:
<https://www2.illinois.gov/dcfs/safekids/reporting/Pages/index.aspx>

PEDIATRIC DROWNING

HPI	Signs & Symptoms	Differential Diagnosis
- Submersion in water - Possibility of trauma - Duration of submersion - Temperature of water - Depth achieved - Type of water	- AMS, unresponsive - Vomiting - Coughing	- Pre-existing medical - Trauma

	EMR	EMT	PM
1. Provide Pediatric Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability.	●	●	●
2. Obtain ECG, monitor Capnography.		●	●
3. Spinal Motion Restriction as indicated	●	●	●

UNSTABLE: Altered Mental Status, abnormal respirations

4. Evaluate for gag reflex	●	●	●
5. Ventilate with BVM on 100% OXYGEN with airway adjuncts, 1 breath every 2-3 sec.	●	●	●
6. Refer to Pediatric Airway Management	●	●	●
7. Consider Hypothermia	●	●	●
8. If normothermic, treat dysrhythmias per protocol			●

STABLE: GCS 15, MAP >65, normal respirations

4. Consider Hypothermia	●	●	●
5. Contact OLMC to consider CPAP		●	●

NOTE:

- Contact **OLMC** to consider CPAP in awake patients in respiratory distress.
- If no pulse present, follow appropriate [cardiac protocol](#).

PEDIATRIC HEAD SPINAL FACIAL INJURIES

HPI	Signs & Symptoms	Differential Diagnosis
- Time of injury - MOI - blunt vs. penetrating - Loss of consciousness - Evidence of multi-system trauma	- AMS - Pain, swelling, bleeding - Cushing's Triad - Unequal, sluggish pupils - Respiratory distress or failure - Vomiting	- Stroke - Diabetic Emergency - Hypovolemic Shock

	EMR	EMT	PM
1. Provide Pediatric Routine Trauma Care - treat life threatening problems with airway, breathing circulation and disability	•	•	•
2. Utilize Spinal Motion Restriction as indicated	•	•	•
3. Obtain ECG, monitor Capnography.		•	•

UNSTABLE: Altered Mental Status, PGCS ≤12, MAP < 65 or Capillary refill >2 sec.

4. Perform Rapid Trauma Assessment and manage airway as indicated	•	•	•
5. Control hemorrhage	•	•	•
6. Obtain blood glucose level. If blood glucose is < 60, see Pediatric Diabetic Emergencies protocol	•	•	•
7. NORMAL SALINE 20 mL/kg increments, may repeat if remain unstable, max. 60mL/kg, titrate to MAP ≥ 65			•
8. If signs of increased ICP (see PEARL 1), ventilate to Capnography of 35-40 mmHg		•	•
9. If condition continues to deteriorate and failure to maintain airway/ventilate/oxygenate, consider Pediatric Airway Management			•
10. For seizures MIDAZOLAM : 0.2 mg/kg mg IM, max. 10 mg or 0.2 mg/kg IVP/IO/IN, max. single dose 2mg, may repeat 2 min., max. 10 mg			•

STABLE: GCS 15, MAP >65, skin warm/dry

4. Perform Focused Exam		•	•
5. Obtain blood glucose level. If blood glucose is < 60, see Pediatric Diabetic Emergencies protocol	•	•	•

PEARLS:

1. Rapid neurologic deterioration (unequal pupils, decerebrate posturing, lateralizing signs) often seen secondary to increased ICP should be treated with relative hyperventilation based on Capnography. Ventilate patient 1 breath/2-3 seconds to Capnography of 35-40 mmHg.
2. A nasal airway should not be used with significant facial injury or possible basilar skull fracture

Use Calculator for dosing

PGCS 14-15	PGCS 9-13	PGCS ≤ 8
- Administer 100% O2 as indicated - Control hemorrhage - Reassess PGCS - Observe	- Administer 100% O2 as indicated - Support ventilations as indicated - Control hemorrhage - Reassess PGCS - Observe	- Administer 100% O2 as indicated - Support ventilations as indicated - Utilize Pediatric Airway Management as indicated - Control hemorrhage

PEDIATRIC HYPERTHERMIA
HEAT EMERGENCIES

HPI	Signs & Symptoms	Differential Diagnosis
- Exposure time to heat - Extremities of age - PMH - Poor PO intake	- AMS - Hot, dry, sweaty skin - Pale, clammy skin - Fatigue, dizziness - Nausea/vomiting - Frontal headache - Seizures	- Heat cramps - Heat exhaustion - Heat stroke - Dehydration - Cocaine, amphetamines - Fever

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing circulation and disability.	•	•	•
2. Move patient to cool environment, remove clothing to facilitate cooling	•	•	•
3. Obtain ECG, monitor Capnography.		•	•

STROKE: Altered Mental Status, flushing, hot skin (dry or moist), temperature ≥104F°

4. Initiate active rapid cooling: Douse towels or sheets with cool water, place on patient, and fan body Apply cold packs to neck, axilla, and groin	•	•	•
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EXHAUSTION: AMS, sweating, weakness, nausea/vomiting, syncope

4. NORMAL SALINE 20 mL/kg increments, may repeat if remain unstable, max. 60mL/kg, titrate to improved patient condition	•
--	---

NOTE:

- Stop active cooling if patient begins to shiver.
- Heat cramps are indicated by normal level of consciousness and muscle cramps or spasms. There is no treatment for these patients.
- If patient seizes, refer to [Pediatric Seizure Protocol](#)

PEARLS:

- Pediatrics are more sensitive to extreme temperatures.
- Sweating generally stops as core temperature rises above 104F° which is indicative of heat stroke

PEDIATRIC HYPOTHERMIA
COLD EMERGENCIES

HPI	Signs & Symptoms	Differential Diagnosis
- Exposure to environment - Extremes of age - Length of exposure - Water involvement - Alcohol, barbiturates use	- Cold, clammy skin - Shivering - AMS - Extremity pain - Sensory abnormality - Bradycardia - Hypotension	- Environmental exposure - Sepsis - Hypoglycemia

	EMR	EMT	PM
1. Provide Pediatric Routine Medical Care - treat life threatening problems with airway, breathing, circulation and disability.	•	•	•
2. Move patient to warm environment. Remove wet clothing.	•	•	•
3. Obtain ECG, monitor Capnography.		•	•

SYSTEMIC HYPOTHERMIA: Whole body involvement

4. Apply heat packs to neck, axilla, groin and thorax	•	•	•
5. Assess for pulse. If no pulse present follow appropriate cardiac protocol a. <u>extremities flex</u>: extend time between medication, repeat defibrillation as core temp. rises b. <u>extremities DO NOT flex</u>: withhold medications, only defibrillate x1			•
			•

FROSTBITE: Isolated body areas of involvement

4. Rapidly re-warm frozen areas with warm water or hot packs wrapped in a towel	•	•	•
5. Apply dry sterile dressing (treat like a burn)	•	•	•
6. Immobilize and elevated frozen areas	•	•	•
7. For Pediatric Pain Management : FENTANYL : 1 mcg/kg IVP/IN/IO/IM, max single dose 100 mcg, repeat every 10 min., max total dose 200 mcg			•

NOTE:

1. [Termination of Resuscitative Efforts](#) does not apply to these patients.

PEARLS:

1. In patients suffering from moderate to severe hypothermia, it is critical to not allow these patients to stand or exercise as this may cause circulatory collapse
2. Shivering typically ceases when core temperature is below 32°C / 90°F.

PEDIATRIC REFERENCE CHARTS

VITALS, EQUIPMENT SIZES, MEDICATIONS

	EMR	EMT	PM
Determine weight (utilize length-based tape) and age	•	•	•

Age	Weight (kg)
<u>Very Pre-Term (28-32 weeks gestation)</u>	1 kg
<u>Moderate Pre-Term (33-36 weeks gestation)</u>	2 kg
<u>Neonate - 2 months</u>	3 kg
	4 kg
	5 kg
<u>3 - 5 months</u>	6 kg
	7 kg
<u>6 - 13 months</u>	8 kg
	9 kg
<u>14 - 18 months</u>	10 kg
	11 kg
<u>18 months - 3 years</u>	12 kg
	13 kg
	14 kg
<u>3.5 - 5 years</u>	16 kg
	18 kg
<u>5.5 - 6.5 years</u>	20 kg
	22 kg
<u>7 - 9 years</u>	24 kg
	26 kg
	28 kg
<u>9.5 - 11 years</u>	30 kg
	32 kg
	34 kg
	36 kg
	38 kg
<u>11.5 - 16 years</u>	40 kg
	42 kg
	44 kg

Very Pre-Term - Moderate Pre-term Neonate

If less then 5kg, contact **OLMC** for medication guidance

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
100-180	30-60	> 60	45-50

Treatment	Dosage	Supplied	Very Pre-term		Moderate Pre-term	
			28-32 weeks		33-36 weeks	
			1 kg		2 kg	
			dose	vol.	dose	vol.
Defibrillation	2 J/kg		2 J		4 J	
	4 J/kg		4 J		8 J	
	6 J/kg		6 J		12 J	
	8 J/kg		8 J		16 J	
	10 J/kg		10 J		20 J	
Cardioversion	1 J/kg		1 J		2 J	
	2 J/kg		2 J		4 J	

Cardiac Meds.						
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.01 mg	0.1 mL	0.02 mg	0.2 mL
Adenosine	Not approved for this weight					
Amiodarone VT	Not approved for this weight					
VT/VF	5 mg/mL	50 mg/mL	5 mg	0.1 mL	10 mg	0.2 mL
Atropine	Not approved for this weight					
Normal Saline (C)	10 mL/kg			10 mL		20 mL

Airway Meds.						
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine	Contact OLMC					
Etomidate	Not approved for this weight					
Fentanyl	1 mcg/kg	50 mcg/mL	1 mcg	0.02 mL	2 mcg	0.04 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (DAI)	Not approved for this weight					
Mag. Sulfate	25 mg/kg	500 mg/mL	25 mg	0.05 mL	50 mg	0.1 mL
Midazolam	0.1 mg/kg	5 mg/mL	0.1 mg	0.02 mL	0.2 mg	0.04 mL
		1 mg/mL		0.1 mL		0.2 mL

Diphenhydramine	Not approved for this weight					
Dextrose 10%	0.5 g/kg	0.1 g/mL	0.2 g	2 mL	0.4 g	4 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.01 mg	0.01 mL	0.02 mg	0.02 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	1 mcg	0.1 mL	2 mcg	0.2 mL
Fentanyl	1 mcg/kg	50 mcg/mL	1 mcg	0.02 mL	2 mcg	0.04 mL
Glucagon	age based	1 mg/mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL
Ketamine (behave)	Not approved for this weight					
Lidocaine	Not approved for this weight					
Midazolam (IM)	0.2 mg/kg	5 mg/mL	0.2 mg	0.04 mL	0.4 mg	0.08 mL
		1 mg/mL		0.2 mL		0.4 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	0.2 mg	0.04 mL	0.4 mg	0.08 mL
		1 mg/mL		0.2 mL		0.4 mL
Naloxone	Not approved for this weight					
Ondansetron	Not approved for this weight					
Normal Saline	20 mL/kg			20 mL		40 mL

Neonate - 2 Months

If less then 5kg, contact **OLMC** for medication guidance

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
100-180	30-60	> 60	45-50

Equipment:		3 kg	4 kg	5 kg
iGel				Blue size 1.5
Suction for iGel				FR 12
Soft suction		FR 8	FR 8	FR 8
ETT - uncuffed		3.5	3.5	3.5
ETT - cuffed		3.0	3.0	3.0
depth		9-10.5 cm	9-10.5 cm	9-10.5 cm

Treatment	Dosage	Supplied	3 kg		4 kg		5 kg	
			dose	vol.	dose	vol.	dose	vol.
Defibrillation	2 J/kg		6 J		8 J		10 J	
	4 J/kg		12 J		16 J		20 J	
	6 J/kg		18 J		24 J		30 J	
	8 J/kg		24 J		32 J		40 J	
	10 J/kg		30 J		40 J		50 J	
Cardioversion	1 J/kg		3 J		4 J		5 J	
	2 J/kg		6 J		8 J		10 J	

Cardiac Meds.								
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.03 mg	0.3 mL	0.04 mg	0.4 mL	0.05 mg	0.5 mL
Adenosine	0.1 mg/kg	3 mg/mL	0.3 mg	0.1 mL	0.4 mg	0.13 mL	0.5 mg	0.17 mL
	0.2 mg/kg		0.6 mg	0.2 mL	0.8 mg	0.27 mL	1 mg	0.33 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	15 mg	0.3 mL	20 mg	0.4 mL	25 mg	0.5 mL
			15 mg	0.3 mL	20 mg	0.4 mL	25 mg	0.5 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.1 mg	1 mL	0.1 mg	1 mL	0.1 mg	1 mL
Normal Saline (C)	10 mL/kg			30 mL		40 mL		50 mL

Airway Meds.								
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine	Contact OLMC							
Etomidate	0.3 mg/kg	2 mg/mL	Not approved for this weight				1.5 mg	0.75 mL
Fentanyl	1 mcg/kg	50 mcg/mL	3 mcg	0.06 mL	4 mcg	0.08 mL	5 mcg	0.1 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (DAI)	1 mg/kg	50 mg/mL	Not approved for this weight				5 mg	0.1 mL
	0.5 mg/kg						2.5 mg	0.05 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	75 mg	0.15 mL	100 mg	0.2 mL	125 mg	0.25 mL
Midazolam	0.1 mg/kg	5 mg/mL	0.3 mg	0.06 mL	0.4 mg	0.08 mL	0.5 mg	0.1 mL
		1 mg/mL		0.3 mL		0.4 mL		0.5 mL

Diphenhydramine	1 mg/kg	50 mg/mL	3 mg	0.06 mL	4 mg	0.08 mL	5 mg	0.1 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	0.6 g	6 mL	0.8 g	8 mL	1 g	10 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.03 mg	0.03 mL	0.04 mg	0.04 mL	0.05 mg	0.05 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	3 mcg	0.3 mL	4 mcg	0.4 mL	5 mcg	0.5 mL
Fentanyl	1 mcg/kg	50 mcg/mL	3 mcg	0.06 mL	4 mcg	0.08 mL	5 mcg	0.1 mL
Glucagon	age based	1 mg/mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL
Ketamine (behave)	Not approved for this weight							
Lidocaine	0.5 mg/kg	20 mg/mL	Not approved		2 mg	0.1 mL	2.5 mg	0.13 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	0.6 mg	0.12 mL	0.8 mg	0.16 mL	1 mg	0.2 mL
		1 mg/mL		0.6 mL		0.8 mL		1 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	0.6 mg	0.12 mL	0.8 mg	0.16 mL	1.0 mg	8.33 mL
		1 mg/mL		0.6 mL		0.8 mL		1.67 mL
Naloxone	Not approved for this weight							
Ondansetron	Not approved for this weight							
Normal Saline	20 mL/kg			60 mL		80 mL		100 mL

3 - 5 Months

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
100-160	30-60	> 70	50-55

Equipment:		6 kg	7 kg
iGel		Blue size 1.5	Blue size 1.5
Suction for iGel		FR 12	FR 12
Soft suction		FR 8	FR 8
ETT - uncuffed		3.5	3.5
ETT - cuffed		3.0	3.0
depth		10-10.5 cm	10-10.5 cm

Treatment	Dosage	Supplied	6 kg		7 kg	
			dose	vol.	dose	vol.
Defibrillation	2 J/kg		12 J		14 J	
	4 J/kg		24 J		28 J	
	6 J/kg		36 J		42 J	
	8 J/kg		48 J		56 J	
	10 J/kg		60 J		70 J	
Cardioversion	1 J/kg		6 J		7 J	
	2 J/kg		12 J		14 J	

Cardiac Meds.						
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.06 mg	0.6 mL	0.07 mg	0.7 mL
Adenosine	0.1 mg/kg	3 mg/mL	0.6 mg	0.2 mL	0.7 mg	0.23 mL
	0.2 mg/kg		1.2 mg	0.4 mL	1.4 mg	0.47 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	30 mg	0.6 mL	35 mg	0.7 mL
			30 mg	0.6 mL	35 mg	0.7 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.12 mg	1.2 mL	0.14 mg	1.4 mL
Normal Saline (C)	10 mL/kg			60 mL		70 mL

Airway Meds.						
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine Normal Saline	0.5 mg/kg	1 mg/mL	3 mg	3 mL	3 mg	3 mL
				2 mL		2 mL
Etomidate	0.3 mg/kg	2 mg/mL	1.8 mg	0.9 mL	2.1 mg	1.05 mL
Fentanyl	1 mcg/kg	50 mcg/mL	6 mcg	0.12 mL	7 mcg	0.14 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (SAI)	1 mg/kg	50 mg/mL	6 mg	0.12 mL	7 mg	0.14 mL
	0.5 mg/kg		3 mg	0.06 mL	3.5 mg	0.07 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	150 mg	0.3 mL	175 mg	0.35 mL
Midazolam	0.1 mg/kg	5 mg/mL	0.6 mg	0.12 mL	0.7 mg	0.14 mL
		1 mg/mL		0.6 mL		0.7 mL

Diphenhydramine	1 mg/kg	50 mg/mL	6 mg	0.12 mL	7 mg	0.14 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	3 g	30 mL	3.5 g	35 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.06 mg	0.06 mL	0.07 mg	0.07 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	6 mcg	0.6 mL	7 mcg	0.7 mL
Fentanyl	1 mcg/kg	50 mcg/mL	6 mcg	0.12 mL	7 mcg	0.14 mL
Glucagon	age based	1 mg/mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL
Ketamine (behave)	Not approved for this weight					
Lidocaine	0.5 mg/kg	20 mg/mL	3 mg	0.15 mL	3.5 mg	0.18 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	1.2 mg	0.24 mL	1.4 mg	0.28 mL
		1 mg/mL		1.2 mL		1.4 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	1.2 mg	0.24 mL	1.4 mg	0.28 mL
		1 mg/mL		1.2 mL		1.4 mL
Naloxone	0.1 mg/kg	1 mg/mL	0.6 mg	0.6 mL	0.7 mg	0.7 mL
Ondansetron	Not approved for this weight					
Normal Saline	20 mL/kg			120 mL		140 mL

6 - 13 Months

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
100-160	30-60	> 70	65

Equipment:		8 kg	9 kg
iGel		Blue size 1.5	Blue size 1.5
Suction for iGel		FR 12	FR 12
Soft suction		FR 8	FR 8
ETT - uncuffed		3.5	3.5
ETT - cuffed		3.0	3.0
depth		10.5-11 cm	10.5-11 cm

Treatment	Dosage	Supplied	8 kg		9 kg	
			dose	vol.	dose	vol.
Defibrillation	2 J/kg		16 J		18 J	
	4 J/kg		32 J		36 J	
	6 J/kg		48 J		54 J	
	8 J/kg		64 J		72 J	
	10 J/kg		80 J		90 J	
Cardioversion	1 J/kg		8 J		9 J	
	2 J/kg		16 J		18 J	

Cardiac Meds.						
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.08 mg	0.8 mL	0.09 mg	0.9 mL
Adenosine	0.1 mg/kg	3 mg/mL	0.8 mg	0.27 mL	0.9 mg	0.30 mL
	0.2 mg/kg		1.6 mg	0.53 mL	1.8 mg	0.60 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	40 mg	0.8 mL	45 mg	0.9 mL
			40 mg	0.8 mL	45 mg	0.9 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.16 mg	1.6 mL	0.18 mg	1.8 mL
Normal Saline (C)	10 mL/kg			80 mL		90 mL

Airway Meds.						
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine	0.5 mg/kg	1 mg/mL	4 mg	4 mL	4 mg	4 mL
Normal Saline				1 mL		1 mL
Etomidate	0.3 mg/kg	2 mg/mL	2.4 mg	1.2 mL	2.7 mg	1.35 mL
Fentanyl	1 mcg/kg	50 mcg/mL	8 mcg	0.16 mL	9 mcg	0.18 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (DAI)	1 mg/kg	50 mg/mL	8 mg	0.16 mL	9 mg	0.18 mL
	0.5 mg/kg		4 mg	0.08 mL	4.5 mg	0.09 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	200 mg	0.4 mL	225 mg	0.45 mL
Midazolam	0.1 mg/kg	5 mg/mL	0.8 mg	0.16 mL	0.9 mg	0.18 mL
		1 mg/mL		0.8 mL		0.9 mL

Diphenhydramine	1 mg/kg	50 mg/mL	8 mg	0.16 mL	9 mg	0.18 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	4 g	40 mL	4.5 g	45 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.08 mg	0.08 mL	0.09 mg	0.09 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	8 mcg	0.8 mL	9 mcg	0.9 mL
Fentanyl	1 mcg/kg	50 mcg/mL	8 mcg	0.16 mL	9 mcg	0.18 mL
Glucagon	age based	1 mg/mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL
Ketamine (behave)	Not approved for this weight					
Lidocaine	0.5 mg/kg	20 mg/mL	4 mg	0.2 mL	4.5 mg	0.23 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	1.6 mg	0.32 mL	1.8 mg	0.36 mL
		1 mg/mL		1.6 mL		1.8 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	1.6 mg	0.32 mL	1.8 mg	0.36 mL
		1 mg/mL		1.6 mL		1.8 mL
Naloxone	0.1 mg/kg	1 mg/mL	0.8 mg	0.8 mL	0.9 mg	0.9 mL
Ondansetron	Not approved for this weight					
Normal Saline	20 mL/kg			160 mL		180 mL

14 - 18 Months

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
100-160	30-60	> 70	> 65

Equipment:		10 kg	11 kg
iGel		Blue size 1.5	Blue size 1.5
Suction for iGel		FR 12	FR 12
Soft suction		FR 8	FR 8
ETT - uncuffed		4.0	4.0
ETT - cuffed		3.5	3.5
depth		11-12 cm	11-12 cm

Treatment	Dosage	Supplied	10 kg		11 kg	
			dose	vol.	dose	vol.
Defibrillation	2 J/kg		20 J		22 J	
	4 J/kg		40 J		44 J	
	6 J/kg		60 J		66 J	
	8 J/kg		80 J		88 J	
	10 J/kg		100 J		110 J	
Cardioversion	1 J/kg		10 J		11 J	
	2 J/kg		20 J		22 J	

Cardiac Meds.						
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.1 mg	1 mL	0.11 mg	1.1 mL
Adenosine	0.1 mg/kg	3 mg/mL	1 mg	0.33 mL	1.1 mg	0.37 mL
	0.2 mg/kg		2 mg	0.67 mL	2.2 mg	0.73 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	50 mg	1 mL	55 mg	1.1 mL
			50 mg	1 mL	55 mg	1.1 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.2 mg	2 mL	0.22 mg	2.2 mL
Normal Saline (C)	10 mL/kg			100 mL		110 mL

Airway Meds.						
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine Normal Saline	0.5 mg/kg	1 mg/mL	5 mg	5 mL	5 mg	5 mL
				0 mL		0 mL
Etomidate	0.3 mg/kg	2 mg/mL	3 mg	1.5 mL	3.3 mg	1.65 mL
Fentanyl	1 mcg/kg	50 mcg/mL	10 mcg	0.2 mL	11 mcg	0.22 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (DAI)	1 mg/kg	50 mg/mL	10 mg	0.2 mL	11 mg	0.22 mL
	0.5 mg/kg		5 mg	0.1 mL	5.5 mg	0.11 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	250 mg	0.5 mL	275 mg	0.55 mL
Midazolam	0.1 mg/kg	5 mg/mL	1 mg	0.2 mL	1.1 mg	0.22 mL
		1 mg/mL		1 mL		1.1 mL

Diphenhydramine	1 mg/kg	50 mg/mL	10 mg	0.2 mL	11 mg	0.22 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	5 g	50 mL	5.5 g	55 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.1 mg	0.1 mL	0.11 mg	0.11 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	10 mcg	1 mL	11 mcg	1.1 mL
Fentanyl	1 mcg/kg	50 mcg/mL	10 mcg	0.2 mL	11 mcg	0.22 mL
Glucagon	age based	1 mg/mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL
Ketamine (behave)	Not approved for this weight					
Lidocaine	0.5 mg/kg	20 mg/mL	5 mg	0.25 mL	5.5 mg	0.28 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2.2 mg	0.44 mL
		1 mg/mL		2 mL		2.2 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL
Naloxone	0.1 mg/kg	1 mg/mL	1 mg	1 mL	1.1 mg	1.1 mL
Ondansetron	0.1 mg/kg	2 mg/mL	1 mg	0.5 mL	1.1 mg	0.55 mL
Normal Saline	20 mL/kg			200 mL		220 mL

18 months - 3 years

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
90-150	24-40	> 70	> 65

Equipment:		12 kg	13 kg	14 kg
iGel		Blue size 1.5	Gray size 2.0	Gray size 2.0
Suction for iGel		FR 12	FR 12	FR 12
Soft suction		FR 8	FR 8	FR 8
ETT - uncuffed		4.5	4.5	4.5
ETT - cuffed		4.0	4.0	4.0
depth		12.5-13.5 cm	12.5-13.5 cm	12.5-13.5 cm

Treatment	Dosage	Supplied	12 kg		13 kg		14 kg	
			dose	vol.	dose	vol.	dose	vol.
Defibrillation	2 J/kg		24 J		26 J		28 J	
	4 J/kg		48 J		52 J		56 J	
	6 J/kg		72 J		78 J		84 J	
	8 J/kg		96 J		104 J		112 J	
	10 J/kg		120 J		130 J		140 J	
Cardioversion	1 J/kg		12 J		13 J		14 J	
	2 J/kg		24 J		26 J		28 J	

Cardiac Meds.								
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.12 mg	1.2 mL	0.13 mg	1.3 mL	1.68 mg	1.4 mL
Adenosine	0.1 mg/kg	3 mg/mL	1.2 mg	0.4 mL	1.3 mg	0.43 mL	1.4 mg	0.47 mL
	0.2 mg/kg		2.4 mg	0.8 mL	2.6 mg	0.87 mL	2.8 mg	0.93 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	60 mg	1.2 mL	65 mg	1.3 mL	70 mg	1.4 mL
			60 mg	1.2 mL	65 mg	1.3 mL	70 mg	1.4 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.24 mg	2.4 mL	0.26 mg	2.6 mL	0.3 mg	2.8 mL
Normal Saline (C)	10 mL/kg			120 mL		130 mL		140 mL

Airway Meds.								
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine	0.5 mg/kg		5 mg		5 mg		5 mg	
Normal Saline			0 mL			0 mL		0 mL
Etomidate	0.3 mg/kg	2 mg/mL	3.6 mg	1.8 mL	3.9 mg	1.95 mL	4.2 mg	2.1 mL
Fentanyl	1 mcg/kg	50 mcg/mL	12 mcg	0.24 mL	13 mcg	0.26 mL	14 mcg	0.28 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (DAI)	1 mg/kg	50 mg/mL	12 mg	0.24 mL	13 mg	0.26 mL	14 mg	0.28 mL
	0.5 mg/kg		6 mg	0.12 mL	6.5 mg	0.13 mL	7 mg	0.14 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	300 mg	0.6 mL	325 mg	0.65 mL	350 mg	0.7 mL
Midazolam	0.1 mg/kg	5 mg/mL	1.2 mg	0.24 mL	1.3 mg	0.26 mL	1.4 mg	0.28 mL
		1 mg/mL		1.2 mL		1.3 mL		1.4 mL

Diphenhydramine	1 mg/kg	50 mg/mL	12 mg	0.24 mL	13 mg	0.26 mL	14 mg	0.28 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	6 g	60 mL	6.5 g	65 mL	7 g	70 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.12 mg	0.12 mL	0.13 mg	0.13 mL	0.14 mg	0.14 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	12 mcg	1.2 mL	13 mcg	1.3 mL	14 mcg	1.4 mL
Fentanyl	1 mcg/kg	50 mcg/mL	12 mcg	0.24 mL	13 mcg	0.26 mL	14 mcg	0.28 mL
Glucagon	age based	1 mg/mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL
Ketamine (behave)	Not approved for this weight							
Lidocaine	0.5 mg/kg	20 mg/mL	6 mg	0.3 mL	6.5 mg	0.33 mL	7 mg	0.35 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	2.4 mg	0.48 mL	2.6 mg	0.52 mL	2.8 mg	0.56 mL
		1 mg/mL		2.4 mL		2.6 mL		2.8 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL		2 mL
Naloxone	0.1 mg/kg	1 mg/mL	1.2 mg	1.2 mL	1.3 mg	1.3 mL	1.4 mg	1.4 mL
Ondansetron	0.1 mg/kg	2 mg/mL	1.2 mg	0.6 mL	1.3 mg	0.65 mL	1.4 mg	2.3 mL
Normal Saline	20 mL/kg			240 mL		260 mL		280 mL

3.5 - 5 years

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
90-150	22-34	> 75	65

Equipment:		16 kg	18 kg
iGel		Gray size 2.0	Gray size 2.0
Suction for iGel		FR 12	FR 12
Soft suction		FR 10	FR 10
ETT - uncuffed		5.0	5.0
ETT - cuffed		4.5	4.5
depth		14-15 cm	14-15 cm

Treatment	Dosage	Supplied	16 kg		18 kg	
			dose	vol.	dose	vol.
Defibrillation	2 J/kg		32 J		36 J	
	4 J/kg		64 J		72 J	
	6 J/kg		96 J		108 J	
	8 J/kg		128 J		144 J	
	10 J/kg		160 J		180 J	
Cardioversion	1 J/kg		16 J		18 J	
	2 J/kg		32 J		36 J	

Cardiac Meds.						
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.16 mg	1.6 mL	0.18 mg	1.8 mL
Adenosine	0.1 mg/kg	3 mg/mL	1.6 mg	0.53 mL	1.8 mg	0.60 mL
	0.2 mg/kg		3.2 mg	1.07 mL	3.6 mg	1.20 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	80 mg	1.6 mL	90 mg	1.8 mL
			80 mg	1.6 mL	90 mg	1.8 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.32 mg	3.2 mL	0.36 mg	3.6 mL
Normal Saline (C)	10 mL/kg			160 mL		180 mL

Airway Meds.						
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine	0.5 mg/kg	1 mg/mL	5 mg	5 mL	5 mg	5 mL
Normal Saline				0 mL		0 mL
Etomidate	0.3 mg/kg	2 mg/mL	4.8 mg	2.4 mL	5.4 mg	2.7 mL
Fentanyl	1 mcg/kg	50 mcg/mL	16 mcg	0.32 mL	18 mcg	0.36 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (DAI)	1 mg/kg	50 mg/mL	16 mg	0.32 mL	18 mg	0.36 mL
	0.5 mg/kg		8 mg	0.16 mL	9 mg	0.18 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	400 mg	0.8 mL	450 mg	0.9 mL
Midazolam	0.1 mg/kg	5 mg/mL	1.6 mg	0.32 mL	1.8 mg	0.36 mL
		1 mg/mL		1.6 mL		1.8 mL

Diphenhydramine	1 mg/kg	50 mg/mL	16 mg	0.32 mL	18 mg	0.36 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	8 g	80 mL	9 g	90 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.16 mg	0.16 mL	0.18 mg	0.18 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	16 mcg	1.6 mL	18 mcg	1.8 mL
Fentanyl	1 mcg/kg	50 mcg/mL	16 mcg	0.32 mL	18 mcg	0.36 mL
Glucagon	age based	1 mg/mL	0.5 mg	0.5 mL	0.5 mg	0.5 mL
Ketamine (behave)	Not approved for this weight					
Lidocaine	0.5 mg/kg	20 mg/mL	8 mg	0.4 mL	9 mg	0.45 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	3.2 mg	0.64 mL	3.6 mg	0.72 mL
		1 mg/mL		3.2 mL		3.6 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL
Naloxone	0.1 mg/kg	1 mg/mL	1.6 mg	1.6 mL	1.8 mg	1.8 mL
Ondansetron	0.1 mg/kg	2 mg/mL	1.6 mg	0.8 mL	1.8 mg	0.9 mL
Normal Saline		20 mL/kg		320 mL		360 mL

5.5 - 6.5 years

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
90-150	22-34	> 75	> 65

Equipment:		20 kg	22 kg
iGel		Gray size 2.0	Gray size 2.0
Suction for iGel		FR 12	FR 12
Soft suction		FR 10	FR 10
ETT - uncuffed		5.5	5.5
ETT - cuffed		5.0	5.0
depth		15.5-16.5 cm	15.5-16.5 cm

Treatment	Dosage	Supplied	20 kg		22 kg	
			dose	vol.	dose	vol.
Defibrillation	2 J/kg		40 J		44 J	
	4 J/kg		80 J		88 J	
	6 J/kg		120 J		132 J	
	8 J/kg		160 J		176 J	
	10 J/kg		200 J		200 J	
Cardioversion	1 J/kg		20 J		22 J	
	2 J/kg		40 J		44 J	

Cardiac Meds.						
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.2 mg	2 mL	0.22 mg	2.2 mL
Adenosine	0.1 mg/kg	3 mg/mL	2 mg	0.67 mL	2.2 mg	0.73 mL
	0.2 mg/kg		4 mg	1.33 mL	4.4 mg	1.47 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	100 mg	2 mL	110 mg	2.2 mL
			100 mg	2 mL	110 mg	2.2 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.4 mg	4 mL	0.44 mg	4.4 mL
Normal Saline (C)	10 mL/kg			200 mL		220 mL

Airway Meds.						
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine Normal Saline	0.5 mg/kg	1 mg/mL	5 mg	5 mL	5 mg	5 mL
				0 mL		0 mL
Etomidate	0.3 mg/kg	2 mg/mL	6 mg	3 mL	6.6 mg	3.3 mL
Fentanyl	1 mcg/kg	50 mcg/mL	20 mcg	0.4 mL	22 mcg	0.44 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (SAI)	1 mg/kg	50 mg/mL	20 mg	0.4 mL	22 mg	0.44 mL
	0.5 mg/kg		10 mg	0.2 mL	11 mg	0.22 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	500 mg	1 mL	550 mg	1.1 mL
Midazolam	0.1 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL

Diphenhydramine	1 mg/kg	50 mg/mL	20 mg	0.4 mL	22 mg	0.44 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	10 g	100 mL	11 g	110 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.2 mg	0.2 mL	0.22 mg	0.22 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	20 mcg	2 mL	22 mcg	2.2 mL
Fentanyl	1 mcg/kg	50 mcg/mL	20 mcg	0.4 mL	22 mcg	0.44 mL
Glucagon	age based	1 mg/mL	1 mg	1 mL	1 mg	1 mL
Ketamine (behave)	4 mg/kg	50 mg/mL	80 mg	1.6 mL	88 mg	1.76 mL
Lidocaine	0.5 mg/kg	20 mg/mL	10 mg	0.5 mL	11 mg	0.55 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	4 mg	0.8 mL	4.4 mg	0.88 mL
		1 mg/mL		4 mL		4.4 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL
Naloxone	0.1 mg/kg	1 mg/mL	2 mg	2 mL	2 mg	2 mL
Ondansetron	0.1 mg/kg	2 mg/mL	2 mg	1 mL	2.2 mg	1.1 mL
Normal Saline		20 mL/kg		400 mL		440 mL

7 - 9 years

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
70-120	18-30	> 80	> 65

Equipment:		24 kg	26 kg	28 kg
iGel		Gray size 2.0	White size 2.5	White size 2.5
Suction for iGel		FR 12	FR 12	FR 12
Soft suction		FR 10	FR 10	FR 10
ETT - cuffed		6.0	6.0	6.0
depth		15.5-16.5 cm	15.5-16.5 cm	15.5-16.5 cm

Treatment	Dosage	Supplied	24 kg		26 kg		28 kg	
			dose	vol.	dose	vol.	dose	vol.
Defibrillation	2 J/kg		48 J		52 J		56 J	
	4 J/kg		96 J		104 J		112 J	
	6 J/kg		144 J		156 J		168 J	
	8 J/kg		192 J		200 J		200 J	
	10 J/kg		200 J		200 J		200 J	
Cardioversion	1 J/kg		24 J		26 J		28 J	
	2 J/kg		48 J		52 J		56 J	

Cardiac Meds.								
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.24 mg	2.4 mL	0.26 mg	2.6 mL	0.28 mg	2.8 mL
Adenosine	0.1 mg/kg	3 mg/mL	2.4 mg	0.8 mL	2.6 mg	0.87 mL	2.8 mg	0.93 mL
	0.2 mg/kg		4.8 mg	1.6 mL	5.2 mg	1.73 mL	5.6 mg	1.87 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	120 mg	2.4 mL	130 mg	2.6 mL	140 mg	2.8 mL
			120 mg	2.4 mL	130 mg	2.6 mL	140 mg	2.8 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.5 mg	4.8 mL	0.5 mg	5 mL	0.5 mg	5 mL
Normal Saline (C)	10 mL/kg			240 mL		260 mL		280 mL

Airway Meds.								
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine Normal Saline	0.5 mg/kg	1 mg/mL	5 mg	5 mL	5 mg	5 mL	5 mg	5 mL
				0 mL		0 mL		0 mL
Etomidate	0.3 mg/kg	2 mg/mL	7.2 mg	3.6 mL	7.8 mg	3.9 mL	8.4 mg	4.2 mL
Fentanyl	1 mcg/kg	50 mcg/mL	24 mcg	0.48 mL	26 mcg	0.52 mL	28 mcg	0.56 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (DAI)	1 mg/kg	50 mg/mL	24 mg	0.48 mL	26 mg	0.52 mL	28 mg	0.56 mL
	0.5 mg/kg		12 mg	0.24 mL	13 mg	0.26 mL	14 mg	0.28 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	600 mg	1.2 mL	650 mg	1.3 mL	700 mg	1.4 mL
Midazolam	0.1 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL		2 mL

Diphenhydramine	1 mg/kg	50 mg/mL	24 mg	0.48 mL	26 mg	0.52 mL	28 mg	0.56 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	12 g	120 mL	13 g	130 mL	14 g	140 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.24 mg	0.24 mL	0.26 mg	0.26 mL	0.28 mg	0.28 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	24 mcg	2.4 mL	26 mcg	2.6 mL	28 mcg	2.8 mL
Fentanyl	1 mcg/kg	50 mcg/mL	24 mcg	0.48 mL	26 mcg	0.52 mL	28 mcg	0.56 mL
Glucagon	age based	1 mg/mL	1 mg	1 mL	1 mg	1 mL	1 mg	1 mL
Ketamine (behave)	4 mg/kg	50 mg/mL	96 mg	1.92 mL	104 mg	2.08 mL	112 mg	2.24 mL
Lidocaine	0.5 mg/kg	20 mg/mL	12 mg	0.6 mL	13 mg	0.65 mL	14 mg	0.7 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	4.8 mg	0.96 mL	5.2 mg	1.04 mL	5.6 mg	1.12 mL
		1 mg/mL		4.8 mL		5.2 mL		5.6 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL		2 mL
Naloxone	0.1 mg/kg	1 mg/mL	2 mg	2 mL	2 mg	2 mL	2 mg	2 mL
Ondansetron	0.1 mg/kg	2 mg/mL	2.4 mg	1.2 mL	2.6 mg	1.3 mL	2.8 mg	1.4 mL
Normal Saline	20 mL/kg			480 mL		520 mL		560 mL

9.5 - 11 years

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
70-120	18-30	> 80	> 65

Equipment:		30 kg	32 kg	34 kg
iGel		White size 2.5	White size 2.5	White size 2.5
Suction for iGel		FR 12	FR 12	FR 12
Soft suction		FR 10	FR 10	FR 10
ETT - cuffed		6.5	6.5	6.5
depth		18.5-19.5 cm	18.5-19.5 cm	18.5-19.5 cm

Treatment	Dosage	Supplied	30 kg		32 kg		34 kg	
			dose	vol.	dose	vol.	dose	vol.
Defibrillation	2 J/kg		60 J		64 J		68 J	
	4 J/kg		120 J		128 J		136 J	
	6 J/kg		180 J		192 J		200 J	
	8 J/kg		200 J		200 J		200 J	
	10 J/kg		200 J		200 J		200 J	
Cardioversion	1 J/kg		30 J		32 J		34 J	
	2 J/kg		60 J		64 J		68 J	

Cardiac Meds.								
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.3 mg	3 mL	0.32 mg	3.2 mL	0.34 mg	3.4 mL
Adenosine	0.1 mg/kg	3 mg/mL	3 mg	1 mL	3.2 mg	1.07 mL	3.4 mg	1.13 mL
	0.2 mg/kg		6 mg	2 mL	6.4 mg	2.13 mL	6.8 mg	2.27 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	150 mg	3 mL	150 mg	3 mL	150 mg	3 mL
			150 mg	3 mL	160 mg	3.2 mL	170 mg	3.4 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.5 mg	5 mL	0.5 mg	5 mL	0.5 mg	5 mL
Normal Saline (C)	10 mL/kg			300 mL		320 mL		340 mL

Airway Meds.								
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine Normal Saline	0.5 mg/kg	1 mg/mL	5 mg	5 mL	5 mg	5 mL	5 mg	5 mL
			0 mL			0 mL		0 mL
Etomidate	0.3 mg/kg	2 mg/mL	9 mg	4.5 mL	9.6 mg	4.8 mL	10.2 mg	5.1 mL
Fentanyl	1 mcg/kg	50 mcg/mL	30 mcg	0.6 mL	32 mcg	0.64 mL	34 mcg	0.68 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (SAI)	1 mg/kg	50 mg/mL	30 mg	0.6 mL	32 mg	0.64 mL	34 mg	0.68 mL
	0.5 mg/kg		15 mg	0.3 mL	16 mg	0.32 mL	17 mg	0.34 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	750 mg	1.5 mL	800 mg	1.6 mL	850 mg	1.7 mL
Midazolam	0.1 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL		2 mL

Diphenhydramine	1 mg/kg	50 mg/mL	30 mg	0.6 mL	32 mg	0.64 mL	34 mg	0.68 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	15 g	150 mL	16 g	160 mL	17 g	170 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.3 mg	0.3 mL	0.3 mg	0.3 mL	0.3 mg	0.3 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	30 mcg	3 mL	32 mcg	3.2 mL	34 mcg	3.4 mL
Fentanyl	1 mcg/kg	50 mcg/mL	30 mcg	0.6 mL	32 mcg	0.64 mL	34 mcg	0.68 mL
Glucagon	age based	1 mg/mL	1 mg	1 mL	1 mg	1 mL	1 mg	1 mL
Ketamine (behave)	4 mg/kg	50 mg/mL	120 mg	2.4 mL	128 mg	2.56 mL	136 mg	2.72 mL
Lidocaine	0.5 mg/kg	20 mg/mL	15 mg	0.75 mL	16 mg	0.8 mL	17 mg	0.85 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	6 mg	1.2 mL	6.4 mg	1.28 mL	6.8 mg	1.36 mL
		1 mg/mL		6 mL		6.4 mL		6.8 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL		2 mL
Naloxone	0.1 mg/kg	1 mg/mL	2 mg	2 mL	2 mg	2 mL	2 mg	2 mL
Ondansetron	0.1 mg/kg	2 mg/mL	3 mg	1.5 mL	3.2 mg	1.6 mL	3.4 mg	1.7 mL
Normal Saline	20 mL/kg			600 mL		640 mL		680 mL

9.5 - 11 years

Equipment:		36 kg	38 kg
iGel		Yellow size 3	Yellow size 3
Suction for iGel		FR 12	FR 12
Soft suction		FR 10	FR 10
ETT - cuffed		6.5	6.5
depth		18.5-19.5 cm	18.5-19.5 cm

Treatment	Dosage	Supplied	36 kg		38 kg	
			dose	vol.	dose	vol.
Defibrillation	2 J/kg		72 J		76 J	
	4 J/kg		144 J		152 J	
	6 J/kg		200 J		200 J	
	8 J/kg		200 J		200 J	
	10 J/kg		200 J		200 J	
Cardioversion	1 J/kg		36 J		38 J	
	2 J/kg		72 J		76 J	

Cardiac Meds.						
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.36 mg	3.6 mL	0.38 mg	3.8 mL
Adenosine	0.1 mg/kg	3 mg/mL	3.6 mg	1.2 mL	3.8 mg	1.27 mL
	0.2 mg/kg		7.2 mg	2.4 mL	7.6 mg	2.53 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	150 mg	3 mL	150 mg	3 mL
			180 mg	3.6 mL	190 mg	3.8 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.5 mg	5 mL	0.5 mg	5 mL
Normal Saline (C)	10 mL/kg			360 mL		380 mL

Airway Meds.						
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine Normal Saline	0.5 mg/kg	1 mg/mL	5 mg	5 mL	5 mg	5 mL
				0 mL		0 mL
Etomidate	0.3 mg/kg	2 mg/mL	10.8 mg	5.4 mL	11.4 mg	5.7 mL
Fentanyl	1 mcg/kg	50 mcg/mL	36 mcg	0.72 mL	38 mcg	0.76 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (SAI)	1 mg/kg	50 mg/mL	36 mg	0.72 mL	38 mg	0.76 mL
	0.5 mg/kg		18 mg	0.36 mL	19 mg	0.38 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	900 mg	1.8 mL	950 mg	1.9 mL
Midazolam	0.1 mg/kg	5 mg/mL	3.6 mg	0.72 mL	3.8 mg	0.76 mL
		1 mg/mL		3.6 mL		3.8 mL

Diphenhydramine	1 mg/kg	50 mg/mL	36 mg	0.72 mL	38 mg	0.76 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	18 g	180 mL	19 g	190 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.36 mg	0.36 mL	0.38 mg	0.38 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	36 mcg	3.6 mL	38 mcg	3.8 mL
Fentanyl	1 mcg/kg	50 mcg/mL	36 mcg	0.72 mL	38 mcg	0.76 mL
Glucagon	age based	1 mg/mL	1 mg	1 mL	1 mg	1 mL
Ketamine (behave)	4 mg/kg	50 mg/mL	144 mg	2.88 mL	152 mg	3.04 mL
Lidocaine	0.5 mg/kg	20 mg/mL	18 mg	0.9 mL	19 mg	0.95 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	7.2 mg	1.44 mL	7.6 mg	1.52 mL
		1 mg/mL		7.2 mL		7.6 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL
Naloxone	0.1 mg/kg	1 mg/mL	2 mg	2 mL	2 mg	2 mL
Ondansetron	0.1 mg/kg	2 mg/mL	3.6 mg	1.8 mL	3.8 mg	1.9 mL
Normal Saline	20 mL/kg			720 mL		760 mL

11.5 - 16 years

Vital Signs:			
Heart Rate	Respirations	Systolic	MAP
60-110	16-Dec	> 90	> 65

Equipment:		40 kg	42 kg	44 kg
iGel		Yellow size 3	Yellow size 3	Yellow size 3
Suction for iGel		FR 12	FR 12	FR 12
Soft suction		FR 12	FR 12	FR 12
ETT - cuffed		6.5	6.5	6.5
depth		18.5-19.5 cm	18.5-19.5 cm	18.5-19.5 cm

Treatment	Dosage	Supplied	40 kg		42 kg		44 kg	
			dose	vol.	dose	vol.	dose	vol.
Defibrillation	2 J/kg		80 J		84 J		88 J	
	4 J/kg		160 J		168 J		176 J	
	6 J/kg		200 J		200 J		200 J	
	8 J/kg		200 J		200 J		200 J	
	10 J/kg		200 J		200 J		200 J	
Cardioversion	1 J/kg		40 J		42 J		44 J	
	2 J/kg		80 J		84 J		88 J	

Cardiac Meds.								
1:10 Epinephrine	0.01 mg/kg	0.1 mg/mL	0.4 mg	4 mL	0.42 mg	4.2 mL	0.44 mg	4.4 mL
Adenosine	0.1 mg/kg	3 mg/mL	4 mg	1.33 mL	4.2 mg	1.4 mL	4.4 mg	1.47 mL
	0.2 mg/kg		8 mg	2.67 mL	8.4 mg	2.8 mL	8.8 mg	2.93 mL
Amiodarone VT VT/VF	5 mg/kg	50 mg/mL	150 mg	3 mL	150 mg	3 mL	150 mg	3 mL
			200 mg	4 mL	210 mg	4.2 mL	220 mg	4.4 mL
Atropine	0.02 mg/kg	0.1 mg/mL	0.5 mg	5 mL	0.5 mg	5 mL	0.5 mg	5 mL
Normal Saline (C)	10 mL/kg			400 mL		420 mL		440 mL

Airway Meds.								
Albuterol	2.5 mg	2.5 mg/3mL	2.5 mg	3 mL	2.5 mg	3 mL	2.5 mg	3 mL
Neb. Epinephrine Normal Saline	0.5 mg/kg	1 mg/mL	5 mg	5 mL	5 mg	5 mL	5 mg	5 mL
				0 mL		0 mL		0 mL
Etomidate	0.3 mg/kg	2 mg/mL	12 mg	6 mL	12.6 mg	6.3 mL	13.2 mg	6.6 mL
Fentanyl	1 mcg/kg	50 mcg/mL	40 mcg	0.8 mL	42 mcg	0.84 mL	44 mcg	0.88 mL
Ipratropium	0.5 mg	0.5 mg/2.5mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL	0.5 mg	2.5 mL
Ketamine (DAI)	1 mg/kg	50 mg/mL	40 mg	0.8 mL	42 mg	0.84 mL	44 mg	0.88 mL
	0.5 mg/kg		20 mg	0.4 mL	21 mg	0.42 mL	22 mg	0.44 mL
Mag. Sulfate	25 mg/kg	500 mg/mL	1000 mg	2 mL	1050 mg	2.1 mL	1100 mg	2.2 mL
Midazolam	0.1 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL		2 mL

Diphenhydramine	1 mg/kg	50 mg/mL	40 mg	0.8 mL	42 mg	0.84 mL	44 mg	0.88 mL
Dextrose 10%	0.5 g/kg	0.1 g/mL	20 g	200 mL	21 g	210 mL	22 g	220 mL
1:1 Epinephrine	0.01 mg/kg	1 mg/mL	0.3 mg	0.3 mL	0.3 mg	0.3 mL	0.3 mg	0.3 mL
Push dose Epi.	1 mcg/kg	10 mcg/mL	40 mcg	4 mL	42 mcg	4.2 mL	44 mcg	4.4 mL
Fentanyl	1 mcg/kg	50 mcg/mL	40 mcg	0.8 mL	42 mcg	0.84 mL	44 mcg	0.88 mL
Glucagon	age based	1 mg/mL	1 mg	1 mL	1 mg	1 mL	1 mg	1 mL
Ketamine (behave)	4 mg/kg	50 mg/mL	160 mg	3.2 mL	168 mg	3.36 mL	176 mg	3.52 mL
Lidocaine	0.5 mg/kg	20 mg/mL	20 mg	1 mL	21 mg	1.05 mL	22 mg	1.1 mL
Midazolam (IM)	0.2 mg/kg	5 mg/mL	8 mg	1.6 mL	8.4 mg	1.68 mL	8.8 mg	1.76 mL
		1 mg/mL		8 mL		8.4 mL		8.8 mL
Midazolam (IV/IO)	0.2 mg/kg	5 mg/mL	2 mg	0.4 mL	2 mg	0.4 mL	2 mg	0.4 mL
		1 mg/mL		2 mL		2 mL		2 mL
Naloxone	0.1 mg/kg	1 mg/mL	2 mg	2 mL	2 mg	2 mL	2 mg	2 mL
Ondansetron	0.1 mg/kg	2 mg/mL	4 mg	2 mL	4 mg	2 mL	4 mg	2.0 mL
Normal Saline	20 mL/kg			800 mL		840 mL		880 mL

ADENOSINE

DRUG CLASS: supraventricular anti-arrhythmic

MECHANISM OF ACTION:

Slows conduction at the AV node, thus it slows conduction and blocks reentry pathways through the AV node.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
2nd or 3rd degree heart block - Sick sinus syndrome - Bradycardia - Caution: 1st degree heart block or bundle branch block - Atrial fib./flutter or ventricular tachycardia - not effective in either	Adult Supraventricular Tachycardia Pediatric Unstable Tachycardia Pediatric Stable Tachycardia	- Facial flushing - Chest pain - Dyspnea - Brief transient dysrhythmias

SUPPLIED

6mg/2mL vial (concentration: 3mg/mL)

DOSAGE:

ADULT: 1st IV/IO: 6 mg
2nd IV/IO: 12 mg

Pediatric: 1st IV/IO: 0.1mg/kg, max. 6mg
2nd IV/IO:0.2mg/kg, max. 12mg
Use calculator for dosing

ADMINISTRATION TIME:

IV/IO: RAPID push

REPEAT DOSING TIME:

2 minutes

ONSET:

IV/IO: 15 seconds

DURATION:

IV/IO: 10 seconds

ALBUTEROL

DRUG CLASS: Beta-2 agonist, sympathomimemic

MECHANISM OF ACTION:

Acts selectively on Beta2 receptor sites in the lungs, relaxing bronchial smooth muscle, decreasing airway resistance, and relief of bronchospasm.

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
- Caution: - PMH of cardiovascular disease - hypertension - tachycardia - prolonged QT syndrome	Adult Asthma/COPD Pediatric Asthma Adult Anaphylaxis Pediatric Anaphylaxis Crush Syndrome	- Tremor - Nervousness - Bronchospasm - Nausea - Palpatations or tachycardia

SUPPLIED

2.5 mg/3 mL plastic ampule

DOSAGE:

ADULT: NEB: 2.5 mg

Pediatric: NEB: 2.5 mg

ADMINISTRATION TIME:

NEB: 8-10 minutes

ONSET:

NEB: 5-15 minutes

DURATION:

3-4 hours

AMIODARONE

DRUG CLASS: Class III anti-dysrhythmic

MECHANISM OF ACTION:

Acts directly on the myocardium to delay repolarization and increase the duration of the action potential.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
2nd or 3rd degree heart block - Iodine allergy - Bradycardia Cardiogenic shock	Adult Ventricular Tachycardia w/ a Pulse Pediatric Unstable Tachycardia Pediatric Stable Tachycardia Adult Ventricular Fibrillation Pediatric Ventricular Fib.	- Hypotension - Bradycardia

SUPPLIED

150 mg/3 mL vial (concentration: 50 mg/mL)

PULSELESS ARREST

DOSAGE:

ADULT: IV/IO: 300 mg
IV/IO: 150 mg IVP/IO

Pediatric: IV/IO: 5 mg/kg, max. 300mg
IV/IO: 5 mg/kg, max. 150mg

ADMINISTRATION TII IV/IO: rapid push

ONSET: IV/IO: 2 - 3 minutes

DURATION: variable

WITH A PULSE

DOSAGE:

CALL OLMC TO CONSIDER

ADULT: IV/IO: 150 mg

Pediatric: IV/IO: 5 mg/kg, max. 150mg

ADMINISTRATION TII IV/IO: over 10 minutes

ONSET: IV/IO: 2 - 3 minutes

DURATION: variable

MIXING INSTRUCTIONS:

- Add dosage of Amiodarone to 100 mL bag D5W
- Rotate bag to mix
- Insert macro drop tubing into bag
- Run at rate based on table below

	TUBING		
	10 gtt	15 gtt	20 gtt
gtt/min	100/min	150/min	200/min
gtt/sec	1.6/sec	2.5/sec	3/sec

ASPIRIN

DRUG CLASS anti-platelet aggregator

MECHANISM OF ACTION:

Aspirin works by blocking the formation of the substance Thromboxane A2, which causes platelets to aggregate and arteries to constrict.

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
- No gag reflex - Active GI bleed - Age ≤ 18	Adult Acute Coronary Syndrome	- GI distress - Nausea - Vomiting

SUPPLIED

81 mg chewable tablets

DOSAGE:

ADULT: PO: 324 mg (81 mg x 4 tablets)

Pediatric: NONE

ONSET:

PO: 5-30 minutes

DURATION:

1-4 hours

PEARLS:

If patient has already taken, augment dosage to 324 mg total



ATROPINE

DRUG CLASS parasympatholytic, anticholinergic agent

MECHANISM OF ACTION:

Atropine is a competitive inhibitor of acetylcholine at muscarinic receptor sites. In the setting of symptomatic bradycardias, atropine decreases vagal effects on the heart resulting in increased chronotropy and dromotropy. Atropine is also used in cholinergic exposures as a direct antidote for the poison.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
- Hypertension - Tachycardia 2nd and 3rd degree heart blocks	Adult Bradycardia Pediatric Bradycardia Acetylcholinesterase Inhibitor Exposure Adult Beta Blocker Adult Ca Channel Blocker	- Tachycardia - Dilated pupils - Dry secretions - Transient dysrhythmias - Flushed skin

SUPPLIED

1 mg/10 mL vial or preload syringe (concentration: 0.1 mg/mL)

DOSAGE:

ADULT: IV/IO: 1 mg, max. 3 mg

CALL OLMC TO CONSIDER

Pediatric: IV/IO: 0.02 mg/kg
max. single dose 0.5 mg
max. total dose 1 mg

ADMINISTRATION TIME:

IV/IO: RAPID push

REPEAT DOSING TIME:

3 minutes

ONSET:

IV/IO: 15 seconds

DURATION:

2 - 6 hours

DEXTROSE 10%

DRUG CLASS monosaccharide

MECHANISM OF ACTION:

Increases serum blood glucose levels

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
- BGL > 60 - IV line not verified patent	Adult Diabetic Emergency Pediatric Diabetic Emergency	- Hyperglycemia - Hyperosmolar syndrome - Infusion pain - Tissue necrosis with infiltration - Vein thrombosis

SUPPLIED

25 gm/250 mL bag (concentration: 0.1 gm/mL)

DOSAGE:

ADULT: IV/IO: 25 gm, max. 50 gm **Pediatric:** IV/IO: 0.5 gm/kg, max. 50 gm

ADMINISTRATION TIME:

IV/IO: SLOW bolus

REPEAT DOSING TIME:

5 minutes

ONSET:

IV/IO: 1 minute

DURATION:

Depends on degree of hypoglycemia

PEARLS:

Because of the long half-life (therapeutic duration) associated with oral hypoglycemic agents and long acting insulin; **often these patients have hypoglycemic relapses and therefore should be carefully monitored in a medical control facility.** To the extent practical, these patients should be transported for further care and monitoring.

DIPHENHYDRAMINE

BENADRYL

DRUG CLASS: antihistamine, H1 antagonist

MECHANISM OF ACTION:

Blocks H1 receptors, causing bronchoconstriction and contraction of the gut, and H2 receptors causing peripheral vasodilation and secretion of gastric acid.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
Acute Asthma / COPD	Pediatric Anaphylaxis Adult Anaphylaxis	- CNS depression - Sedation - Dizziness - Seizures - Paradoxical CNS stimulation

SUPPLIED

50 mg/mL vial

DOSAGE:

ADULT:

Unstabl IV/IO/IM: 50 mg
Stable: IV/IO/IM: 25 mg

Pediatric:

Unstable: IV/IO/IM: 1 mg/kg, max. 50
Stable: IV/IO/IM: 1 mg/kg, max. 25

ADMINISTRATION TIME:

IV/IO: SLOW over 2 minutes

ONSET:

IV/IO: immediate
IM: 30 minutes

DURATION:

IV/IO/IM: 4 - 7 hours

DUODOTE

ATROPINE 2-PAM CHLORIDE

DRUG CLASS: anticholinergic

MECHANISM OF ACTION:

Atropine blocks acetylcholine effect on smooth/cardiac muscle, CNS, and secretory gland cells, reducing parasympathetic overstimulation. Decreases secretions, relieves airway constriction, attenuates bradycardia. 2-PAM reactivates acetylcholinesterase, which hydrolyzes excess acetylcholine to restore cholinergic neural function.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
- To treat prophylactically - None in poisoning	Acetylcholinesterase Inhibitor Exposure	- Dysrhythmias - Pupil dilation - Confusion, dizziness - Headache, HTN - Nausea/vomiting - Rash/dry skin

SUPPLIED

Preload: 2.1mg Atropine and 600 mg/2mL Pralidoxine Chloride

MILD:	ATROPINE auto-injector	OR	ATROPINE	600mg auto-injector	OR	2-PAM
0-2yrs (<13kg)	0.5 mg		0.05 mg/kg			
3-7yrs (13-25 kg)	1 mg		1 mg			
8-14yrs (26-50 kg)	2 mg		2 mg			
>14 yrs	2 mg		2 mg			
Pregnant	2 mg		2 mg			
Geriatric	1 mg		1 mg			

MODERATE:	ATROPINE auto-injector	OR	ATROPINE	600mg auto-injector	OR	2-PAM
0-2yrs (<13kg)	0.5 mg		0.05 mg/kg			15 mg/kg
3-7yrs (13-25 kg)	1 mg		1 mg	1		15 mg/kg
8-14yrs (26-50 kg)	2 mg		2 mg	1		15 mg/kg
>14 yrs	2-4 mg		2-4 mg	1		600 mg
Pregnant	2-4 mg		2-4 mg	1		600 mg
Geriatric	2 mg		2 mg	1		10 mg/kg

SEVERE:	ATROPINE auto-injector	OR	ATROPINE	600mg auto-injector	OR	2-PAM
0-2yrs (<13kg)	0.5 mg		0.1 mg/kg			45 mg/kg
3-7yrs (13-25 kg)	2 mg		0.1 mg/KG	1		45 mg/kg
8-14yrs (26-50 kg)	4 mg		4 mg	2		45 mg/kg
>14 yrs	6 mg		6 mg	3		1800 mg
Pregnant	6 mg		6 mg	3		1800 mg
Geriatric	2-4 mg		2-4 mg	2-3		25 mg/kg

EPINEPHRINE

DRUG CLASS: Adrenergic catecholamine, sympathomimetic

MECHANISM OF ACTION:

Alpha effects: vasoconstriction

Beta effects: increases automaticity, conductivity, contractility, dilation of bronchial tree

CONTRAINDICATIONS	PROTOCOLS USED		SIDE EFFECTS
- None - Cautions: - elderly - CVD - Sensitivity to epi/sulfites - In active labor	Adult Pulseless Arrest Adult Bradycardia Adult Anaphylaxis Adult Sepsis Adult Cardio. Shock ROSC Croup - Pediatric	Pediatric Arrest Pediatric Bradycardia Pediatric Anaphylaxis Pediatric Sepsis Pediatric Cardio Shock Pediatric Shock Epiglottitis - Pediatric	- Tachyarrhythmia - tremors - Anxiety - nausea - headache

CARDIAC EPINEPHRINE 1mg/10mL	
SUPPLIED 1 mg/10 mL vial or preload syringe (concentration: 0.1 mg/mL)	
DOSAGE: ADULT: IV/IO: 1 mg, no max. Pediatric: IV/IO: 0.01 mg/kg max. single dose 1 mg	
ADMINISTRATION TIME: IV/IO: rapid push	
REPEAT DOSING TIME: 3-5 minutes	
ONSET: IV/IO: 1-2 minutes	
DURATION: IV/IO: 5 - 10 minutes	

PUSH DOSE EPINEPHRINE	
SUPPLIED Must mix to concentration of 10mcg/mL	
DOSAGE: ADULT: IV/IO: 50 mcg max. total 100 mcg Pediatric: IV/IO: 1 mcg/kg max. single dose 50 mcg max. total dose 100 mcg	
ADMINISTRATION TIME: IV/IO: push	
REPEAT DOSING TIME: 5 minutes	
ONSET: IV/IO: 1 - 2 minutes	
DURATION: IV/IO: 5 - 10 minutes	
MIXING INSTRUCTIONS: Supplies: 1mg/10mL cardiac epinephrine 10mL syringe - transfer device 10mL flush - Attach pre-load 1mg/10mL epinephrine to transfer device - Attach 10mL syringe to other side of transfer device - Draw up 1mL of epinephrine into the 10mL syringe - Disconnect epinephrine. - Draw up 9mL of NS into 10mL syringe Result: 10mcg/mL epinephrine	

IM EPINEPHRINE 1mg/mL

SUPPLIED

1 mg/mL vial or ampule

DOSAGE:

ADULT:

IM: 0.3 mg
max dependent on

Pediatric:

IM: 0.01 mg/kg
max. single dose 0.3 mg

ADMINISTRATION LOCATION:

anterolateral thigh: max 5mL
shoulder: max 2mL

REPEAT DOSING TIME:

5 minutes

ONSET:

IM: 5-10 minutes

DURATION:

IM: 5 - 10 minutes

NEBULIZED EPINEPHRINE

SUPPLIED

1 mg/mL vial or ampule

DOSAGE: PEDIATRICS ONLY - 0.5 mg/kg NEB, may repeat, max total dose 5 mg

	3kg	4kg	5kg	6kg	7kg	8kg	9kg		≥ 10kg	
				dose	vol.	dose	vol.		dose	vol.
1:1 Epinephrine	Contact			3mg	3mL	4mg	4mL		5mg	5mL
Normal Saline	OLMC				2mL		1mL			0mL

ADMINISTRATION TIME:

NEB: 8-10 minutes

ONSET:

NEB: 5-10 minutes

DURATION:

NEB: 60-90 minutes

REPEAT DOSING TIME:

5 minutes

MIXING INSTRUCTIONS:

- Add dosage of 1:1 Epinephrine to nebulizer chamber
- Add dosage of normal saline to nebulizer chamber (if required)
- hook up to 8-10 L of oxygen

ETOMIDATE

DRUG SHORTAGES ONLY

DRUG CLASS: non-barbiturate hypnotic

MECHANISM OF ACTION:

Acts at the level of the reticular-activating system to produce anesthesia and depresses CNS function via inhibition of GABA (inhibitory) receptors. Etomidate does not induce analgesia but induces unconsciousness within one circulation time. Recovery is rapid as a result of extensive redistribution and rapid metabolism.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
- MAP < 65 - Cautions: - renal failure, liver cirrosis, pregnancy, seizure disorder	Drug Shortages	- Trismus if pushed too fast - Involuntary muscle movement - Nausea/vomiting - Arrhythmias

SUPPLIED

40mg/20mL (concentration 2mg/mL)

DOSAGE:

ADULT: IV/IO: 0.3mg/kg max. dose 40mg				
lbs.	kg.	dose	vol.	
100	45	14 mg	6.8	mL
110	50	15 mg	8	mL
120	55	16 mg	8.2	mL
130	59	18 mg	9	mL
140	64	19 mg	9.5	mL
150	68	20 mg	10	mL
160	73	22 mg	10.9	mL
170	77	23 mg	12	mL
180	82	25 mg	12.3	mL
190	86	26 mg	13	mL
200	91	27 mg	13.6	mL
210	95	29 mg	14	mL
220	100	30 mg	15	mL
230	105	31 mg	16	mL
240	109	33 mg	16.4	mL
250	114	34 mg	17	mL
260	118	35 mg	17.7	mL
270	123	37 mg	18	mL
280	127	38 mg	19.1	mL
290	132	40 mg	20	mL

Pediatric: IV/IO: 0.3mg/kg max. dose 40mg			
lbs.	kg.	dose	vol.
11	5	1.5 mg	0.75 mL
13	6	1.8 mg	0.9 mL
15	7	2.1 mg	1.05 mL
18	8	2.4 mg	1.2 mL
20	9	2.7 mg	1.35 mL
22	10	3 mg	1.5 mL
24	11	3.3 mg	1.65 mL
26	12	3.6 mg	1.8 mL
29	13	3.9 mg	1.95 mL
31	14	4.2 mg	2.1 mL
35	16	4.8 mg	2.4 mL
40	18	5.4 mg	2.7 mL
44	20	6 mg	3 mL
48	22	6.6 mg	3.3 mL
53	24	7.2 mg	3.6 mL
57	26	7.8 mg	3.9 mL
62	28	8.4 mg	4.2 mL
66	30	9 mg	4.5 mL
70	32	9.6 mg	4.8 mL
75	34	10 mg	5.1 mL
79	36	11 mg	5.4 mL
84	38	11 mg	5.7 mL
86	39	11.7 mg	5.9 mL
88	40	12 mg	6 mL
90	41	12.3 mg	6.14 mL

ADMINISTRATION TIME:

IV/IO: SLOW push over 1 minute

ONSET:

IV/IO: 1 minute

DURATION:

IV/IO: 3-5 minutes

FENTANYL

DRUG CLASS: synthetic opiate

MECHANISM OF ACTION:

Narcotic agonist-analgesic of opiate receptors; inhibits ascending pain pathways, thus altering response to pain; increases pain threshold; produces analgesia, respiratory depression, and sedation.

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
Hypersensativity to opiates	Adult Pain Management Pediatric Pain Management Drug Shortages	- Chest wall rigidity - Respiratory distress/failure - Laryngospasm - Bronchoconstriction

SUPPLIED

100mcg/2mL (concentration 50mcg/mL)

DOSAGE:

ADULT: IV/IO/IN/IM: 1mcg/kg
max. single dose 100mcg
max. total dose 200mcg

Pediatric: IV/IO/IN/IM: 1mcg/kg
max. single dose 100mcg
max. total dose 200mcg

ADMINISTRATION TIME:

IV/IO: SLOW push

REPEAT DOSING TIME:

10 minutes

ONSET:

IV/IO/IN: 1-3 minutes
IM: 10-20 minutes

DURATION:

1-2 hours

GLUCAGON

DRUG CLASS: pancreatic hormone

MECHANISM OF ACTION:

Glucagon increases blood glucose by stimulating glycogenolysis and inhibiting conversion of glucose to glycogen. This process stimulates gluconeogenesis (metabolism of glucose in the liver), relaxes the smooth muscle of the GI tract, and produces positive inotropic and chronotropic effects

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
- Chronic hypoglycemia (no stores) - Adrenal insufficiency - Allergy to protein - Starvation	Adult Diabetic Emergencies Pediatric Diabetic Emergencies Beta-blocker Overdose	- Nausea/vomiting - Allergic reaction - protein

SUPPLIED

1mg vial to reconstitute with 1mL normal saline (kit)

DIABETIC EMERGENCIES

DOSAGE:

ADULT: IM/IN: 1 mg

Pediatric: IM/IN: < 20 kg: 0.5 mg
> 20 kg: 1 mg

ADMINISTRATION LOCATION

anterolateral thigh: max 5mL
shoulder: max 2mL

ONSET:

IM/IN: 8-10 minutes

DURATION:

IM/IN: 1-2 hours

BETA-BLOCKER OVERDOSE

DOSAGE:

ADULT: IV: 1 mg, max 2 mg

Pediatric: NONE

ADMINISTRATION TIME

IV: slow push over 1 minute

REPEAT DOSING TIME:

5 minutes

ONSET:

IV: 45 seconds

DURATION:

IV: 1-2 hours

ORAL GLUCOSE

DRUG CLASS: monosaccharide

MECHANISM OF ACTION:

After absorption from GI tract, glucose is distributed in the tissues and provides a prompt increase in circulating blood sugar

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
• Unable to protect airway • No gag reflex • Unable to tolerage oral route	• Adult Diabetic Emergencies • Pediatric Diabetic Emergencies	• Nausea

SUPPLIED

15 gram tube

DOSAGE:

ADULT: ORAL: 15 gm

Pediatric: ORAL: > 2 years old: 15 gm

ADMINISTRATION TIME

based on toleration of patient

ONSET:

ORAL: 10 minutes

DURATION:

ORAL: varies based on patient metabolism

HYDRALAZINE

DRUG CLASS: vasodilator

MECHANISM OF ACTION:

Lowers blood pressure by exerting a peripheral vasodilating effect through direct relaxation of vascular smooth muscle. Dilates arterioles with little effect on veins and decreases systemic vascular resistance

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
- Coronary artery disease - Mitral valve disease - Renal damage	Pre-Eclampsia/Eclampsia	- Nausea/vomiting/diarrhea - Headache, hypotension - Palpatations, dyspnea - Rash

SUPPLIED

20 mg/mL vial

DOSAGE:

ADULT: 1st IV/IO: 5 mg (0.25mL)
2nd IV/IO: 10 mg (0.5mL)

Pediatric: NONE

ADMINISTRATION TIME:

IV/IO: slow push over 1 minute

REPEAT DOSING TIME:

10 minutes

ONSET:

IV/IO: 5-20 minutes

DURATION:

IV/IO: 1-4 hours

IPRATROPIUM BROMIDE

DRUG CLASS: anticholinergic, parasympatholytic

MECHANISM OF ACTION:

Antagonizes the action of acetylcholine by blocking muscarinic cholinergic receptors, resulting in bronchodilation and drying of respiratory tract secretions.

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
- Glaucoma - Prostate hypertrophy	Adult Asthma/COPD Pediatric Asthma Adult Anaphylaxis Pediatric Anaphylaxis	- Nausea - Bitter taste in mouth - Blurred vision, dilated pupils

SUPPLIED

0.5mg/2.5mL plastic ampule

DOSAGE:

ADULT: NEB: 0.5 mg

Pediatric: NEB: 0.5 mg

ADMINISTRATION TIME:

NEB: 8-10 minutes

ONSET:

NEB: 5-10 minutes

DURATION:

NEB: 4-6 hours

KETAMINE

DRUG CLASS: general anesthetic

MECHANISM OF ACTION:

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
<u>Patient Restraint</u> ≥ 65 years old	Behavioral Emergencies	Disorientation, delirium, psychosis
<u>Drug Assisted Intubation</u> - none	Adult Drug Assisted Intubation Pediatric Advanced Airway Mangement	- Auditory/visual hallucinations - Rigidity, dystonic reactions - Respiratory arrest, increased secretions

SUPPLIED 500mg/10mL (50mg/1mL)

DRUG ASSISTED INTUBATION		
ADULT:		Pediatric: Induction 1 IV/IO: 1mg/kg, max 300mg Induction 2 IV/IO: 0.5mg/kg, max total 450mg Sedation IV/IO: 0.5mg/kg, no repeat

BEHAVIORAL / PATIENT RESTRAINT		
ADULT:		Pediatric: IM: 4mg/kg, max single 300mg IM: 2mg/kg, max total 450mg
IDEAL BODY WEIGHT		
ADULT MALE:		
	Initial Dose:	
height	dose	vol.
< 5'6"	200 mg	4 mL
5'6"-6"	250 mg	5 mL
> 6'	300 mg	6 mL
	Repeat Dose:	
height	dose	vol.
< 5'6"	100 mg	2 mL
5'6"-6"	125 mg	2.5 mL
> 6'	150 mg	3 mL
ADULT FEMALE:		
	Initial Dose:	
height	dose	vol.
< 5'	180 mg	3.6 mL
5'-5'6"	200 mg	4 mL
> 5'6"	250 mg	5 mL
	Repeat Dose:	
height	dose	vol.
< 5'	90 mg	1.8 mL
5'-5'6"	100 mg	2 mL
> 5'6"	125 mg	2.5 mL
ADMINISTRATION LOCATION: anterolateral thigh: max 5mL shoulder: max 2mL ONSET: 3-4 minutes REPEAT DOSING TIME: 5 minutes DURATION: 12 - 25 minutes		

LIDOCAINE

DRUG CLASS: local anesthetic

MECHANISM OF ACTION:

Acts on pain nerve cells' voltage-gated sodium channels to lock open, preventing depolarization, and therefore pain reception.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
- AV blocks - Idioventricular or escape rhythms - Bradycardic dysrhythmias	Intraosseous Infusion	- Hypotension - Nausea - Bradycardia leading to arrest - Seizures, muscle twitching - widening QRS complex

SUPPLIED

100mg/5mL (concentration 20mg/mL)

DOSAGE:

ADULT: IO: 1mg/kg, max 50mg
may repeat x1

Pediatric: IO: 0.5 mg/kg, max 40mg
repeat 0.25mg/kg

ADMINISTRATION TIME:

IO: SLOW push over 1-2 minutes

REPEAT DOSING TIME:

PRN

ONSET:

IO: 1 minute

DURATION:

1-2 hours

MAGNESIUM SULFATE

DRUG CLASS: anti-dysrhythmic

MECHANISM OF ACTION:

Acts directly on the myocardium to delay repolarization and increase the duration of the action potential.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
2nd or 3rd degree heart block - Women in active labor - Hypotension - Renal failure, dialysis	Adult Ventricular Tachycardia w/ a Pulse Adult Ventricular Fibrillation Pediatric Ventricular Fib. Pre-eclampsia/Eclampsia Adult Asthma/COPD Pediatric Asthma	- Hypotension - Flushing, diaphoresis - Reflex depression - Flaccid paralysis - Drowsiness - Respiratory paralysis - Hypothermia

SUPPLIED

1 gm/2 mL vial (concentration: 0.5 gm/mL)

ASTHMA / SUSPECTED TORSADES DE POINTS

DOSAGE:

ADULT: IV/IO: 2 gm

Pediatric: IV/IO: 25 mg/kg, max. 2gm

ADMINISTRATION TIME:

Asthma: in 100mL D5W over 15 minutes

Torsades de Points: in 100mL D5W over 5 minutes

Use Calculator for Dosage

ONSET:

IV/IO: immeidate

DURATION:

IV/IO: 30-60 minutes

MIXING INSTRUCTIONS:

- Add dosage of Magnesium to 100 mL D5W
- Rotate bag to mix
- Insert macro drop tubing
- Run at rate based on table below

		TUBING		
		10 gtt	15 gtt	20 gtt
Asthma over 15 min.	gtt/min	67/min	100/min	133/min
	gtt/sec	1.1/sec	1.7/sec	2.2/sec

		TUBING		
		10 gtt	15 gtt	20 gtt
Torsades over 5 min.	gtt/min	200/min	300/min	400/min
	gtt/sec	3.3/sec	5/sec	6.7/sec

PRE-ECLAMPSIA / ECLAMPSIA	
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DOSAGE:

ADULT: IV/IO: 4 gm

Pediatric: NONE

ADMINISTRATION TIME:

in 100mL D5W over **15 minutes**

ONSET:

IV/10: immeddate

DURATION:

IV/IO: 30-60 minutes

MIXING INSTRUCTIONS:

- Add dosage of Magnesium to 100 mL D5W
- Rotate bag to mix
- Insert macro drop tubing
- Run at rate based on table below

		TUBING		
		10 gtt	15 gtt	20 gtt
OB over 15 min.	gtt/min	67/min	100/min	133/min
	gtt/sec	1.1/sec	1.7/sec	2.2/sec

MIDAZOLAM VERSED

DRUG CLASS: Benzodiazepine

MECHANISM OF ACTION:

Enhances the effects of GABA (inhibitory neurotransmitter) to decrease nerve cell activity in all regions of the CNS. As an anticonvulsant, midazolam augments pre-synaptic inhibitions of neurons, limiting the spread of electrical activity

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
- Hypotension, shock - Pregnancy unless seizing - Taking narcotics or depressants - Debilitated patient, chronic disease	Adult Seizures Pediatric Seizures Drug Shortages Behavioral Emergencies Eclampsia Head/Spine Injuries Stimulant Overdose	- Hypotension - Drowsiness, sedation - Confusion, amnesia - Ataxia - Respiratory depression/arrest - Crosses placental barrier

CHECK VIAL CONCENTRATION PRIOR TO ADMINISTRATION
SUPPLIED

10mg/2mL vial (5mg/1mL)
1mg/1mL vial

SEIZURES - IM DOSING
DOSAGE: ADULT: IM: 10 mg Pediatric: IM: 0.2 mg/kg, max. total 10 mg
ONSET: IM: 5-15 minutes
DURATION: 40 minutes

SEIZURES - IV/IO/IN DOSING
DOSAGE: ADULT: IV/IO/IN: 2 mg, max. 10mg Pediatric: IV/IO/IN: 0.2 mg/kg max. single dose 2mg max. total dose 10mg
ADMINISTRATION TIME: IV/IO: push IN: rapid push, max 1mL each nostril
ONSET: IV/IO/IN: 1-3 minutes
DURATION: 40 minutes

DRUG ASSISTED INTUBATION - DRUG SHORTAGES ONLY
DOSAGE: ADULT: Induction IV/IO: 0.2mg/kg, max Sedation IV/IO: 2mg, max 10mg Pediatric: Induction IV/IO: 0.1mg/kg, max 10mg Sedation IV/IO: 0.1mg/kg, max 10mg

NALOXONE

NARCAN

DRUG CLASS Narcotic antagonist

MECHANISM OF ACTION:

Naloxone binds competitively to opiate receptor sites, displacing narcotics and synthetic narcotics.
Naloxone also antagonizes all actions of narcotics.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
Caution in patients with cardiac irritability and narcotic addiction	Adult Depressant Overdose Pediatric Depressant Overdose	- Seizures - withdrawl symptoms - Nausea/vomiting

SUPPLIED

2 mg/mL preload syringe

DOSAGE:

ADULT: IV/IO/IN/IM: 2 mg
max. 10mg

Pediatric: IV/IO/IN/IM: 0.1 mg/kg
max. 2mg

ADMINISTRATION TIME:

IV/IO: push
IN: rapid push, max 1mL each nostril
IM: anterolateral max 5mL, shoulder max 2mL

NOTE:
Do not administer to patients
< 5kg

REPEAT DOSING TIME:

3 minutes

ONSET:

IV/IO/IN: 1-2 minutes
IM: 2-8 minutes

DURATION:

30-60 minutes

NITROGLYCERIN

DRUG CLASS: vasodilator

MECHANISM OF ACTION:

Decreases blood return to right heart, decreases preload and afterload and oxygen consumption.

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
- Hypersensitivity to nitrates - Head trauma - Hypotension < 90 SBP - Use of ED drugs within 48 hours - Brain bleed	Adult Acute Coronary Syndrome Adult Congestive Heart Failure	- Headache - Hypotension - Nausea/vomiting - Palptations - Sublingual burning

SUPPLIED

0.4 mg dissovable tablets or spray

DOSAGE:

ADULT: SL: 0.4 mg
max protocol dependant

Pediatric: NONE

REPEAT DOSING TIME:

5 minutes

ONSET:

SL: 1-3 minutes

DURATION:

30 minutes

NORMAL SALINE

ADULT:		Cardiogenic Anaphylactic Neurogenic	Trauma	Septic
		20mL/kg	20mL/kg	30mL/kg
		volume	volume	volume
lbs.	kg.			
100	45	909 mL	909 mL	1364 mL
110	50	1000 mL	1000 mL	1500 mL
120	55	1091 mL		1636 mL
130	59	1182 mL		1773 mL
140	64	1273 mL		1909 mL
150	68	1364 mL		2045 mL
160	73	1455 mL		2182 mL
170	77	1545 mL		2318 mL
180	82	1636 mL		2455 mL
190	86	1727 mL		2591 mL
200	91	1818 mL		2727 mL
210	95	1909 mL		2864 mL
220	100	2000 mL		3000 mL
230	105	2091 mL		3136 mL
240	109	2182 mL		3273 mL
250	114	2273 mL		3409 mL
260	118	2364 mL		3545 mL
270	123	2455 mL		3682 mL
280	127	2545 mL		3818 mL
290	132	2636 mL		3955 mL
300	136	2727 mL		4091 mL
310	141	2818 mL		4227 mL
320	145	2909 mL		4364 mL
330	150	3000 mL		4500 mL
340	155	3091 mL		4636 mL
350	159	3182 mL		4773 mL
360	164	3273 mL		4909 mL
370	168	3364 mL		5045 mL
380	173	3455 mL		5182 mL
390	177	3545 mL		5318 mL
400	182	3636 mL		5455 mL

Pediatric:		Cardiogenic	Hypovolemic Sepsis Anaphylactic
		10 mL/kg	20 mL/kg
		volume	volume
lbs	kg		
6.6	3	30 mL	60 mL
8.8	4	40 mL	80 mL
11	5	50 mL	100 mL
13	6	60 mL	120 mL
15	7	70 mL	140 mL
18	8	80 mL	160 mL
20	9	90 mL	180 mL
22	10	100 mL	200 mL
24	11	110 mL	220 mL
26	12	120 mL	240 mL
29	13	130 mL	260 mL
31	14	140 mL	280 mL
35	16	160 mL	320 mL
40	18	180 mL	360 mL
44	20	200 mL	400 mL
48	22	220 mL	440 mL
53	24	240 mL	480 mL
57	26	260 mL	520 mL
62	28	280 mL	560 mL
66	30	300 mL	600 mL
70	32	320 mL	640 mL
75	34	340 mL	680 mL
79	36	360 mL	720 mL
84	38	380 mL	760 mL
88	40	400 mL	800 mL
92	42	420 mL	840 mL
97	44	440 mL	880 mL

ONDANSETRON

ZOFRAN

DRUG CLASS Anti-emetic, selective serotonin (5-ht3) receptor antagonist

MECHANISM OF ACTION:

Blocks the action of serotonin, a natural substance that may cause nausea and vomiting.

<u>CONTRAINDICATIONS</u>	<u>PROTOCOLS USED</u>	<u>SIDE EFFECTS</u>
Pregnancy	Pediatric Nausea/Vomiting Overdose - Stimulant Adult Nausea/Vomiting	- Blurred vision - Diarrhea in pediatrics

SUPPLIED

4mg/2mL vial (concentration 2mg/mL)
4mg dissolvable tablet

DOSAGE:

ADULT: **IV/IO:** 4 mg
ORAL: 4 mg, max 8mg

Pediatric: **IV/IO:** 0.1 mg/kg, max.
ORAL: ≥ 40kg: 4 mg
max 8mg

ADMINISTRATION TIME:

IV/IO: push over 30 seconds
ORAL: let dissolve on tongue

NOTE:
Do not administer to patients
< 10kg

REPEAT DOSING TIME:

5 minutes

ONSET:

IV/IO/ORAL: 1-2 minutes

DURATION:

2-4 hours

SODIUM BICARBONATE 8.4%

DRUG CLASS: Alkalinizing agent

MECHANISM OF ACTION:

In the presence of hydrogen ions, sodium bicarbonate dissociates to sodium and carbonic acid, the carbonic acid picks up a hydrogen ion changing to bicarbonate and then dissociates into water and CO2, functioning as an effective buffer and alkalinizing the blood.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
- Metabolic or respiratory alkalosis - Severe vomiting/GI suctioning - In cardiac arrest - non-dialysis patients	Adult Ventricular Fibrillation / Tachycardia w/o Pulse Crush Syndrome Adult Asystole/PEA Adult Beta Blocker OD	- Edema - Hypocalcemia - Tetany - Gastric distension - Metabolic alkalosis

SUPPLIED

50mEq/50mL preload syringe (concentration 1mEq/mL)

DOSAGE:

ADULT: IV/IO: 50mEq

Pediatric: NONE

ONSET:

IV: 2-10 minutes

DURATION:

30 - 60 minutes

NOTE:

This is a OLMC order for [Crush Syndrome](#)

TRANEXAMIC ACID

TXA

DRUG CLASS: Antifibrinolytic

MECHANISM OF ACTION:

Inhibits plasminogen activation and plasma activity. Helps prevent the breakdown of clots. Is a synthetic equivalent of the amino acid lysine. Helps to reduce mortality in the acutely hemorrhaging adult trauma and OB patient if given within three hours of injury or as soon as post-partum hemorrhage is apparent.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
< 16 years old - Injury > 3 hours old - Isolated head injury - Active intravascular clotting (DIC) - PMH of thromboembolism PMH - of renal failure	Musculoskeletal Trauma Head / Spinal Trauma Delivery Complications - Bleeding	- Anaphylaxis - Thrombosis - Nausea/vomiting/diarrhea - Visual disturbances: blurred vision, changes in color - Hypotension with rapid administration

ON LINE MEDICAL CONTROL ORDER ONLY

SUPPLIED

1 gm/10mL vial (concentration 0.1gm/mL)

DOSAGE:

ADULT: IV/IO: 1 gm

Pediatric: NONE

ADMINISTRATION TIME:

in 100mL D5W over **10 minutes**

ONSET:

IV/IO: 5-15 minutes

DURATION:

IV/IO: 3 hours

MIXING INSTRUCTIONS:

- Add dosage of TXA to 100 mL D5W
- Rotate bag to mix
- Insert macro drop tubing
- Run at rate based on table below

	TUBING		
	10 gtt	15 gtt	20 gtt
gtt/min	100/min	150/min	200/min
gtt/sec	1.6/sec	2.5/sec	3.3/sec

NOTE:

Trauma patients receiving TXA should be a Category I Trauma and transported to a Level 1 Trauma Center. The receiving facility must be prepared to continue the course of treatment.

VERAPAMIL

DRUG CLASS: Calcium channel blocker

MECHANISM OF ACTION:

Inhibits transmembrane influx of extracellular calcium ions across membranes of myocardial cells and vascular smooth muscle cells without changing serum calcium concentrations, resulting in inhibition of cardiac and vascular smooth muscle contraction and thereby dilating main coronary and systemic arteries. Decreases conduction through SA and AV nodes.

CONTRAINDICATIONS	PROTOCOLS USED	SIDE EFFECTS
2nd or 3rd degree heart block - Hypotension - Severe CHF exacerbation - Sick sinus syndrome - WPW Syndrome Wide Complex Tachycardia	Adult Supraventricular Tachycardia Adult Atrial Fibrillation / Flutter	- Headache - Dizziness - Bradycardia, palpitations - Hypotension - AV block - Nausea

SUPPLIED

10mg/4mL (concentration 2.5mg/mL)

DOSAGE:

ADULT: IV/IO: 5 mg, max. 10 mg

Pediatric: NONE

ADMINISTRATION TIME:

IV/IO: SLOW push over 5 minutes

REPEAT DOSING TIME:

15 minutes

ONSET:

IV/IO: 1-5 minutes

DURATION:

IV/IO: 10-20 minutes

DRUG SHORTAGE PROCEDURES

Unless otherwise specified, all medications are IV/IO slow push over 60 seconds.

Drug Assisted Intubation		
Drug Shortage: <u>KETAMINE</u>		
	Induction	Post-Intubation Sedation
Adult	ETOMIDATE 0.3 mg/kg (max 40 mg)	MIDAZOLAM 2 mg (max 10 mg) AND FENTANYL 50 mcg (max 200 mcg) Repeat combination every 15 mins as needed to achieve Target: RASS -2
Pediatric	(≥ 10 y/o) ETOMIDATE 0.3 mg/kg (max 20 mg) (< 10 y/o) MIDAZOLAM 0.1 mg/kg (max 10 mg)	MIDAZOLAM 0.1 mg/kg (max 10 mg) AND FENTANYL 0.1 mcg/kg (max 100 mcg) Repeat combination every 15 mins as needed to achieve Target: RASS -2
Drug Shortage: <u>ETOMIDATE and KETAMINE</u>		
	Induction	Post-Intubation Sedation
Adult	MIDAZOLAM 0.2 mg/kg (max 20 mg)	MIDAZOLAM 2 mg (max 10 mg) AND FENTANYL 50 mcg (max 200 mcg) Repeat combination every 15 mins as needed to achieve Target: RASS -2
Pediatric	MIDAZOLAM 0.1 mg/kg (max 10 mg)	MIDAZOLAM 0.1 mg/kg (max 10 mg) AND Fentanyl 0.1 mcg/kg (max 100 mcg) Repeat combination every 15 mins as needed to achieve Target: RASS -2

BEHAVIORAL EMERGENCIES

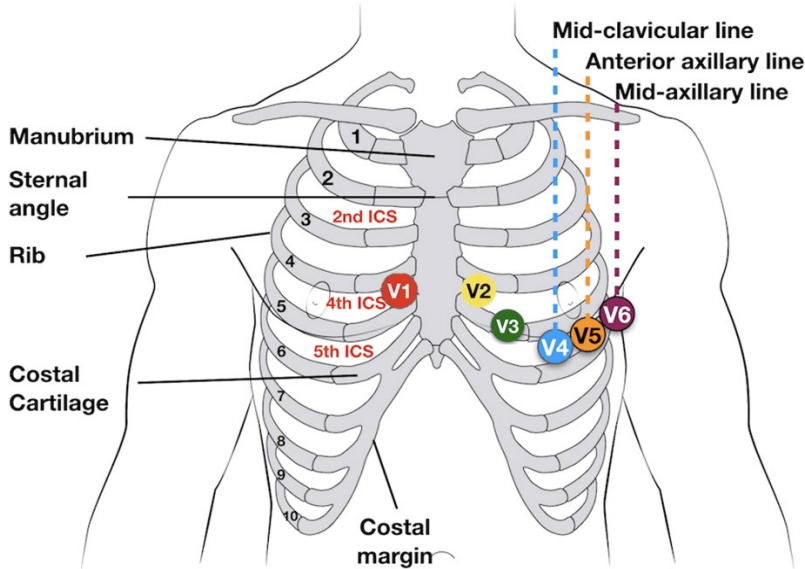
Drug Shortage: <u>KETAMINE</u>	
Adult	MIDAZOLAM 10mg IM
Pediatric	MIDAZOLAM 0.2 mg/kg IM (max single dose 10 mg)

12 LEAD ECG

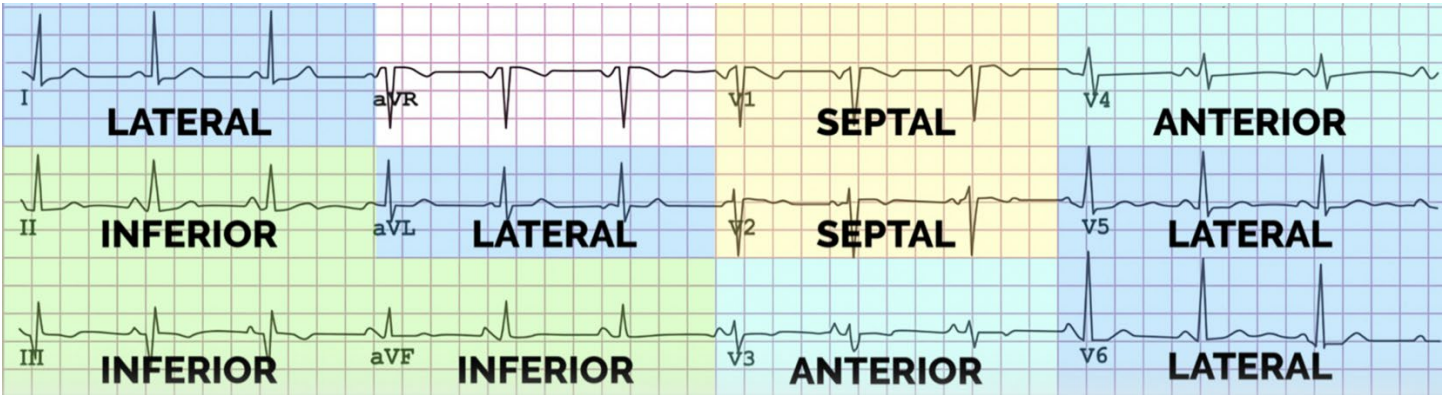
INDICATIONS / CONSIDERATIONS:

- 1. Discomfort/Pain:
chest, jaw, neck, shoulder, arm, back, epigastric
- 2. Anginal Equivalent / Atypical Presentation:
SOB, diaphoresis, dizziness, fatigue, weakness, syncope, nausea, vomiting, palpitations
- 3. PMH and Risk Factors
Age, diabetes, heart failure, high cholesterol, hypertension, previous MI, smoking

LEFT SIDE LEAD PLACEMENT:



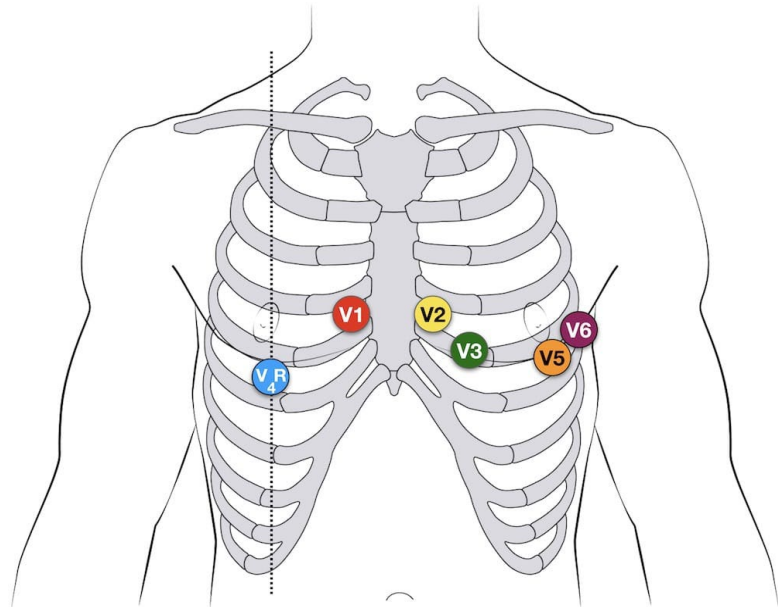
V1:	4th intercostal space, right of sternum
V2:	4th intercostal space, left of sternum
V3:	Midway between V2 & V4
V4:	5th intercostal space, midclavicular
V5:	Midway between V4 & V6
V6:	5th intercostal space, midaxillary



MI LOCATION	ELEVATION	DEPRESSION
Lateral	I, aVL, V5, V6	II, III, aVF
Inferior	II, III, aVF	I, aVL, V5, V6
Septal	V1, V2	None
Anterior	V3, V4	None
Posterior	None	V1, V2, V3, V4

Ischemia:	Hyperacute T waves, T wave inversion, ST depression
Injury:	ST elevation > 1mm in 2 or more contiguous leads
Infarction:	Injury ST elevation and reciprocal changes, deep wide Q waves

RIGHT SIDED PLACEMENT: (V4R)



V1:	4th intercostal space, right of sternum
V2:	4th intercostal space, left of sternum
V3:	Midway between V2 & V4
V4:	5th intercostal space, right midclavicular line
V5:	Midway between V4 & V6
V6:	5th intercostal space, midaxillary

INDICATIONS:

- ST elevation in inferior leads (II, III, aVF) - this indicates blockage in the Right Coronary Artery. Moving V4 to the right side chest evaluates for right ventricular injury/infarction. Consider right ventricular involvement if ST elevation of > 1mm observed in V4 in a right side 12 lead.

AMPUTATED AND AVULSED PARTS

1. Adult/Pediatric Routine Trauma Care
2. If gross contamination, gently rinse stump and amputated part with NORMAL SALINE.
3. Control bleeding with direct pressure. Utilize hemostatic gauze if available. If needed, apply a tourniquet.
4. Cover stump with sterile NS moistened dressing, additional sterile dressing, and wrap with elastic bandage (ACE wrap) providing uniform pressure over entire stump.
5. Amputated part:
 - A. In case of incomplete amputation, splint in physiologic position.
 - B. Wrap part in NS moistened gauze and place in a plastic bag/container.
 - C. Place bag/container in in a larger bag or container with cold packs or ice and water.
 - D. Do not allow parts to come in direct contact with ice/cold packs or submerge in NS.

CRICOTHYROTOMY - NEEDLE

PROTOCOLS USED:

- [Pediatric Advanced Airway](#)
- [Pediatric Airway Obstruction](#)

INDICATIONS:

- Failure to intubate, failure to ventilate with a BVM
- Other devices or rescue techniques have failed/not available
- Pediatric < 10 years old

CONTRAINDICATIONS:

- Ability to secure airway with less invasive methods
- Airway trauma that makes cricothyroid membrane access futile - laryngeal fracture, tracheal transection

EQUIPMENT:

- 10-14G IV catheter
 - 3.0 ETT adapter
- 10mL syringe

	EMR	EMT	PM
1. Continue ALS care as indicated by specific protocol			•
2. Place patient in supine position. Hyperextend neck to make anatomy accessible. Consider towel roll under neck			•
3. Stabilize larynx/trachea with non-dominant hand using the thumb and middle finger. Locate and palpate the cricothyroid membrane between the thyroid and cricoid cartilage.			•
4. Prepare the area with betadine or similar antiseptic.			•
5. Attach 10mL syringe to 10-14G IV catheter. Insert IV catheter at a 45-degree angle through cricothyroid membrane into the trachea.			•
6. Aspirate air through syringe to confirm entry of needle into trachea			•
7. Hold needle/syringe firmly. Slide plastic cannula forward into trachea until flush against neck			•
8. Remove needle and syringe			•
9. Attach 3.0 ETT adapter to syringe hub			•
10. Connect BVM to ETT adapter.			•
11. Confirm placement with EtCO2. Secure hub to neck.			•

NOTE:

1. The Gold Standard in confirmation of placement is Capnography. Secondary methods include auscultation, bilateral chest rise and fall, fogging of tube, pulse oximetry.

CRICOTHYROTOMY - SURGICAL

PROTOCOLS USED:

- [Adult Drug Assisted Intubation](#)
- [Adult Airway Obstruction](#)
- [Pediatric Advanced Airway](#)
- [Pediatric Airway Obstruction](#)

INDICATIONS:

- Airway cannot be secured by non-surgical methods
- Other devices or rescue techniques have failed/not available

CONTRAINDICATIONS:

- Pediatric patient < 10 years old
- Ability to secure airway with less invasive methods
- Airway trauma that makes cricothyroid membrane access futile - laryngeal fracture, tracheal transection

EQUIPMENT:		OPTIONAL:
10 blade scalpel 5.5-6.0 ETT - Lubricant	- ETT holder 10mL syringe - Gauze	- Bougie - Tracheal hook - Pediatric Magill forceps

	EMR	EMT	PM
1. Continue ALS care as indicated by specific protocol			•
2. Place patient in supine position. Extend neck to make anatomy accessible			•
3. Stabilize larynx/trachea with non-dominant hand using the thumb and middle finger. Locate and palpate the cricothyroid membrane between the thyroid and cricoid cartilage.			•
4. Prepare the area with betadine or similar antiseptic.			•
5. Using the dominant hand, make a 0.5-1 inch vertical incision through the skin, exposing the cricothyroid membrane. Do not puncture through the membrane yet.			•
6. Using the non-dominant hand (pointer finger) re-palpate the cricothyroid membrane.			•
7. Using dominant hand, make a horizontal incision through the center of the cricothyroid membrane. Extend the horizontal incision to each interior edge of the cricoid cartilage.			•
8. Place tool into incision (may be pediatric Magill forceps, curved hemostat, bougie, tracheal hook, or non-dominant finger). Once placed, remove scalpel			•
9. Confirm tool is properly placed in tracheal lumen - ensuring it is not in false lumen or through posterior trachea.			•
10. Pass ET tube (over bougie if using) and intubate trachea			•
11. Remove bougie if using, connect BVM with EtCO2 and verify			•
12. Inflate balloon and secure with commercial device. Continue to monitor with Capnography throughout.			•

NOTE:

1. Ignore any bleeding until ET tube has been placed. Placement typically has a tamponade effect.
2. Take care not to penetrate through the posterior wall of the trachea during the horizontal or vertical incision.
3. If using a bougie, insert and aim towards feet. Verify it is not inserted into a false lumen. Bougie should stop at carina < 4 inches from incision.
4. The Gold Standard in confirmation of placement is Capnography. Secondary methods include auscultation, bilateral chest rise and fall, fogging of tube, pulse oximetry.

I-GEL AIRWAY

INDICATIONS: Adult or pediatric patient, unresponsive medical or trauma without a gag reflex. BVM ventilation ineffective.

CONTRAINDICATIONS: Intact gag reflex. Known esophageal disease, caustic substance ingestion.

PROCEDURE:

1. Pre-oxygenate patient with BVM with 100% oxygen and appropriate airway adjunct (OPA or NPA).
2. Suction airway as needed.
3. Choose the correct size based on patient weight:

I-gel	Patient size	Patient weight (kg)
1.5	Infant	5-12
2	Small pediatric	10-25
2.5	Large pediatric	25-35
3	Small adult	30-60
4	Medium adult	50-90
5	Large adult	90+

4. Assemble and check equipment. Apply water-soluble lubricant to back, sides, and front of the cuff.
 - a. **NOTE:** Be sure that there is only a thin layer of lubricant on the end of the i-gel to avoid blowing it into the lungs.
5. Place patient in “sniffing” position or “jaw thrust” if cervical injury suspected. The chin should be pressed down before proceeding to insert the i-gel.
6. Grasp the lubricated i-gel firmly along the bite block. Position the device so the i-gel cuff outlet is facing toward the chin of the patient.
7. Introduce the leading soft tip into the mouth in a direction toward the hard palate. Glide the device downwards and backwards along the hard palate with a continuous, but gentle push until a definitive resistance is felt.
8. Teeth (incisors) should be resting on the integral bite block.
9. Confirm placement by auscultating breath sounds, check for chest rise and confirm placement with EtCO₂ monitoring and SpO₂ monitoring.
10. Secure the tube with head strap or (tape from maxilla to maxilla).
11. If required, a lubricated NG or suction catheter may be passed down gastric channel. Apply suction to decompress the stomach.
12. Continue to monitor, sedate per protocol as necessary.

INTRAOSSIOUS INFUSION EZ-IO®

Indications:

- Shock, arrest, or impending arrest; Unresponsive or conscious critical patient
- 2 unsuccessful IV attempts or 90 second duration or no visible sites

Contraindications:

- Fracture in same extremity or infection at insertion site
- Previous orthopedic procedures (knee replacement, previous IO within 48 hours)
- Pre-existing medical condition (tumor near site, peripheral vascular disease)
- Inability to locate landmarks (significant edema)

1. Locate and prep insertion site (proximal medial tibia or proximal humerus).
2. Prime EZ-connect tubing, leaving 9 mL of fluid in syringe.
3. Prepare EZ-IO driver and select appropriate needle set.

Patient	3-39 kg	40 kg	Humeral & Bariatric
Needle	15mm Pink	25mm Blue	45mm Yellow

4. Insert EZ-IO needle at 90 degrees through skin until needle stops at bone. (The line on the needle *closest to the hub* must remain visible. If the line is not visible, remove needle from skin and place band-aid over site. EZIO may not be utilized if the line is not visible.)
5. Activate driver by depressing trigger.
6. Once decreased resistance, or needle flange at skin (whichever first), release trigger.
7. While stabilizing needle hub, remove driver from needle set.
8. Remove stylet from needle by rotating counterclockwise.
9. Connect primed EZ-connect tubing.
10. Using syringe, aspirate then flush with remaining 9 mL of NS to confirm placement.
11. If conscious patient:
 - **ADULT:** inject LIDOCAINE 1mg/kg (max 50 mg) over 1-2 minutes. Wait 60 seconds before beginning IVF infusion. May repeat lidocaine x1 for pain control.
 - **PEDIATRIC:** inject LIDOCAINE 0.5mg/kg (max 40mg) over 1-2 minutes. Wait 60 seconds before beginning IVF infusion. May repeat lidocaine 0.25mg/kg over 30-60 seconds (max 20mg).
12. Remove the syringe, attach IV tubing, and begin infusion.
13. Apply pressure bag to IVF to facilitate infusion.
14. Secure tubing to extremity with tape.
15. If utilizing Proximal Humerus site, immobilize arm to limit movement.
16. Frequently reassess IV bag pressure, amount of fluid infused, site, and patient.

NEEDLE DECOMPRESSION, CHEST

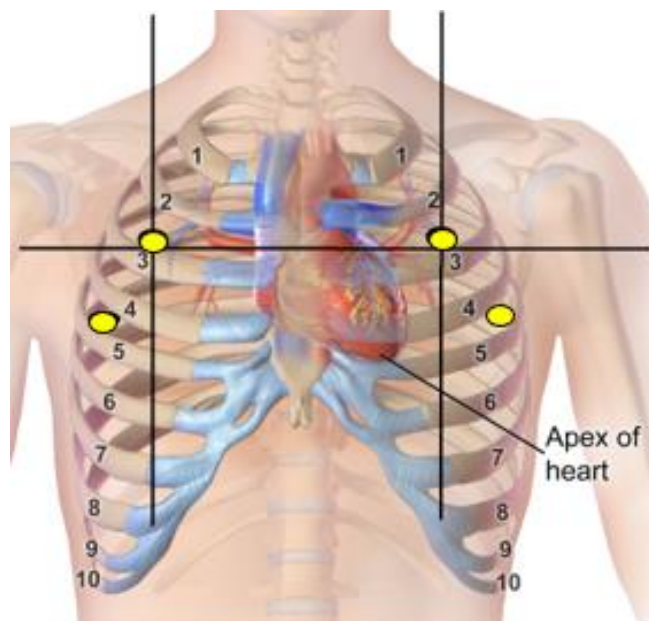
Indications

- Tension pneumothorax (hypotension with absent breath sounds during ventilation)
- Traumatic arrest with chest trauma

Contraindication

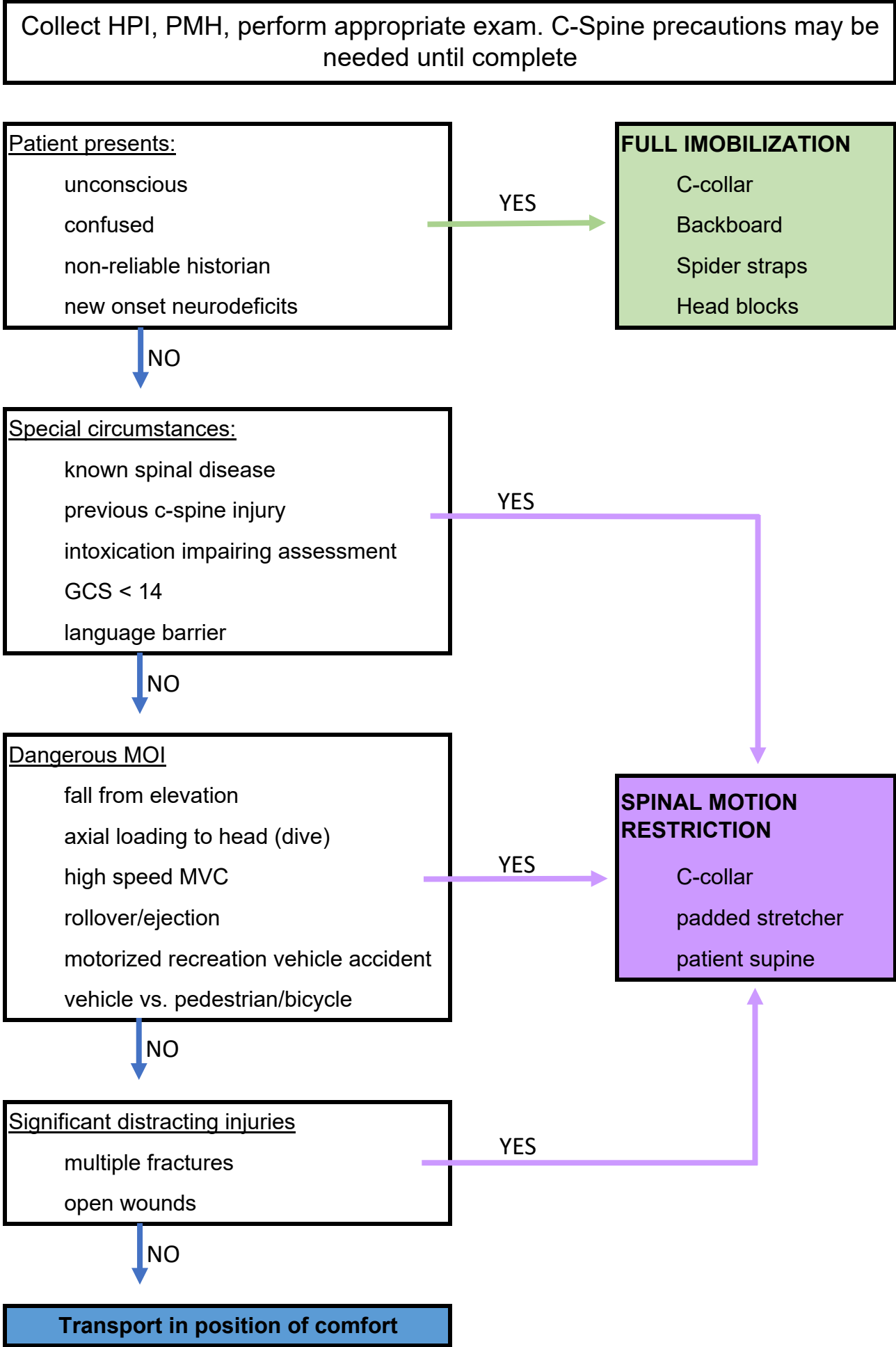
- Pneumothorax without hypotension

1. Locate and palpate site at 2nd intercostal space (above 3rd rib), mid-clavicular line.
2. Prep site appropriately.
3. Insert 10-14 gauge (2.5-3.25 inch) IV catheter @ 90° angle to chest wall, until air is released.
4. While holding needle steady, advance catheter over needle, until catheter hub at skin
5. Leave catheter in place and remove needle.
6. Assess for improvement in clinical status (e.g., increased BP).
7. Frequently reassess patient (esp. BP, lung sounds) and catheter patency; may need to repeat procedure if clinical deterioration or catheter occlusion occurs.
8. IF FAILED attempt OR PATIENT IS MARKEDLY OBESE, use 4th intercostal space lateral to inferior mammary fold mid axillary.



SPINAL MOTION RESTRICTION

(SMR)



NOTE:

- 1. A backboard may be used as a transfer device to remove the patient from the environment to the stretcher. If the patient does not meet the conditions for full immobilization, the backboard should be removed as soon as feasible.

SYNCHRONIZED CARDIOVERSION

PROTOCOLS USED:

- [Adult Supraventricular Tachycardia](#)
- [Pediatric Unstable Tachycardia](#)
- [Pediatric Stable Tachycardia](#)
- [Adult Rapid Atrial Fibrillation](#)
- [Adult Ventricular Tachycardia with a Pulse](#)

	EMR	EMT	PM
1. Continue ALS care as indicated by specific protocol			•
2. Place limb leads and combo pads in the anterior-posterior positioning Anterior (-): V4 positioning Posterior (+): just under (L) scapula, between spine and (L) scapula			•
3. Turn on Synch. Ensure indicators are shown above each QRS complex.			•
4. Select appropriate joule dosage based on chart below.			•
5. Once charged and patient is cleared, press and hold to shock			•
6. Document settings in patient care report			•

SUPRAVENTRICULAR TACHYCARDIA / NARROW COMPLEX TACHYCARDIA

ADULT:

<u>Zoll</u>	<u>LifePak</u>	<u>Phillips</u>
70J	50J	50J
120J	100J	100J
150J		150J
200J		

PEDIATRIC:

Initial dose:	1 J/kg
2nd dose:	2 J/kg

RAPID ATRIAL FIBRILLATION / FLUTTER

ADULT:

<u>Zoll</u>	<u>LifePak</u>	<u>Phillips</u>
120J	100J	100J
150J	200J	150J
200J	300J	200J
	360J	

VENTRICULAR TACHYCARDIA WITH A PULSE

ADULT:

<u>Zoll</u>	<u>LifePak</u>	<u>Phillips</u>
120J	100J	100J
150J	200J	150J
200J	300J	200J
	360J	

PEDIATRIC:

Initial dose:	1 J/kg
2nd dose:	2 J/kg

TARP ASSISTED COOLING WITH OSCILLATION

TACO METHOD

PROTOCOLS USED:

- [Exertional Heat Stroke](#)

INDICATIONS:

- AMS following strenuous athletic activities
- Rectal temperature > 104F following strenuous athletic activities

EQUIPMENT:

- Large tarp or body bag
- Ice
- Water

	EMR	EMT	PM
1. Continue care as indicated by specific protocol	•	•	•
2. Remove excess clothing or equipment from the patient	•	•	•
3. Unfold the tarp/body bag. Place patient in the middle of the tarp/body bag which will act as a make-shift tub	•	•	•
4. Pick up each corner of the tarp/body bag to form a sling or "taco" shape around the patient.	•	•	•
5. Slowly pour cool water and ice into the tarp/body bag up to the patient's chest	•	•	•
6. Gently move the tarp/body bag back and forth to agitate the water and start the cooling process	•	•	•
7. Monitor the patients vitals and rectal temperature. Only remove the patient once their rectal temperature reaches <102F or a resolution of AMS (if rectal thermometer is available).	•	•	•
8. Once patient is cooled to <102F, transport to the closest appropriate comprehensive emergency department.	•	•	•





Vertical line separator

TOURNIQUET USE

INDICATIONS:

- Severe/life-threatening extremity hemorrhage that continues after direct pressure and/or pressure dressing.
 - May be used first line for extremity amputation.
1. Place as far distally on extremity as possible, 2-3 inches proximal to wound.
 2. Place on bare skin if possible.
 3. Tighten windlass rod until bleeding stops and pulse is no longer palpable.
 4. Monitor for further bleeding, tighten tourniquet only if necessary.
 - If bleeding not controlled, consider additional tourniquet applied proximal to first tourniquet.
 5. Record placement time and notify MEDICAL CONTROL of tourniquet use.
 6. Consider pain management.

Tourniquet Removal: If placed prior to EMS arrival and determined to be unnecessary,

Contact MEDICAL CONTROL to consider Tourniquet removal.

NOTE:

- Do not place tourniquet over a joint.
- Do not cover the tourniquet with a dressing or splint.
- Do not cover wound, impaled foreign body, or open fracture.
- Lower leg injuries may require thigh placement for adequate compression.

TRANSCUTANEOUS PACING

PROTOCOLS USED:

- [Adult Bradycardia](#)
- [Pediatric Bradycardia](#)
- [Calcium Channel Blocker OD](#)
- [Beta Blocker OD](#)

	EMR	EMT	PM
1. Continue ALS care as indicated by specific protocol			•
2. Place limb leads and combo pads in the anterior-posterior positioning Anterior (-): V4 positioning Posterior (+): just under (L) scapula, between spine and (L) scapula			•
3. Ensure pacer is set to DEMAND at a RATE 80/minute			•
4. Turn pacer on			•
5. Increase mA until electrical capture on monitor Electrical capture: 1 pacer spike/wide QRS complex			•
6. Verify mechanical capture at the same rate seen on the monitor Mechanical capture: femoral or (R) brachial/radial pulse corresponding to paced beats			•
7. Document settings in patient care report			•

NOTE:

1. Do not assess carotid or (L) brachial/radial pulses to determine mechanical capture due to interference of muscle contraction.

2020 AHA GUIDELINES

Component	Neonate 0-28 days	Infant 28 days - 1 year	Children 1 year - puberty	Adults & Adolecents
Verify Scene Safety	Make sure the environment is safe for rescuers and victim			
Recognition	Check for responsiveness, absent/abnormal breathing Treat agonal gasps as absent breathing			
Compression: Ventilation NO advanced airway	3:1	1 rescuer: 30:2 ≥ 2 rescuers 15:2		30:2
Compression: Ventilation with advanced airway	Continuous compressions 1 breath/2-3 sec.			Continuous compressions, 1 breath/6 sec.
Compression rate	100 - 120 / min.			
Hand Placement	Lower 1/3 sternum, using 2 thumbs encircling the chest	1 rescuer 2 fingers in the center of the chest, just below the nipple line 2 rescuers 2 thumb-encircling the hands in the center of the chest, just below the nipple line	1-2 hands on lower half of sternum	2 hands on the lower half of the sternum
Compression depth	1/3 diameter of chest	1/3 diameter of the chest		
		~1.5 in.	~ 2 inchest	
Chest Recoil	Hands/thumbs should not leave the chest, allow complete recoil, do not lean after each compression			
Minimize interuptions	Limit interruptions in chest compressions to less than 10 seconds with a CCF goal of 80%			
Pediatric AED w/o attenuator	If manual defib and pediatric attenuator not available - utilize adult AED			
Inadequate breathing w/ pulse	40-60 breath/min. 1 breath/ 1-2 sec.	20-30 breaths/ min. 1 breath / 2-3 sec.		10 breath/min. 1 breath/ 6 sec.

AIRWAY OBSTRUCTION AHA GUIDELINES

	Neonate	Infant	Children	Adults/Adolecents
Conscious Cannot speak	Cycles of 5 backblows/ slaps, & 5 chest compression		Abdominal thrusts, in rapid sequence until relieved. Obese/Late Preg/Abd thrust ineffective: chest thrusts.	
Unresponsive	Begin CPR starting with chest compressions (no pulse check). Each time airway is opened, check mouth. Attempt removal only if obstruction is visualized. If unrelieved by manual maneuvers, suction, magill forceps, and unable to ventilate, consider cricothyrotomy.			

APGAR SCORING

	0	1	2
Appearance (skin color)	Blue; Pale	Blue hands or feet	Completely pink
Pulse	Absent	< 100/min	> 100/min
Grimace (reflex irritability)	Absent	Grimace	Cry or Active Withdrawal
Activity (muscle tone)	Limp	Some extremity flexion	Active motion
Respirations	Absent	Weak cry; Hypoventilation	Good, Strong Cry

Reference: American Academy Pediatrics, <https://pediatrics.aappublications.org/content/136/4/819>

The Apgar score is reported at 1 minutes and 5 minutes after birth for all infants, and at 5-minute intervals thereafter until 20 minutes for infants with a score less than 7.

ABBREVIATIONS, ACRONYMS and SYMBOLS

AAAabdominal aortic aneurysm
 Abd..... Abdomen
 ACS.....acute coronary syndrome
 AEDautomated external defibrillator
 AHAAmerican Heart Association
 AIDS acquired immune deficiency syndrome
 ALSAdvanced Life Support
 ALTE Apparent Life Threatening Event
 AMAagainst medical advice
 Amb.....ambulance
 AMI..... acute myocardial infarction
 Amp..... ampule
 AMS altered mental status
 A&O.....alert & oriented
 APGAR .. appearance, pulse, grimace, activity, respirations
 ASA aspirin
 ASAPas soon as possible
 AV atrioventricular
 AVPU..... mental status: alert, verbal, pain, unresponsive
 BB backboard
 BCP birth control pills
 BLSBasic Life Support
 bm bowel movement
 BOW Bag of water
 BP.....blood pressure
 BPM or bpm.....beats per minute
 BRUE Brief Resolved Unexplained Event
 bs breath sounds
 BSA.....body surface area
 BSIbody substance isolation
 BVM bag valve mask
 C..... Celsius or centigrade
 CA cancer
 c&a..... conscious and alert
 CABG coronary artery bypass graft
 CAD coronary artery disease
 CC chief complaint
 C-Collar..... cervical collar
 CHBcomplete heart block
 CHF.....congestive heart failure
 CID.....cervical immobilization device
 cm..... centimeter
 CMScirculation, motor, sensation
 CNS..... central nervous system
 c/o..... complains of
 CO carbon monoxide
 CO2 carbon dioxide
 COPD.....chronic obstructive pulmonary disease
 CP..... chest pain
 CPAP..... continuous positive airway pressure
 CPRcardiopulmonary resuscitation
 CPSS Cincinnati Prehospital Stroke Scale
 Cric..... cricothyrotomy
 C-Section.....caesarian section

CSF cerebral spinal fluid
 C-spine..... cervical spine
 CV..... cardiovascular
 CVA..... cerebral vascular accident
 D&C dilatation and curettage
 D/Cdiscontinue
 D5W5% dextrose in water
 DBPdiastolic blood pressure
 DCAPdeformity, contusion, abrasion, penetration
 DCFS Department of Children and Family Services
 Dept department
 Dig Digoxin
 DKA diabetic ketoacidosis
 DM..... diabetes mellitus
 DNA does not apply
 DNR..... do not resuscitate
 DOA dead on arrival
 DOE..... dyspnea on exertion
 Drsgdressing
 DTs delirium tremens
 DVT..... deep vein thrombosis
 Dx diagnosis
 ECG or EKG..... electrocardiogram
 ECRN.....Emergency Communications RN
 ED..... emergency department
 EDC estimated date of confinement
 EDD esophageal detector device
 EMSEmergency Medical Services
 EMT.....Emergency Medical Technician
 ENT ear, nose and throat
 ET endotracheal
 EtCO2end tidal carbon dioxide
 ETA estimated time of arrival
 ETOH.....alcohol
 Exam.....examination
 F Fahrenheit
 FB foreign body
 FBO foreign body obstruction
 Fib..... fibrillation
 FHT..... fetal heart tones
 FiO2..... fraction of inspired O2 (% O2 delivered)
 Fr.....french (suction catheter diameter)
 Fx..... fracture
 GCSGlasgow Coma Score
 GERDgastro-esophageal reflux disease
 GI..... gastrointestinal
 gm gram
 gsw..... gunshot wound
 Gtt drops
 GUgenitourinary
 Gyngynecological
 h or hr..... hour
 H/A headache
 H2O water

HCO₃ bicarbonate
 HEPAhigh efficiency particulate airborne mask
 HFheart failure
 HHNhand held nebulizer
 HIV human immunodeficiency virus
 HLDhyperlipidemia
 HPIhistory of present illness
 HR heart rate
 HTN hypertension
 Hxhistory
 ICPintracranial pressure
 ICUintensive care unit
 i.d. internal diameter
 IDinfectious disease
 IDPH Illinois Department of Public Health
 IM intramuscular
 IN intranasal
 INH inhalation
 IO intraosseous
 IV intravenous
 IVF intravenous fluids
 IVPintravenous push
 IVPBintravenous piggy back
 IVR idioventricular rhythm
 j joules
 JVD jugular venous distension
 Kpotassium
 KEDKendrick extrication device
 Kg kilogram
 KVOkeep vein open
 L liter
 Lbplow back pain
 Lbs pounds
 LLE left lower extremity
 LLQleft lower quadrant
 L/minuteliters per minute
 LMPlast menstrual period
 LOC level of consciousness
 L or Ltleft
 LUE left upper extremity
 LUQleft upper quadrant
 LVleft ventricle
 LVAD left ventricular assist device
 mAmilliamperes (pacing)
 MADTM Mucosal Atomization Device
 MAEWmoves all extremities well
 MAPmean arterial pressure
 mcg microgram
 mcgtsmicrodrops
 MCI mass casualty incident
 MERCI Medical Emergency Radio Comm. of Illinois
 mEq milliequivalents
 mg milligram(s)
 minminute
 MImyocardial infarction
 mLmilliliter(s)
 mmHg millimeters of mercury
 MOI mechanism of injury
 MPMPmultiple patient management plan
 m/s musculoskeletal

MS morphine sulfate
 MVCmotor vehicle crash
 n/a not applicable
 NADno apparent distress
 NCnasal cannula
 NEB nebulizer
 NKA no known allergies
 NP/NPAnasopharyngeal airway
 NPO nothing by mouth
 NRBM non-rebreather mask
 NSnormal saline
 NSAID nonsteroidal anti-inflammatory drug
 NSR normal sinus rhythm
 NTG nitroglycerine
 N/V nausea/vomiting
 O₂oxygen
 OBobstetric
 ODoverdose
 OLMCOn-Line Medical Control
 OP/OPAoropharyngeal airway
 Oriented X 1oriented to person
 Oriented X 2 oriented to person, place
 Oriented X 3 oriented to person, place, time
 Oriented X 4oriented to person, place, time, event
 Ppulse
 PAC premature atrial contraction
 PALS Pediatric Advanced Life Support
 PCN penicillin
 pCO₂ or PaCO₂partial pressure of carbon dioxide
 PCRpatient care record
 PEA pulseless electrical activity
 PEEP positive end expiratory pressure
 PERL pupils equal and reactive to light
 PID pelvic inflammatory disease
 PJC premature junctional contraction
 Pmh past medical history
 PMSpulses, motor, sensory
 POper os (by mouth)
 pO₂partial pressure of oxygen
 POLST Practitioner Order for Life Sustaining Treatment
 PPE personal protective equipment
 PPV positive pressure ventilation
 PRI P-R interval
 Prn pro re nata or as needed
 Pt patient
 PTA prior to arrival
 PVC premature ventricular contraction
 Q every
 Rrespirations
 RAroom air
 Resp respiratory
 RLEright lower extremity
 RLQright lower quadrant
 RNRegistered Nurse
 R/O rule out
 ROM range of motion
 ROSC return of spontaneous circulation
 RR respiratory rate
 RSVrespiratory syncytial virus
 R or Rtright

RTS revised trauma score
 RUE..... right upper extremity
 RUQ right upper quadrant
 SA sinoatrial node
 SAMPLE symptoms, allergies, medications,
 past history, last oral intake, events leading up to illness
 SB..... sinus bradycardia
 SBP systolic blood pressure
 SCI spinal cord injury
 SCUBA self-contained underwater breathing apparatus
 SIDS sudden infant death syndrome
 SL sublingual
 SMV sensation, motor, vascular
 SOB..... shortness of breath
 SOP Standard Operating Procedure
 SpO2 pulse oximetry
 SQ or SC subcutaneous
 S&S..... signs & symptoms
 STAT..... immediately
 STD sexually transmitted disease
 SubQ or SQ subcutaneous
 SVT supraventricular tachycardia
 T temperature
 T1DM type 1 diabetes mellitus
 T2DM type 2 diabetes mellitus
 Tab tablet
 TB tuberculosis
 TBI traumatic brain injury
 TIA transient ischemic attack
 TIC tenderness, instability, crepitus
 TKO/KVO..... to keep open
 Tx treatment
 Unk..... unknown
 URI..... upper respiratory infection
 UTI..... urinary tract infection
 V-fib or VF..... ventricular fibrillation
 VS vital signs
 VSD..... ventricular septal defect
 V-tach or VT ventricular tachycardia
 w/..... with
 w/d..... warm and dry
 WPW..... Wolff-Parkinson White Syndrome
 Wt..... weight
 WNL within normal limits
 w/o..... without
 WOB work of breathing
 y/o year old
 @ at
 ° degree
 #..... number
 ↑ or ↓ increased or decreased
 < less than
 > greater than
 ≥ equal to or greater than
 ≤ equal to or less than
 + positive or plus

APPENDIX III: AREA-WIDE HOSPITALS

REGION X RHCC

Hospital	Medical Control Phone Number		ER Phone Number	Carepoint Fax
Highland Park Hospital	(847) 432-2294	(847) 432-2295	(847) 480-3751	(847) 926-5325

REGION X RESOURCE HOSPITALS

Hospital	Medical Control Phone Number		ER Phone Number	Carepoint Fax
Condell Medical Center	(847) 362-2963	(847) 573-4258	(847) 990-5300	(847) 990-2992
Highland Park Hospital	(847) 432-2294	(847) 432-2295	(847) 480-3751	
Lake Forest Hospital	(847) 535-7375		(847) 535-6150	(847) 535-7376
St. Francis Hospital	(847) 864-6564	(847) 864-8550	(847) 316-4000	
Vista Medical Center East	(847) 360-4234		(847) 360-4181	

LEVEL I TRAUMA CENTERS / COMPREHENSIVE EMERGENCY ROOM

Hospital	Medical Control Phone Number		ER Phone Number	Carepoint Fax
Condell Medical Center	(847) 362-2963	(847) 573-4258	(847) 990-5300	
Evanston Hospital	(847) 492-9453	(847) 492-1457	(847) 570-2111	(847) 570-2932
Lutheran General Hospital	(847) 696-0743	(847) 696-9073	(847) 723-7722	
Saint Francis Hospital	(847) 864-6564	(847) 864-8550	(847) 316-4000	

LEVEL II TRAUMA CENTERS / COMPREHENSIVE EMERGENCY ROOM

Hospital	Medical Control Phone Number		ER Phone Number	Carepoint Fax
Alexian Brothers Medical Ctr.	(847) 437-8118	(847) 952-7454	(847) 981-3599	
Glenbrook Hospital	(847) 729-9260	(847) 657-6010	(847) 657-5632	(847) 657-5960
Good Shepherd Hospital	(847) 385-9525		(847) 842-4444	
Highland Park Hospital	(847) 432-2294	(847) 432-2295	(847) 480-3751	(847) 926-5325
Lake Forest Hospital	(847) 295-1440		(847) 535-6150	
Northwest Community Hospital	(847) 259-8720		(847) 618-3913	
Saint Alexius Medical Center	(847) 843-5308		(847) 843-5309	(847) 490-6930
Vista Medical Center East	(847) 360-4234			(847) 360-4181
Froedtert Pleasant Prairie (WI)	(262) 697-5563			(262) 577- 8202

COMPREHENSIVE EMERGENCY ROOM

Hospital	Medical Control Phone Number		ER Phone Number	Carepoint Fax
Aurora Medical Center (WI)	(262) 694-1968	(262) 694-1973	(262) 948-5640	
Captain James A. Lovell FHCC	(224) 610-1442	(224) 610-1076	(224) 610-5505	(224) 610-5306
Resurrection Medical Center	(773) 774-8455		(773) 990-5255	
Skokie Hospital	(847) 674-2665	(847) 674-2694	(847) 933-6950	

STANDBY EMERGENCY ROOM

Hospital	Medical Control Phone Number		ER Phone Number	Carepoint Fax
Midwestern Regional Med Ctr.			(847) 872-6220	.

REGION X FREESTANDING EMERGENCY CENTERS

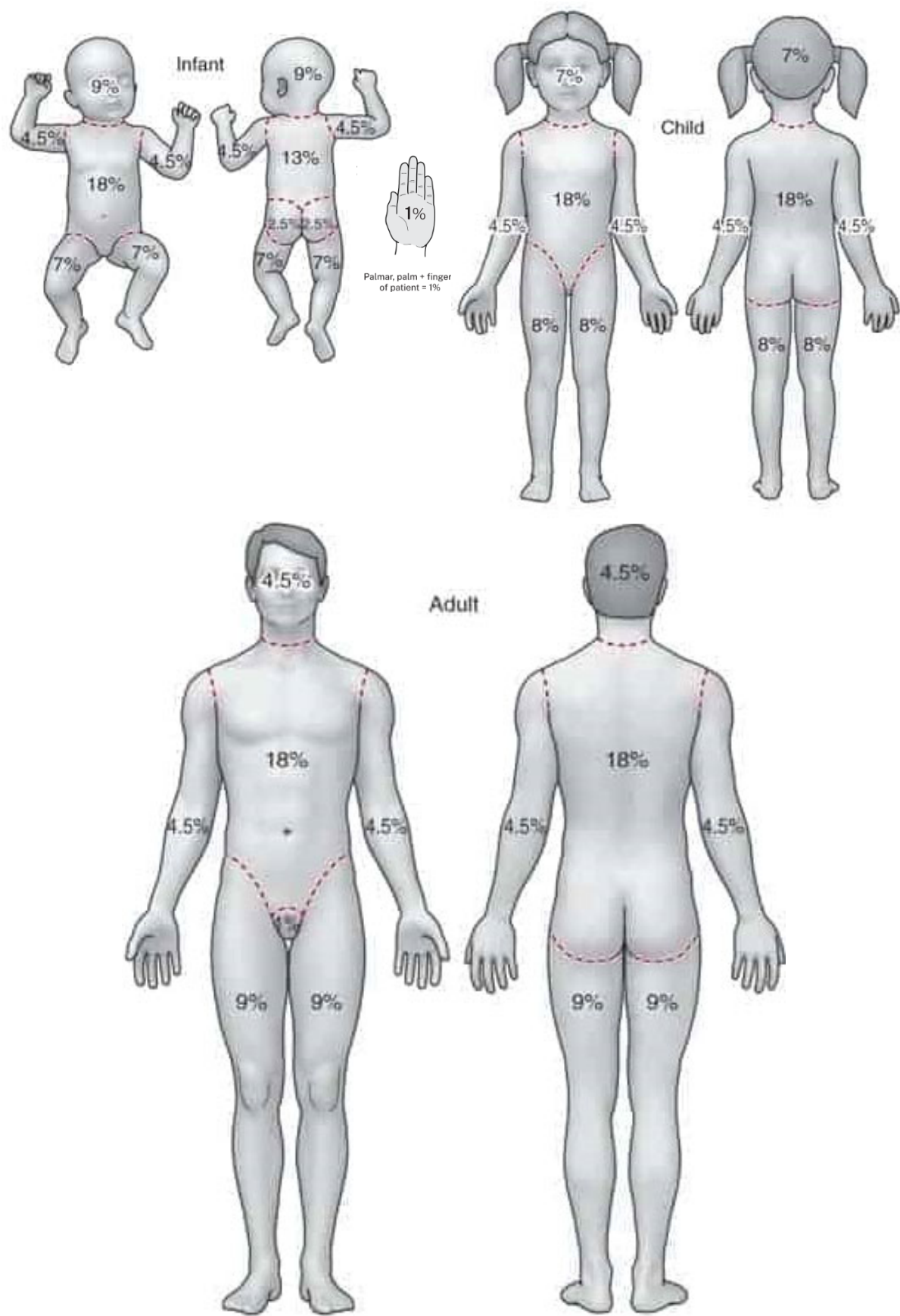
Hospital	Medical Control Phone Number		ER Phone Number	Carepoint Fax
Northwestern Grayslake	(847) 535-8736		(847) 535-8950	
Vista Lindenhurst	(847) 356-4782		(847) 356-4705	

REGION 9 RHCC HOSPITAL (Information Only)

Hospital	Medical Control Phone Number		ER Phone Number	Carepoint Fax
Sherman Hospital			(847) 429-2950	

BODY SURFACE BURN
PERCENTAGE CALCULATION

Body Location	Infant 28 days - 1 year	Children 1 year - puberty	Adults & Adolescents
Full head & neck	18%	14%	9%
Upper back	9%	9%	9%
Lower back	9%	9%	9%
Anterior chest	9%	9%	9%
Anterior abdomen	9%	9%	9%
Full upper extremity	9%	9%	9%
Full lower extremity	13.50%	16%	18%
Genitalia	1%		1%
patient's palm = 1% BSA			



CYANOKIT

INDICATIONS: Adult with cyanide exposure (inhalation, ingestion, dermal) with symptoms of:

CNS: Headache, Dilated pupils, AMS, Confusion, Sensorium change, Seizures

Resp: Dyspnea, Tachypnea/early, Bradypnea/late

CV: Chest tightness, Hypertension/early, Hypotension/late, Shock

GI: Nausea, Vomiting

1. Adult Routine Medical Care. **NOTE:** pulse oximetry may be inaccurate.
2. Cyanokit is not compatible with many other medications; 2 IV lines required. Do not administer other drugs in the same intravenous line as Cyanokit.
3. Decontamination should be concurrent with initial resuscitation.
4. Cyanokit contains 1 vial of HYDROXOCOBALAMIN 5 gm, place in upright position.
5. Reconstitute: Using transfer spike add 200 mL of NORMAL SALINE to vial.
6. Mix: Invert or rock vials for at least 60 seconds to mix contents, do not shake.
7. Administer vial IV over 15 minutes, approximately 15 mL per minute.
8. Do not delay transport.
9. Notify ED as early as possible.

FUNCTIONAL NEEDS

CARE OF PATIENTS WITH FUNCTIONAL NEEDS

Patients with special health care (functional) needs may require reasonable modification to policies, practices, and procedures. Patients may be dependent on durable medical equipment, supplies, and assistance services. They may have physical, sensory, mental health, cognitive and/or intellectual disabilities affecting their ability to function independently; others that may benefit include women in late stages of pregnancy, geriatric patients, and those needing bariatric equipment. Communicate with caregiver/parent for information, and to assist with care. Confirm patient's baseline assessment with the caregiver.

Tracheostomy:

- Evaluate for displacement, obstruction, pulmonary problem, equipment issue.
- May ventilate/oxygenate via a BVM with a tracheostomy adapter or mask over stoma.
- If unable to ventilate, cover opening and ventilate with BVM over nose and mouth.
- Suction as needed.

Stoma:

Consider infant/child mask over stoma for ventilation; seal mouth and nose if air escaping.

Left Ventricular Assist Device (LVAD): Battery operated, mechanical pump surgically implanted next to the heart. Tube pulls blood from LV into pump to send blood directly into aorta.

- Patient Assessment
 - Evaluate perfusion based on mental status, skin condition, EtCO₂
 - May (or may not) have peripheral pulse/BP
 - ECG waveform may have artifact and or be flat
 - SpO₂ may be unreliable
 - Usually taking anticoagulant medication
- Patient Treatment
 - Treat altered mental status, respiratory distress, and hypoperfusion per SOP.
 - If unconscious and non-breathing, CHEST COMPRESSIONS are ALLOWED
 - May be defibrillated without disconnecting pump; do not defibrillate over pump.
 - Defibrillate at nipple line or above. Anterior-posterior pad placement preferred
 - Often have pacemaker and/or Implantable Cardioverter Defibrillator (ICDs)
 - LVAD Coordinator may be contacted for further information
- Equipment
 - NEVER remove both sources of power/batteries at same time.
 - Avoid water submersion or contact with strong magnets or magnetic fields
 - Transport specialized equipment, extra battery pack, charger and cords with patient

Other technology-assisted special needs:

If possible, transport specialized equipment, emergency information, and medications to emergency department with patient.

GLASGOW COMA SCALE

	Adult	> 2 years	< 2 years	
EYE Opening	Spontaneous	Spontaneous	Spontaneous	4
	To Voice	To Voice	To Voice	3
	To Pain	To Pain	To Pain	2
	None	None	None	1
VERBAL Response	Oriented	Oriented, Appropriate words	Coos, Babbles, Appropriate words	5
	Confused	Confused	Irritable, Cries but consolable	4
	Inappropriate	Inappropriate words, Persistent cry	Cries to pain, Inconsolable	3
	Incomprehensible	Incomprehensible sounds	Moans to pain	2
	None	None	None	1
MOTOR Response	Obeys Commands	Obeys Commands	Normal Spontaneous Movements	6
	Localizes pain	Localizes pain	Withdraws from Touch	5
	Withdraws from pain	Withdraws from pain	Withdraws from Pain	4
	Flexion to pain	Abnormal Flexion	Abnormal Flexion	3
	Extension to pain	Abnormal Extension	Abnormal Extension	2
	None	None	None	1

CARE OF PATIENTS WITH GRAFTS OR FISTULAS

Arteriovenous (AV) Fistula. Most common type of access, created surgically by connecting an artery directly to a vein, bypassing the capillaries. Pressure from arterial flow causes enlargement of the veins and a bulging vessel often seen through the skin. This allows easy, dependable access for hemodialysis.

Arteriovenous (AV) Graft. A synthetic plastic tube is used to surgically connect an artery directly to a vein, bypassing the capillaries.

Central Venous Catheter (CVC). Usually a temporary device, a catheter often inserted in the internal jugular vein, that has two ports, one to remove blood and the other to return blood to the body, during hemodialysis. Catheter related infection is a common, serious complication.

AV Graft or fistula

1. Do not take a blood pressure or start an IV on the arm where a graft or fistula is present.
2. If a graft or fistula is bleeding, apply direct pressure and transport the patient.
3. In cardiac arrest, graft or fistula may be used for IV access. **Contact MEDICAL CONTROL** for further direction.

Peritoneal dialysis (also called continuous ambulatory peritoneal dialysis/CAPD)

Dialysis fluid is instilled into peritoneal (abdominal) cavity through a surgically implanted catheter. The fluid is kept in the abdominal cavity for a prescribed period of time, and is then drained out, carrying out waste products.

1. Do not disconnect the CAPD bags from catheter.
2. Do not infuse any fluids or medications directly into catheter.
3. Transport patient with CAPD intact, maintaining drainage bag lower than waist height.

PATIENT CONTAMINATED WITH A HAZARDOUS MATERIAL

Use proper PPE and containment procedures during entire contact with the patient(s), equipment and environment. Avoid self-injury.

Contact MEDICAL CONTROL early to allow receiving hospital(s) time to prepare for the contaminated patient(s). The hospital staff treatment of the patient(s) may be performed separate from the main Emergency Department area, possibly in the ambulance.

All attempts are to be made to decontaminate the patient prior to moving into ambulance.

1. If warranted, contact the Department/Regional HazMat Response Team/Illinois Poison Center for assistance.
2. Remove as much of the outermost layer of clothing as possible.
3. The usual decontamination solution is soap and water.
 - a. Refer to reference material for any variation to this solution i.e.: alkali.
4. If powdered/dry agent, brush excess before irrigating.
5. If possible, bring copy of the MSDS with the patient to the hospital.

CHEMICAL WEAPONS (vapor or liquid):

<u>NERVE AGENTS</u>	<u>BLISTER AGENTS</u>
Symptoms: Salivation, Lacrimation, Urination, Defecation, Gastrointestinal Distress, Emesis, Breathing Difficulty with Bronchospasm and Copious Secretions, Arrhythmias and Myosis (constricted pupils) (SLUDGE BAM)	Symptoms: Reddened skin, blistering, tearing, itching, CNS effect and respiratory failure
↓	
Duodote Auto-Injector Refer to Nerve Agent Exposure (see pg.77)	
Follow hazmat response protocols as above. Treat these materials as extremely toxic substances.	

BIOLOGICAL AGENTS (Anthrax/Botulism/Plague/Smallpox) Symptoms may include: Fever, chills, diarrhea, sore throat, swollen lymph nodes, malaise, cough, respiratory insufficiency or distress, and jaundice.

1. For all possible exposures wear appropriate PPE
2. If the patient is coughing, all rescuers to wear N95 mask and surgical mask on patient.
3. Cover any lesions with dressings to avoid spread of contaminant.

CYANIDE POISONING Symptoms may include hypotension, apnea and seizures.

1. **Contact MEDICAL CONTROL** as soon as possible.
2. Administer Cyanide antidote if available.

NOTE: Illinois Poison Center (800) 222-1222

HIGH RISK REFUSAL

Follow your System Policy - Use for Reference

DETERMINING DECISIONAL CAPACITY

1. Understanding: Is able to state understanding and meaning of information such as diagnosis, risk/benefits, and care options
2. Appreciation: Is able to explain how the information applies to self
3. Reasoning: Compares information and consequences
4. Expressing choice: Is able to express a decision

PATIENTS WITH DECISIONAL CAPACITY

- Patients with decisional capacity have the right to consent to, or refuse, some or all assessment, treatment, and/or transport.
- In all situations when a decisional patient refuses assessment, treatment, or transportation:
 1. Attempt to medically assess patient, including obtaining vitals
 2. Explain why treatment/transport is recommended
 3. Explain risks of refusal

HIGH RISK REFUSALS

EMS personnel should initiate contact with OLMC for approval of refusal for the following types of high-risk patients:

1. Received ALS assessment/treatment or would have met criteria for ALS transport, including L1-2 trauma
2. Patients under age 18, regardless of presence of parents/guardian on scene
3. Patients 65 years old and older
4. Altered mental status, under the influence of drugs/alcohol, or behavioral/psychological complaints
5. Obstetric patients
6. Any complex call where EMS personnel wish to seek additional consultation

EMS personnel will be given a log number by the ECRN or ECP who takes the call. This number must be documented in the patient care report.